# Simple signal plots for tuning

Source: supplied saved simulation output; recorded end time 34.790 s.

Angles use degrees; body angular rates use deg/s; wheel angular velocity uses rad/s.

Body accelerations use recorded inertial acceleration in vehicle axes: g converted with 9.80665 m/s^2 per g.

Vehicle axes rotate with the body (Z-down); tire-frame and suspension-frame signs are retained.

Suspension deflection is the block's recorded relative height, not a measured coil-spring displacement.

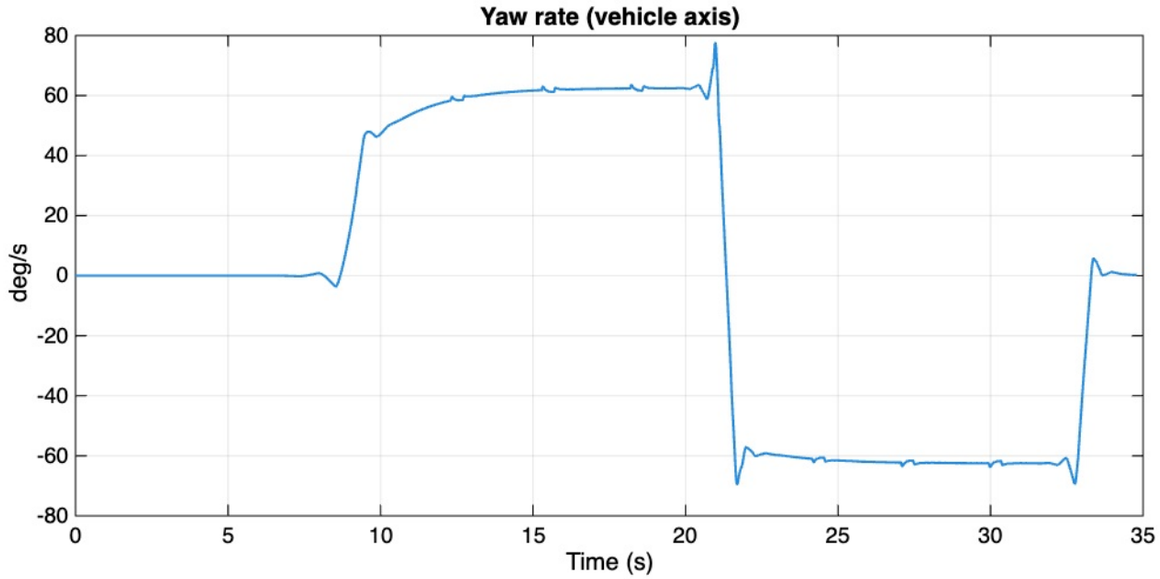
Each wheel metric has a comparison, four individual wheel plots, and a first-3-second startup view.

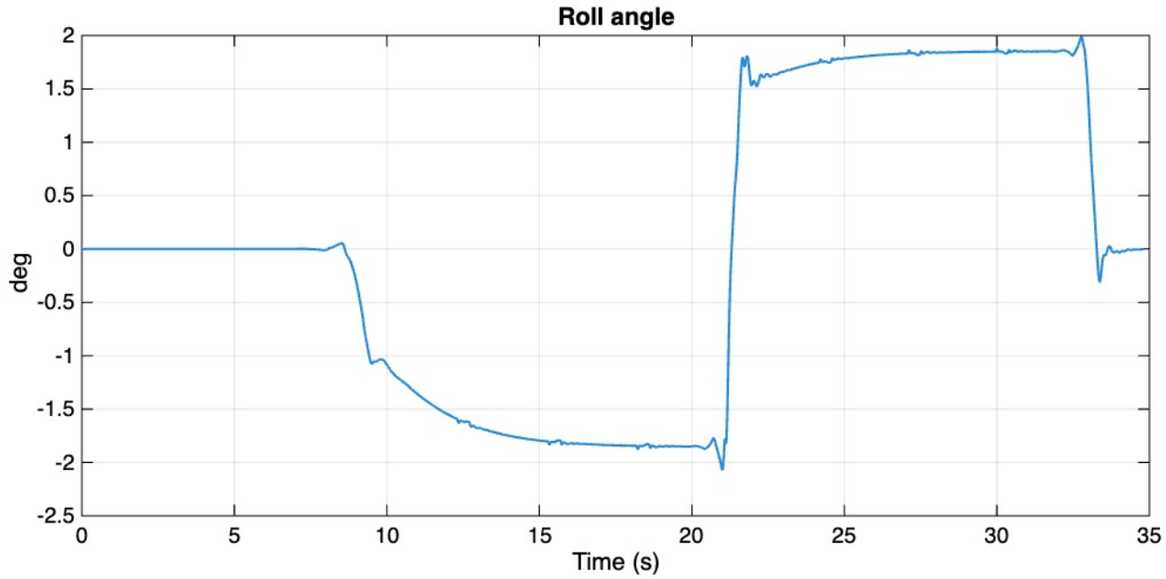
Steering-rack / handwheel torque feedback is NOT recorded. Tire aligning moment is shown separately and is not equivalent.

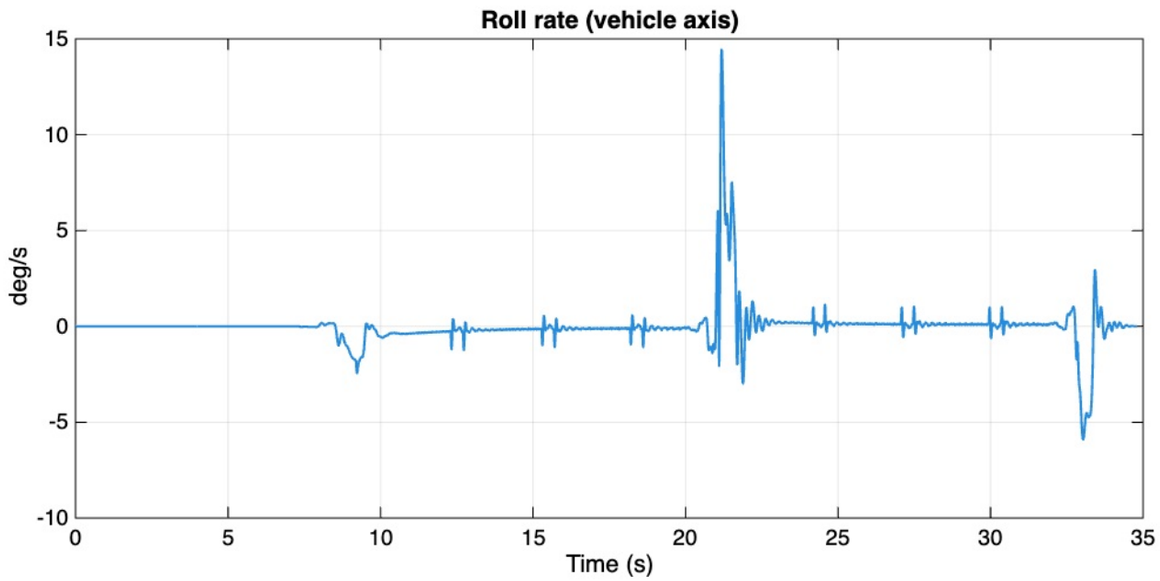
Roll/pitch/yaw rates are body-axis angular velocities, not numerical derivatives of Euler angles.

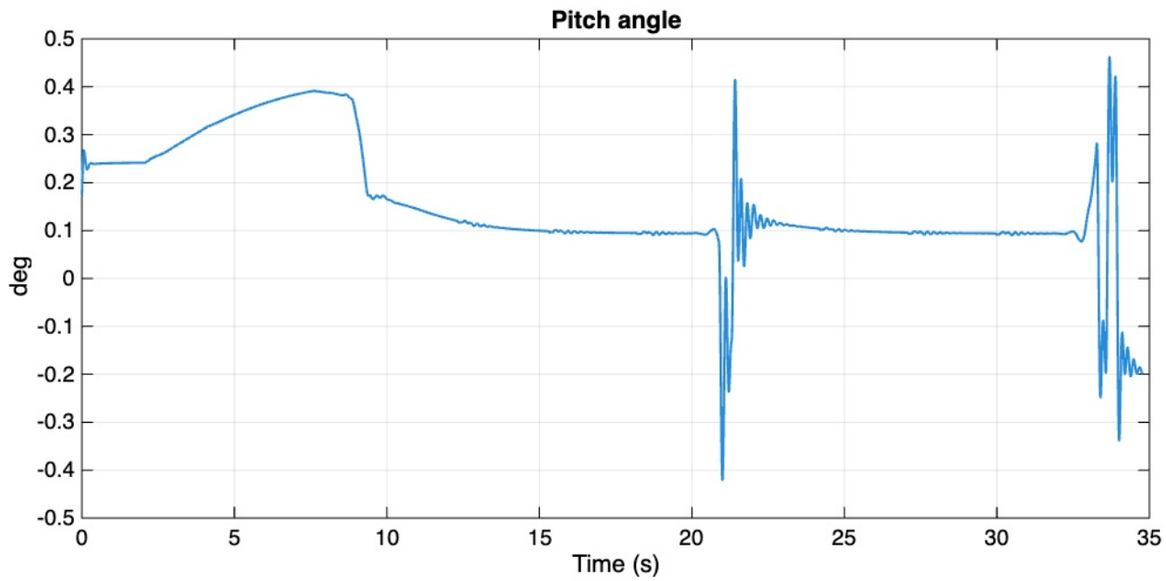
Motor curves are labeled Motor 1-4; their wheel assignment has not been inferred. CSV index retains source names.

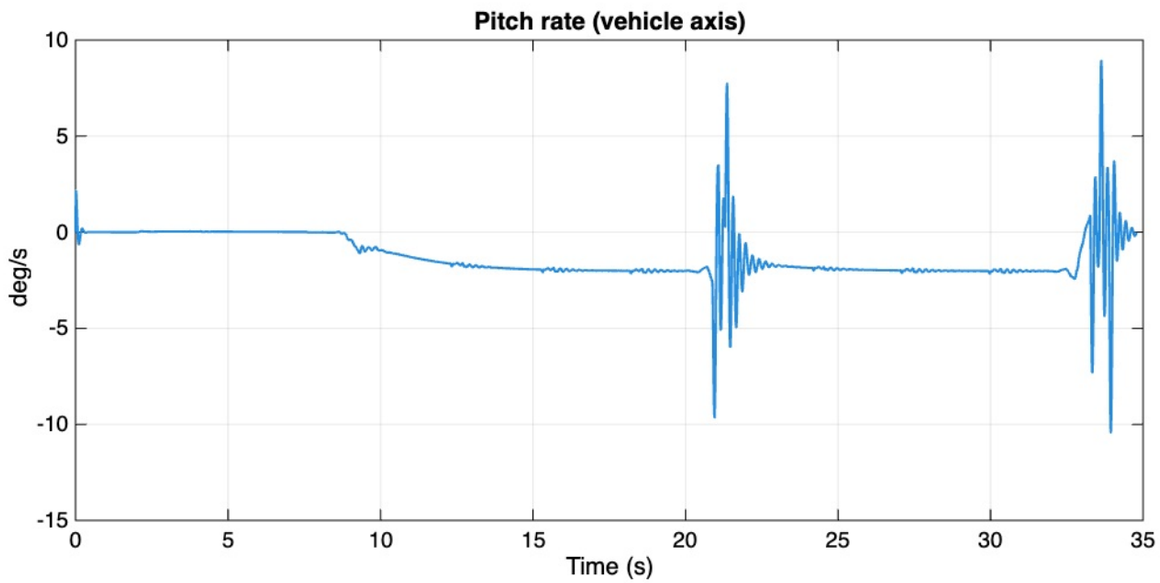
Simple signal plots for tuning - plot page 1

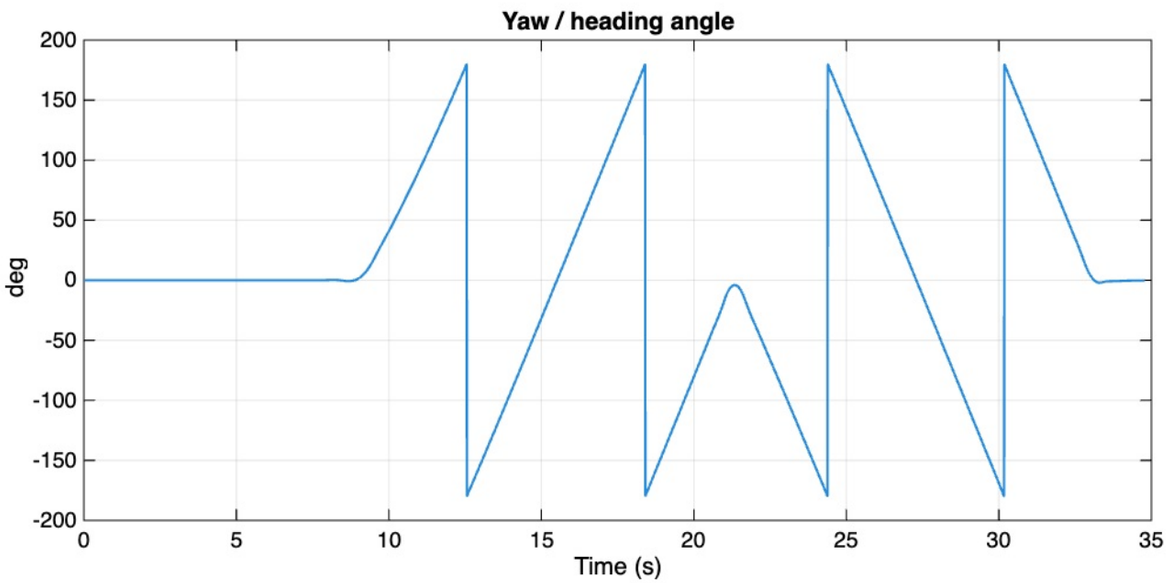


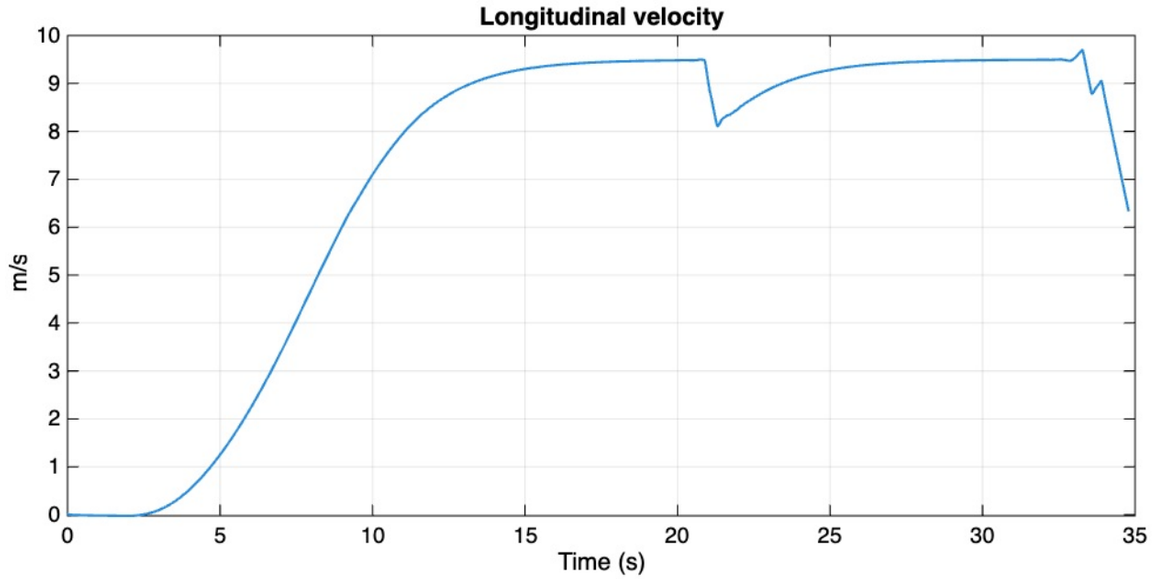



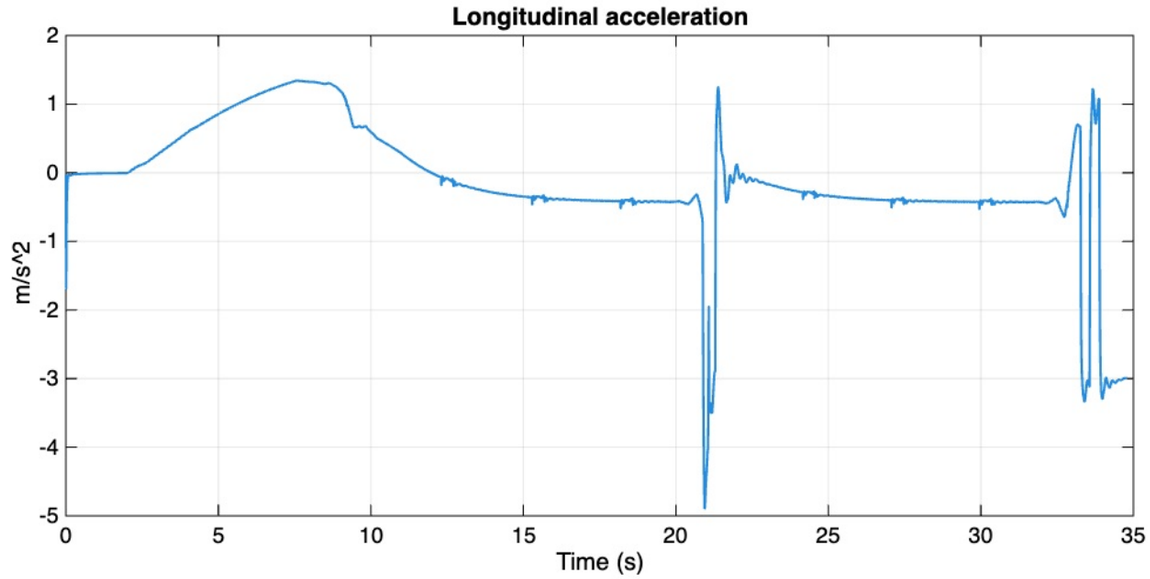



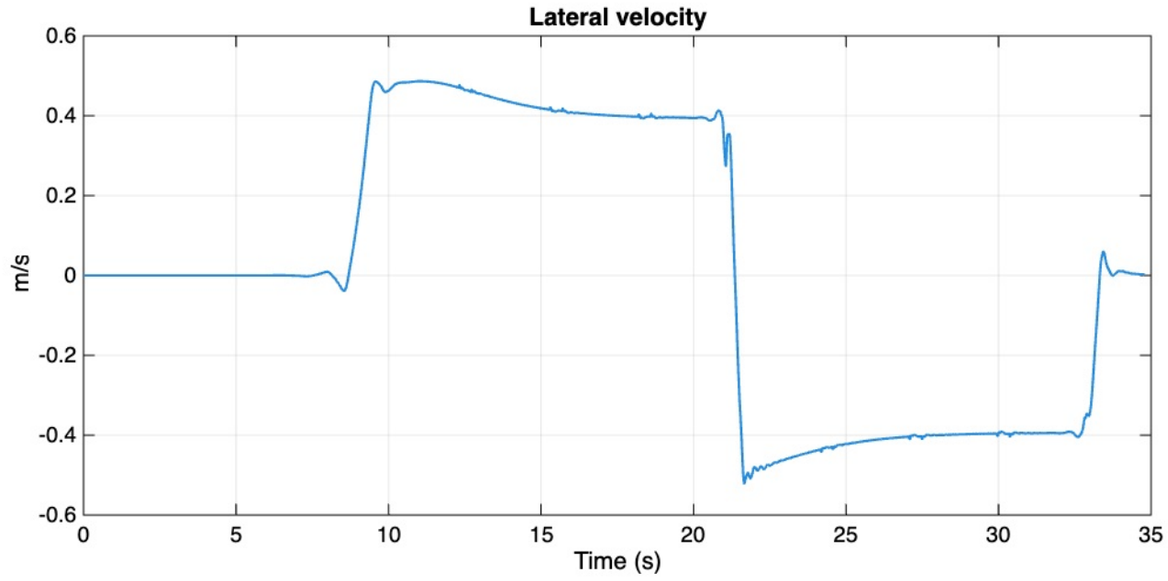



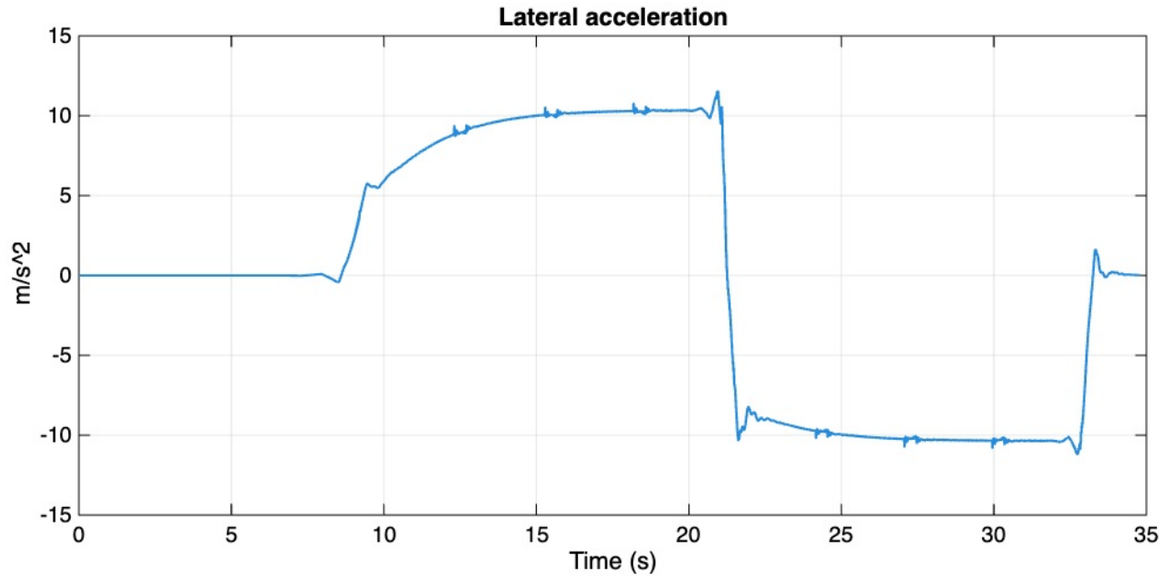



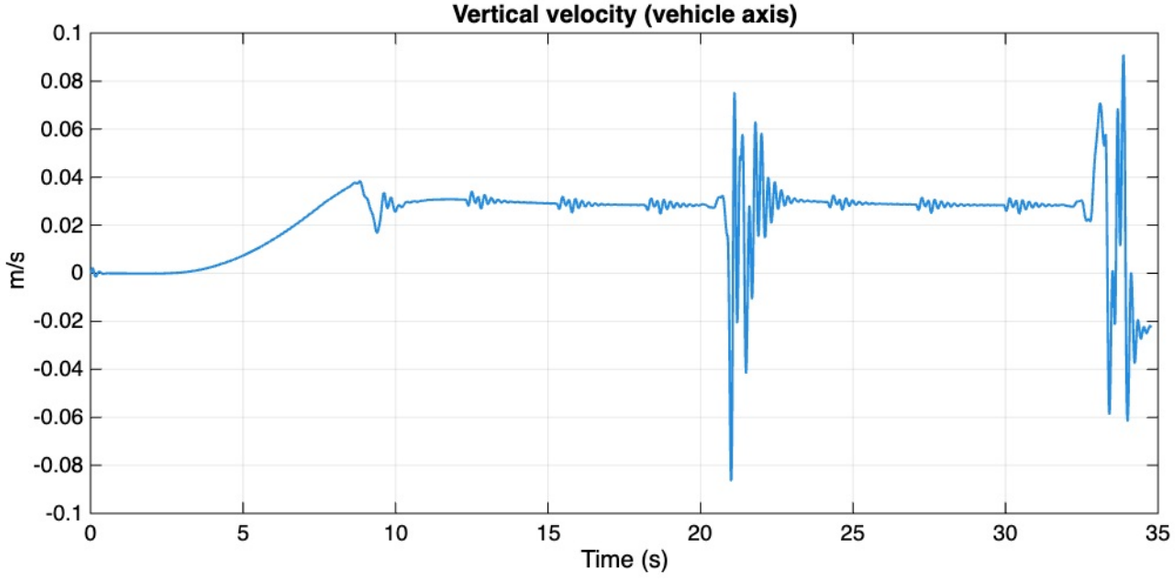
Simple signal plots for tuning - plot page 2

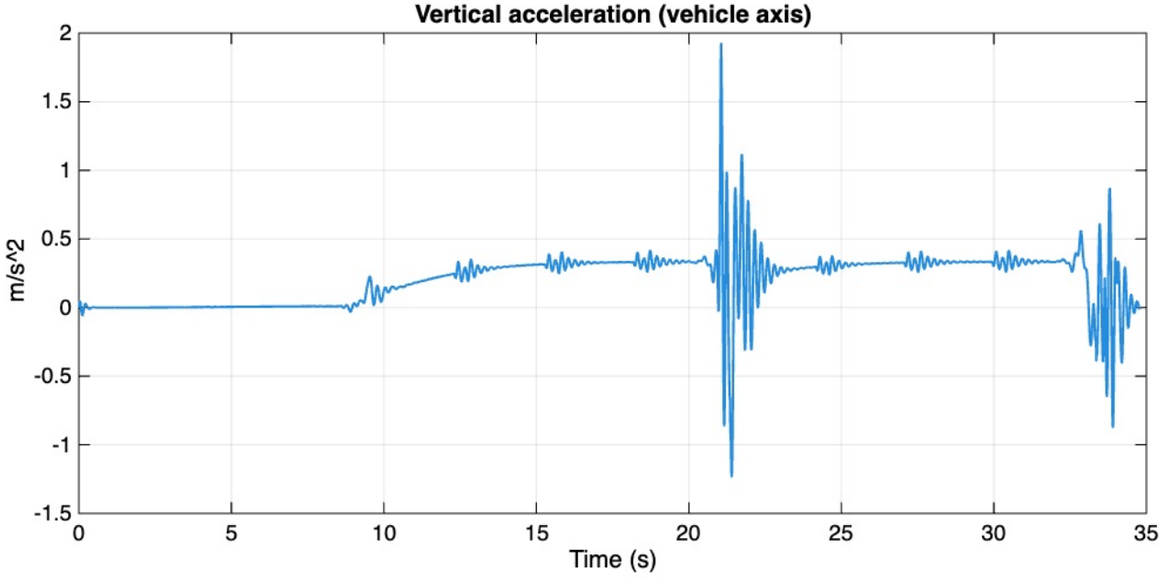


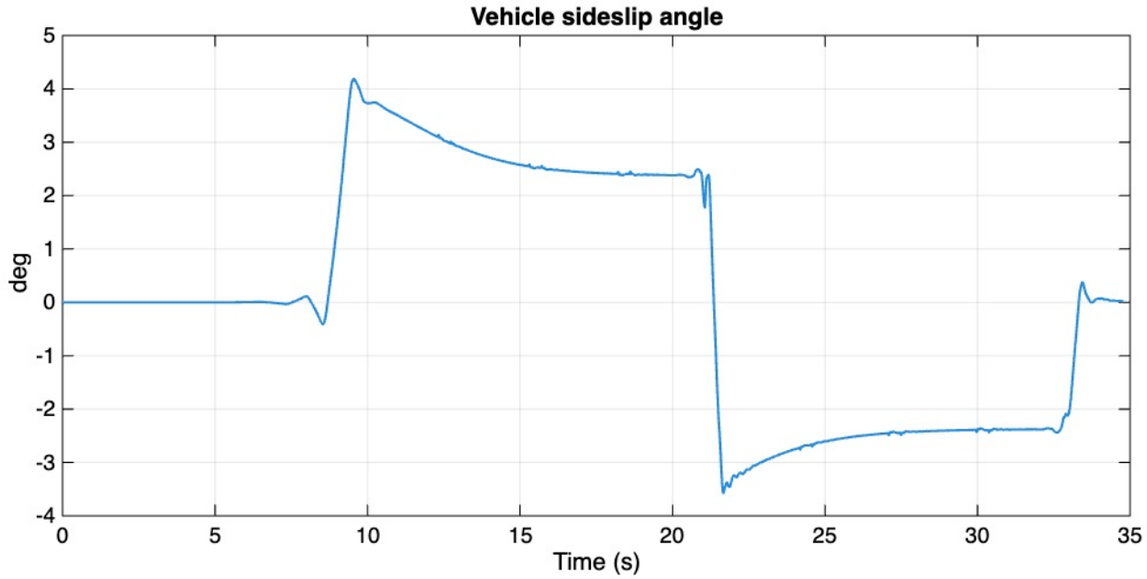



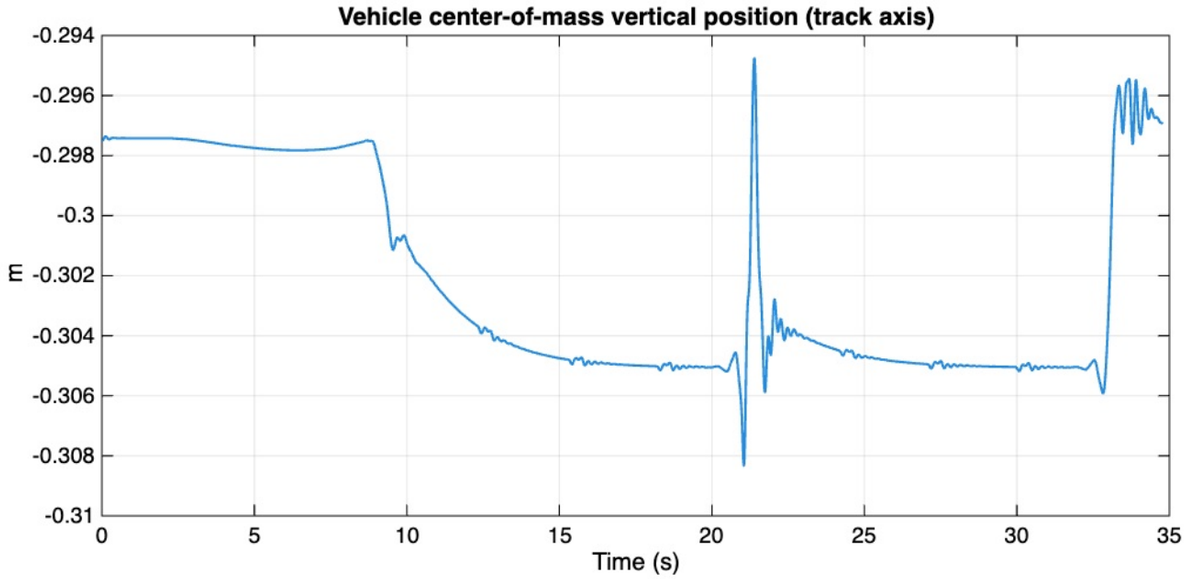



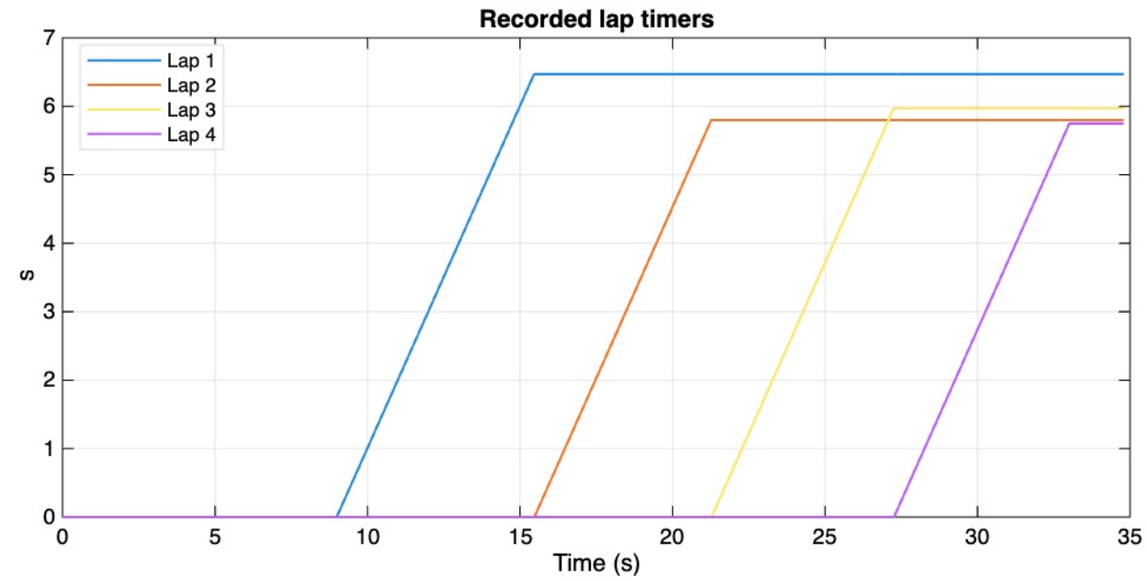



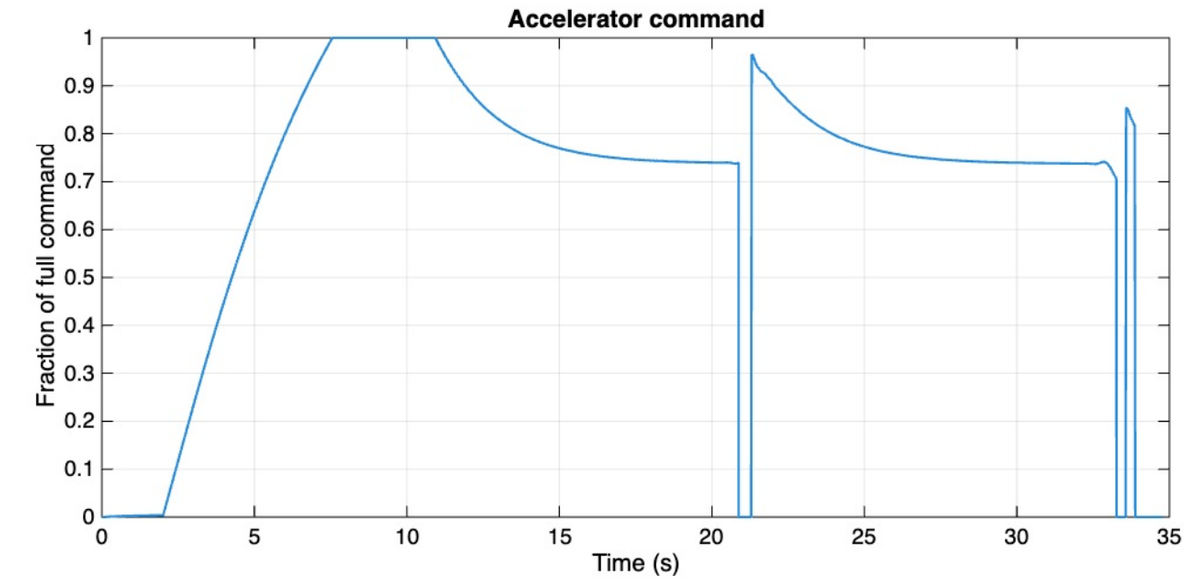



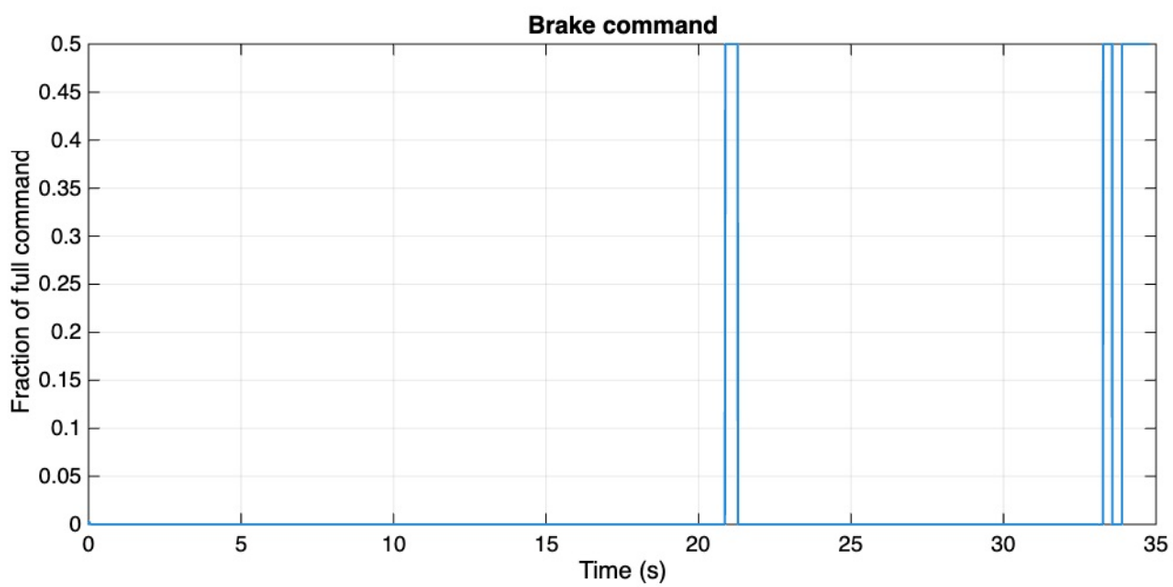
Simple signal plots for tuning - plot page 3

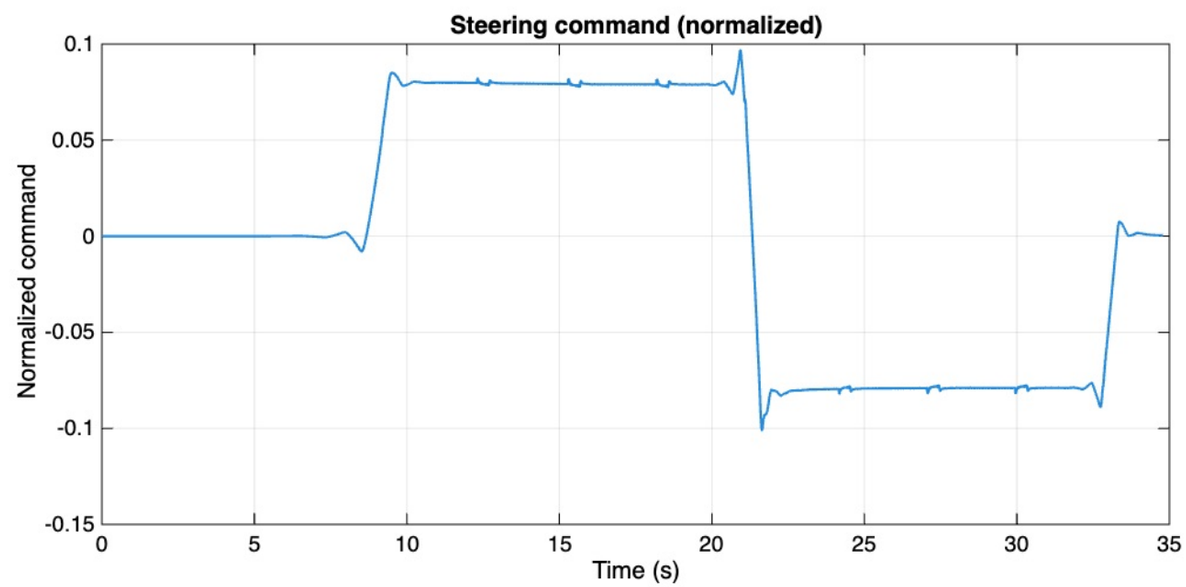


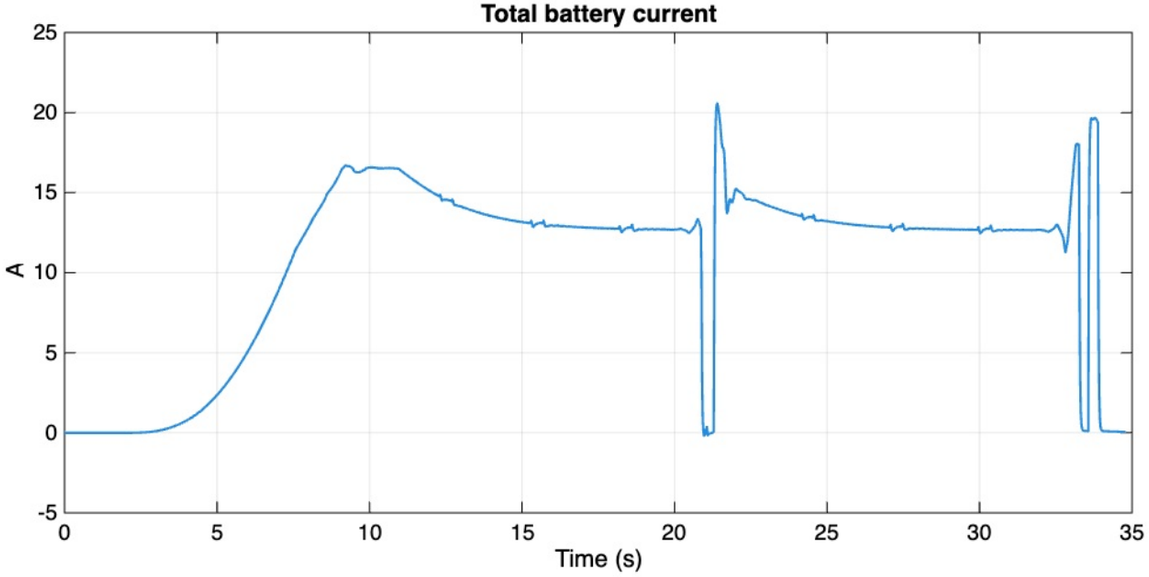



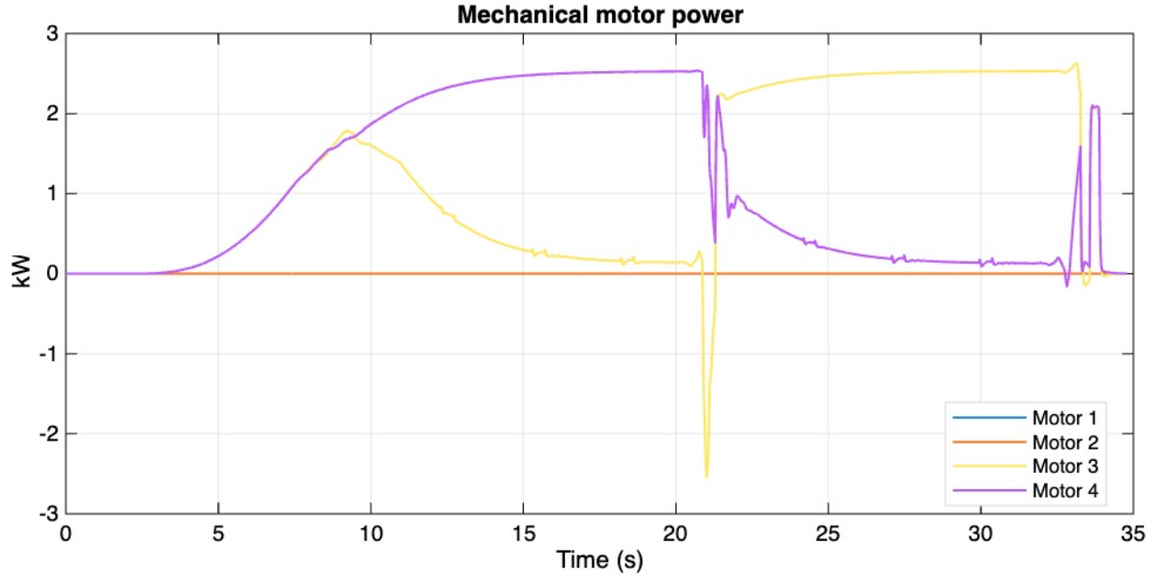



Simple signal plots for tuning - plot page 4

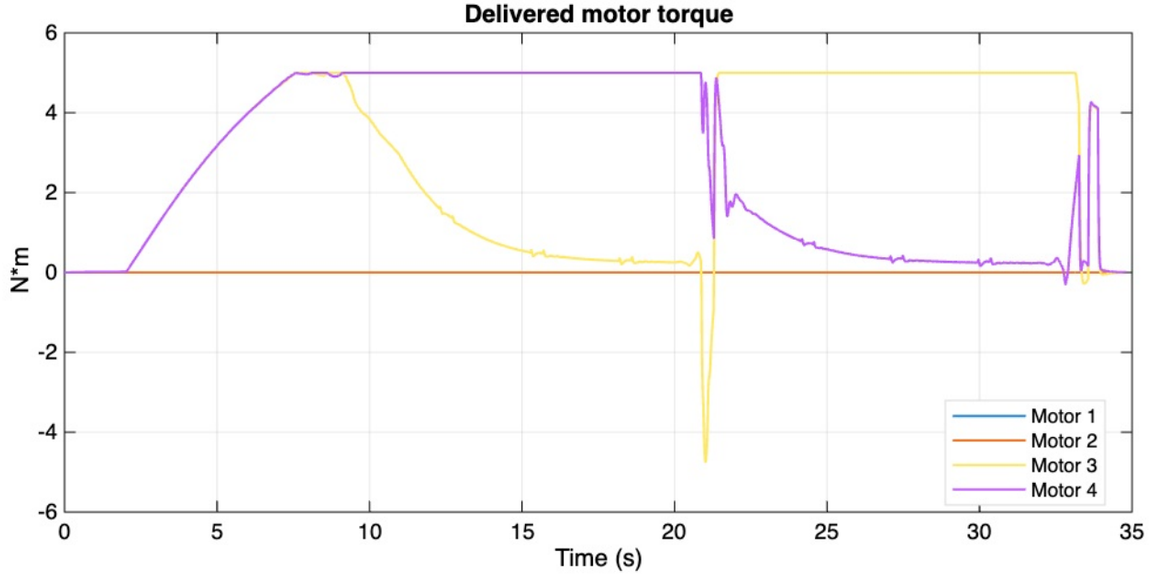
**Total battery current**



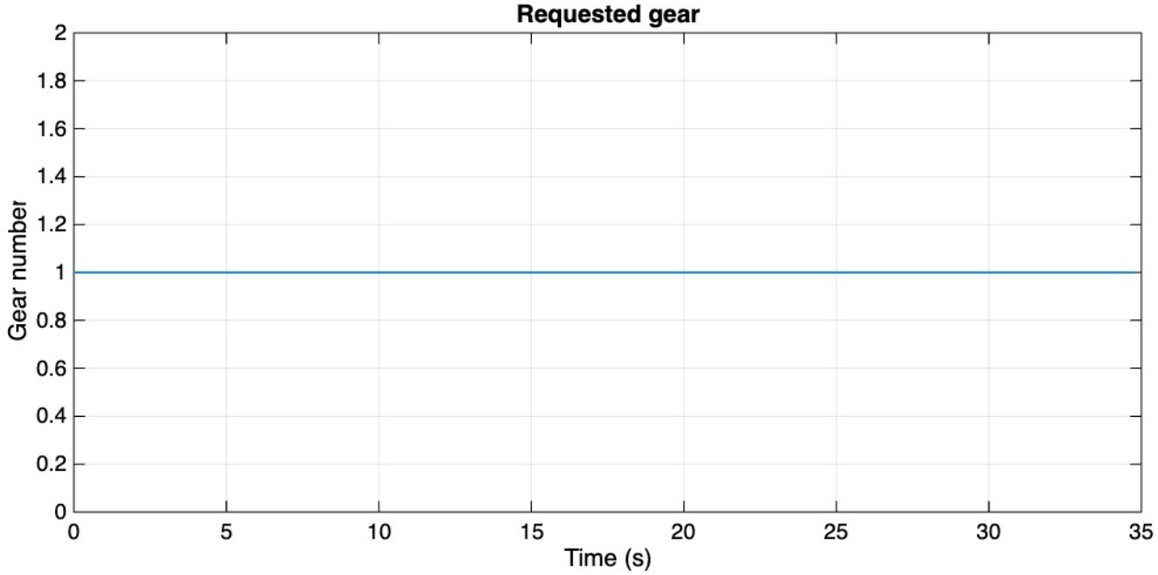

**Mechanical motor power**



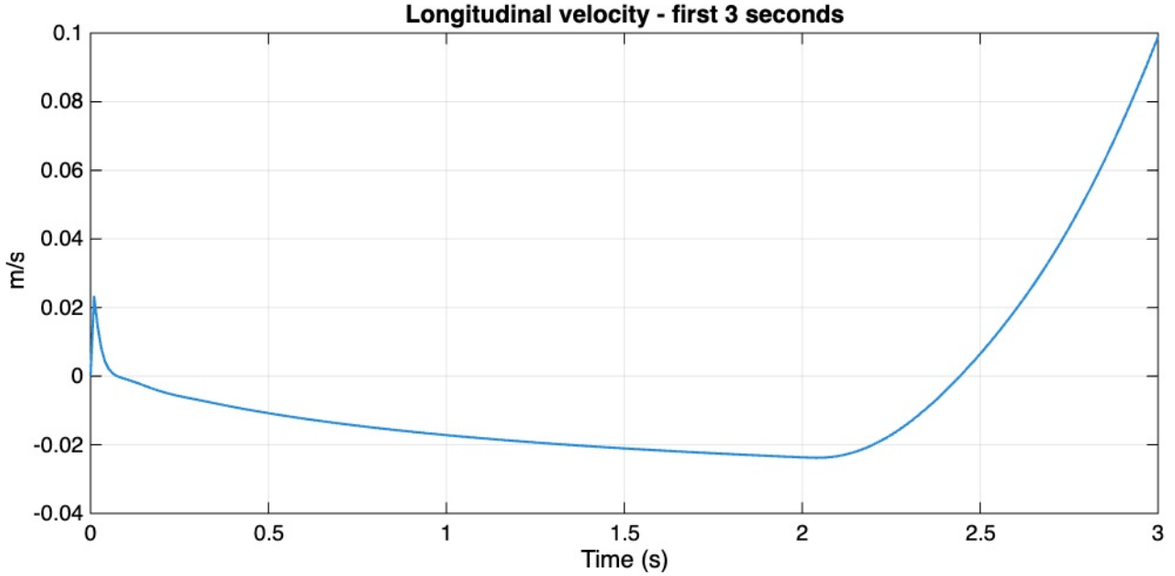

**Delivered motor torque**



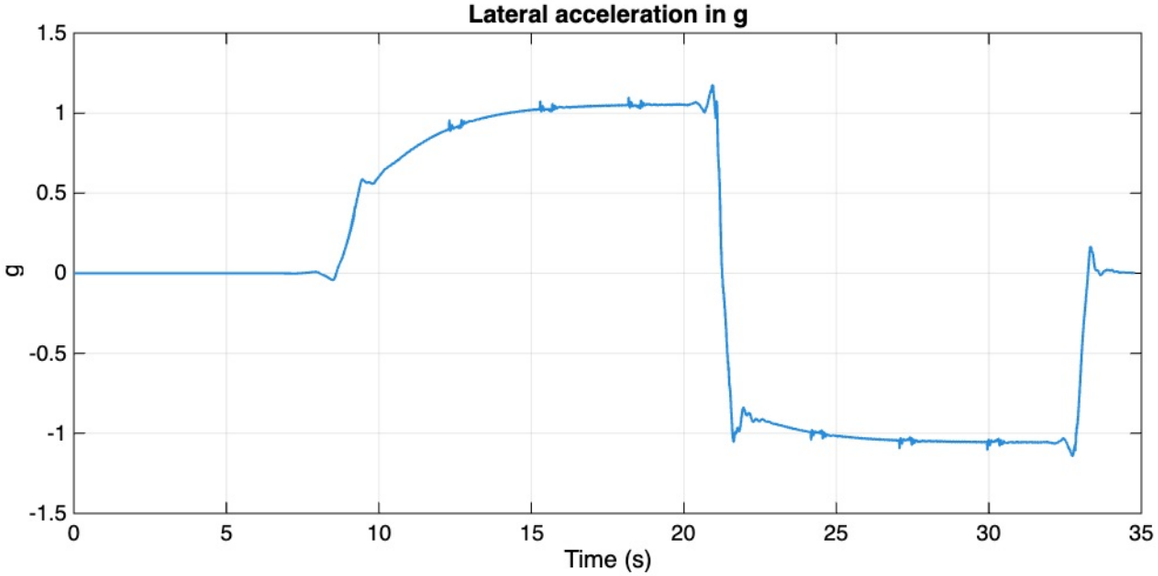

**Requested gear**



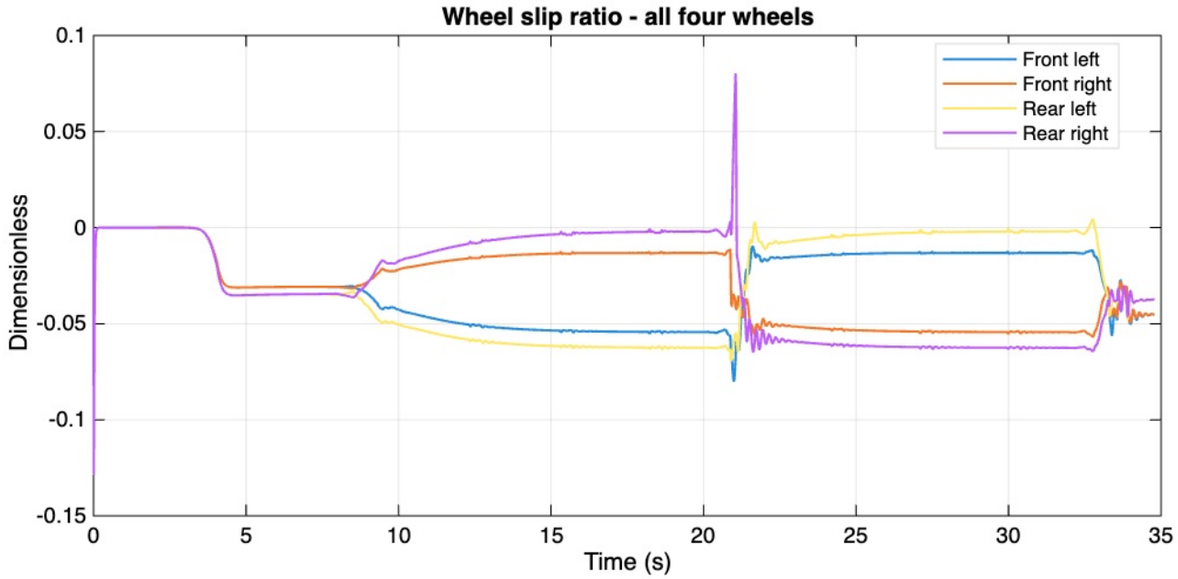

**Longitudinal velocity - first 3 seconds**

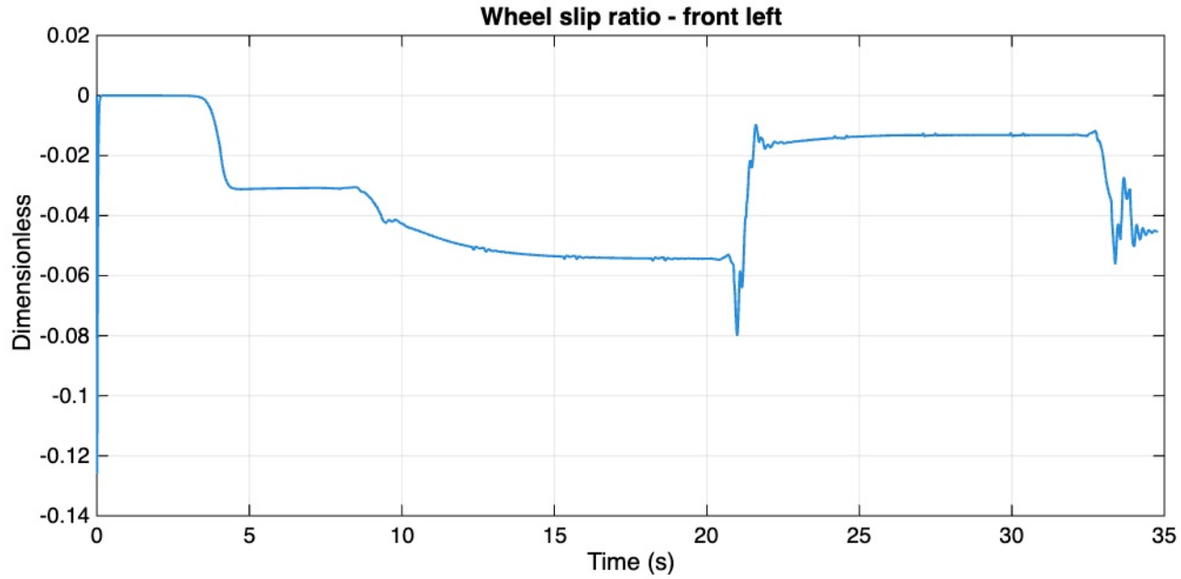


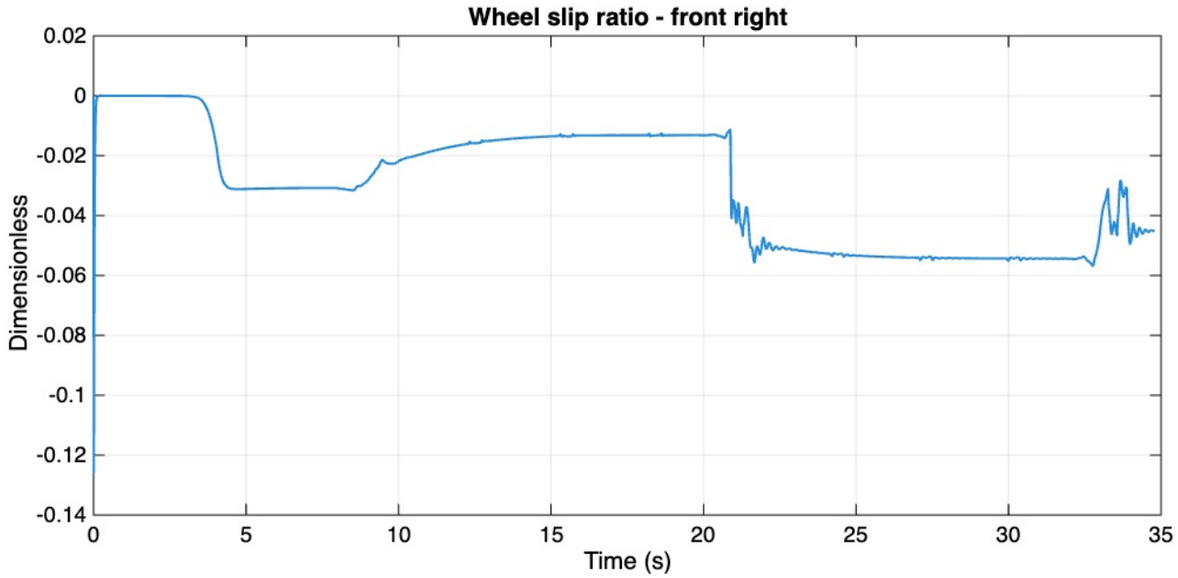

**Lateral acceleration in g**

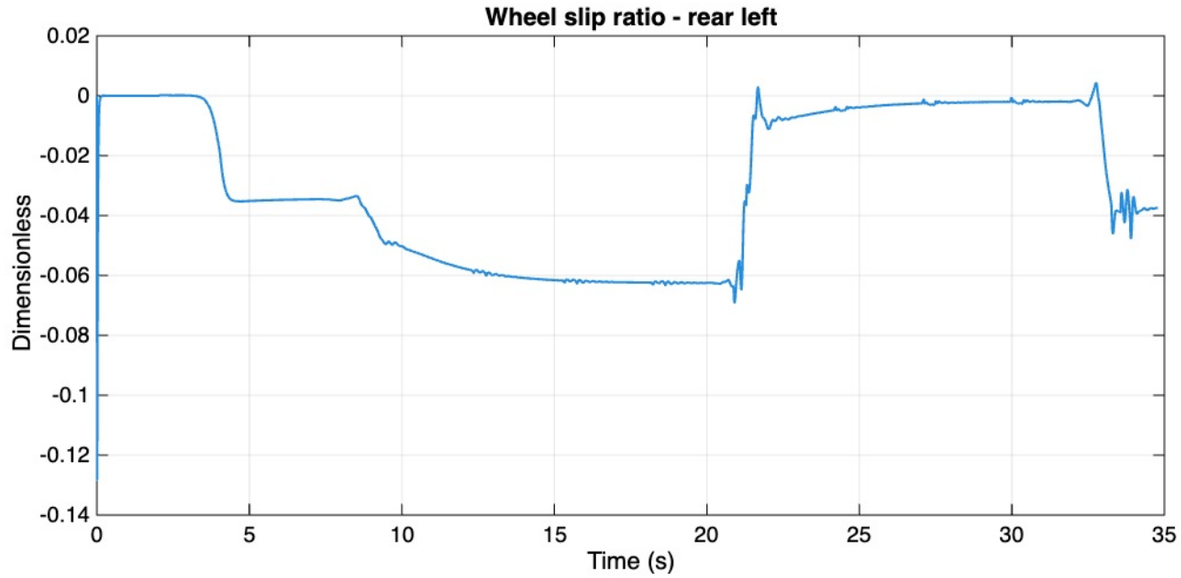


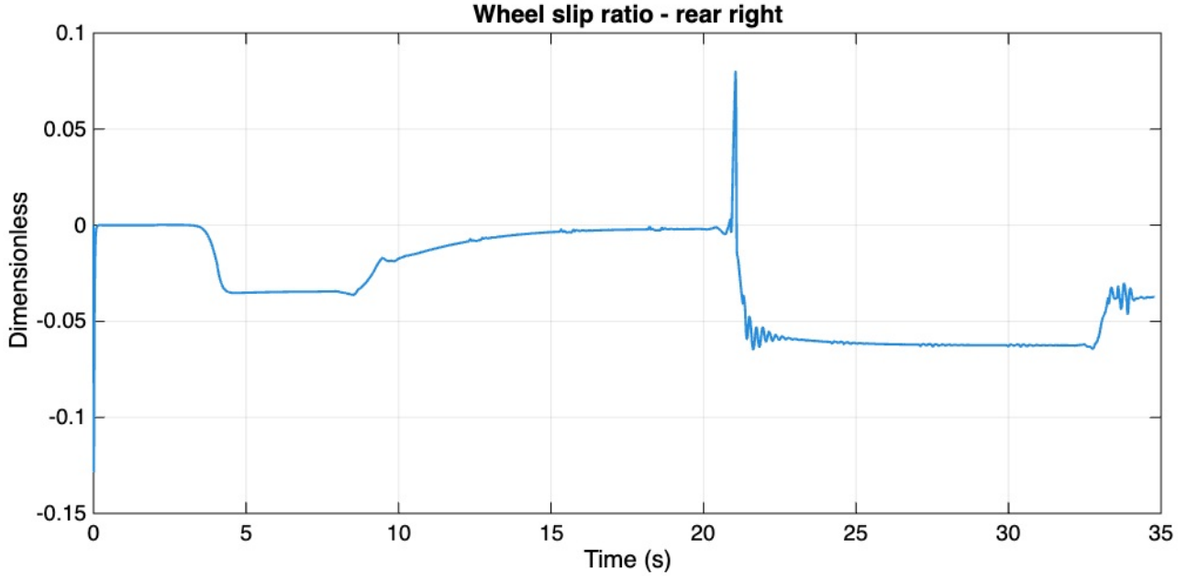
Simple signal plots for tuning - plot page 5

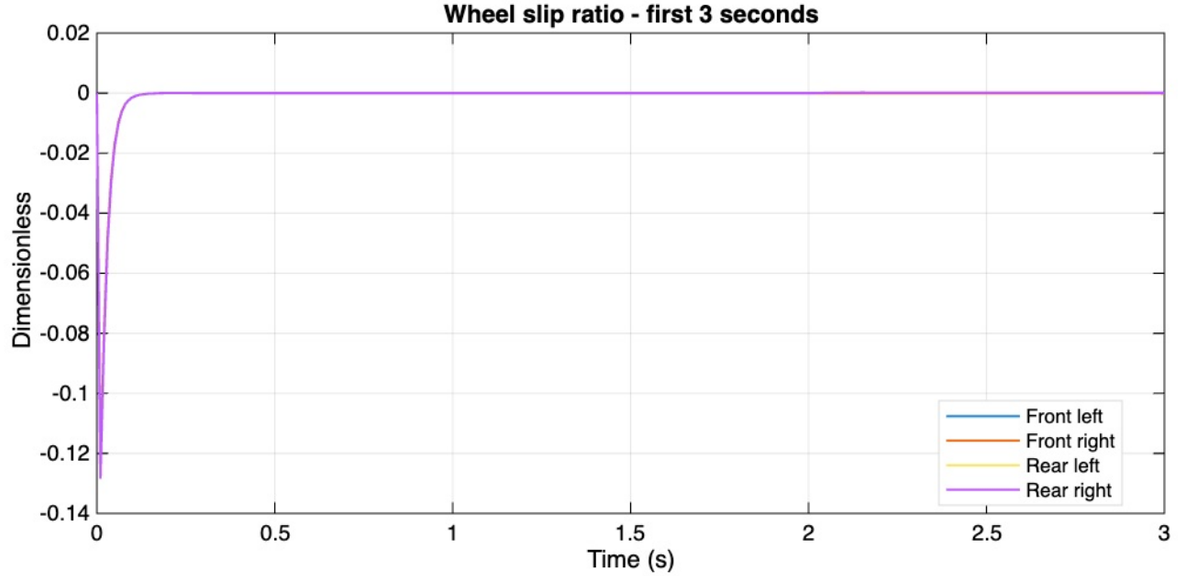


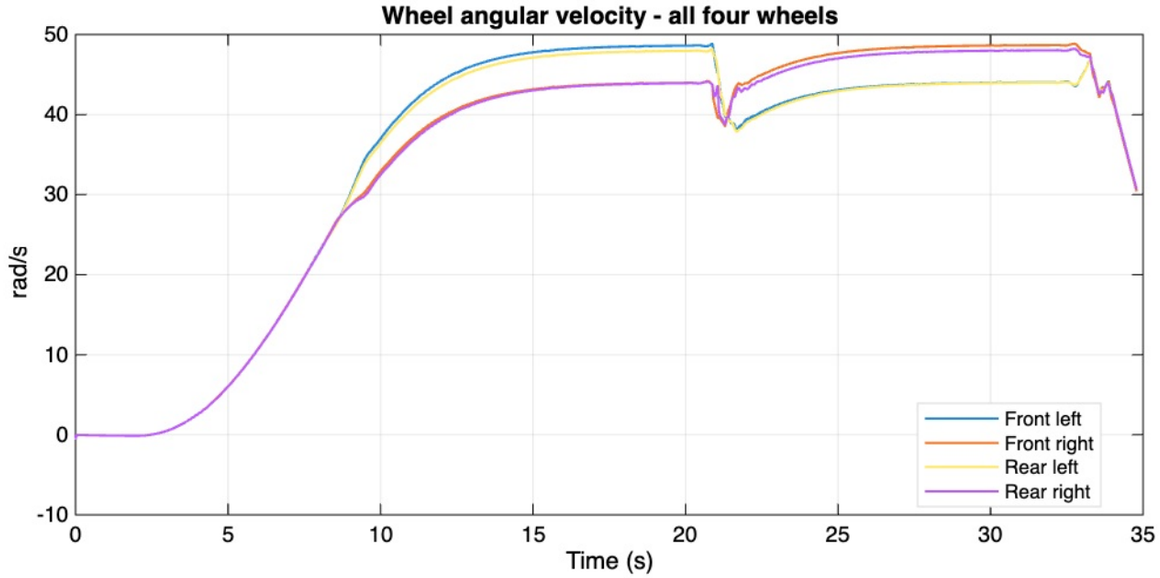



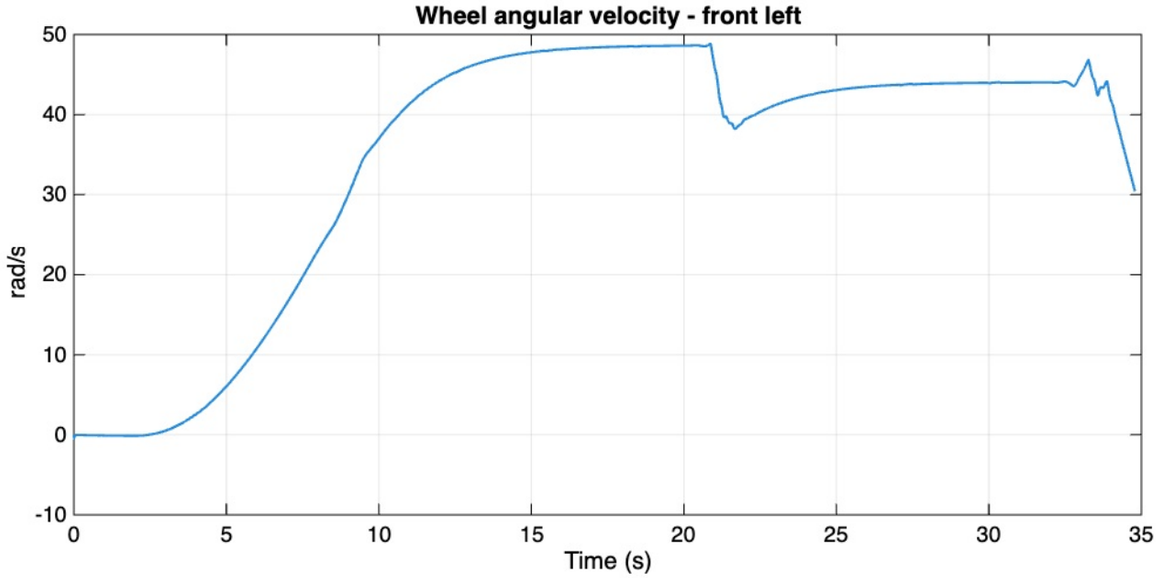



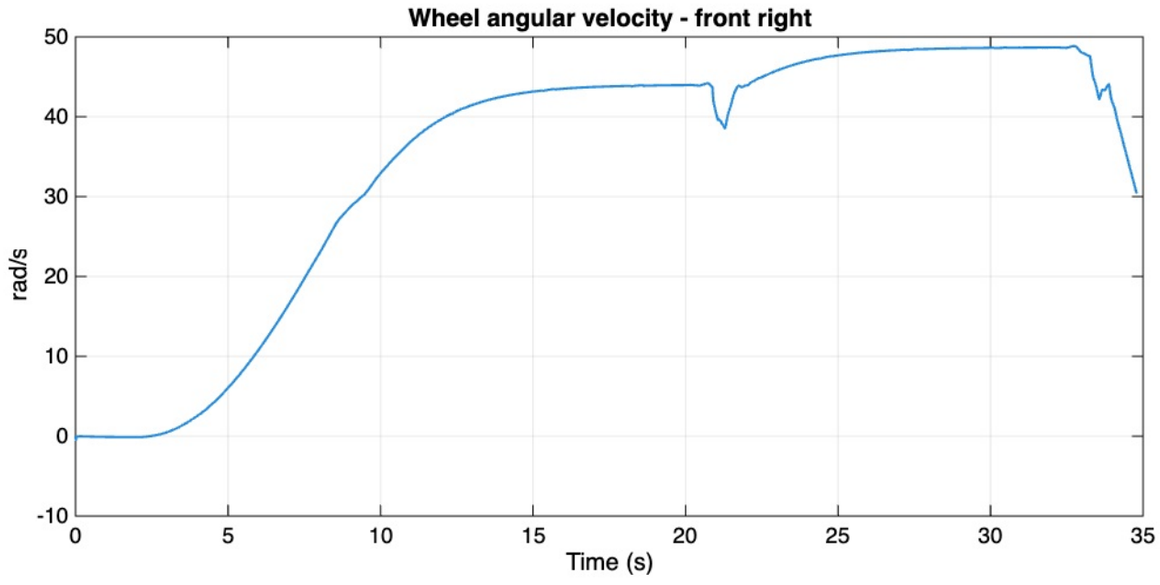



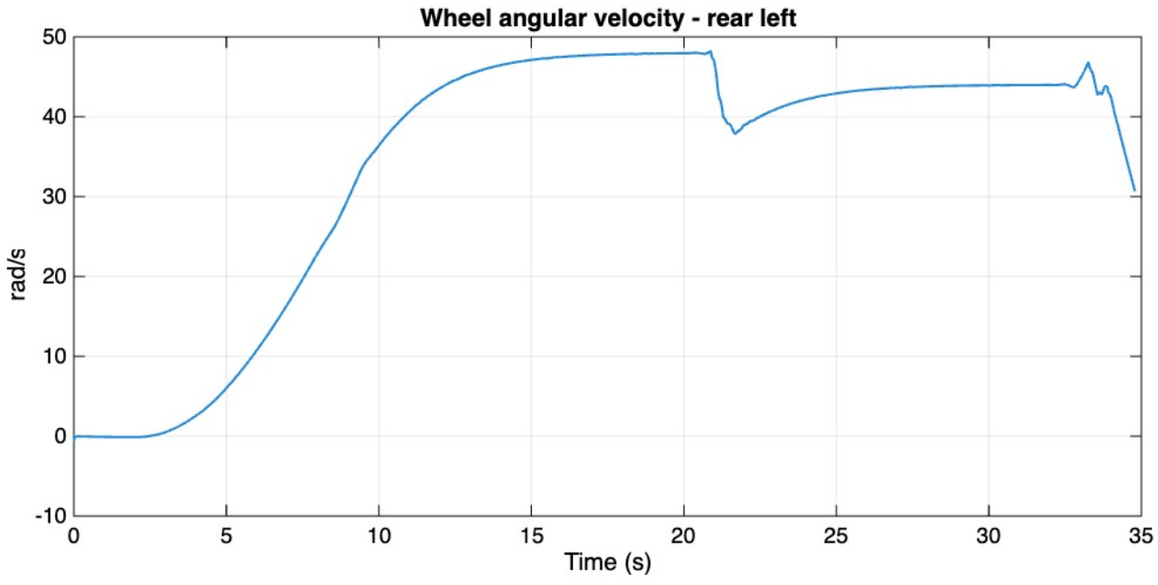



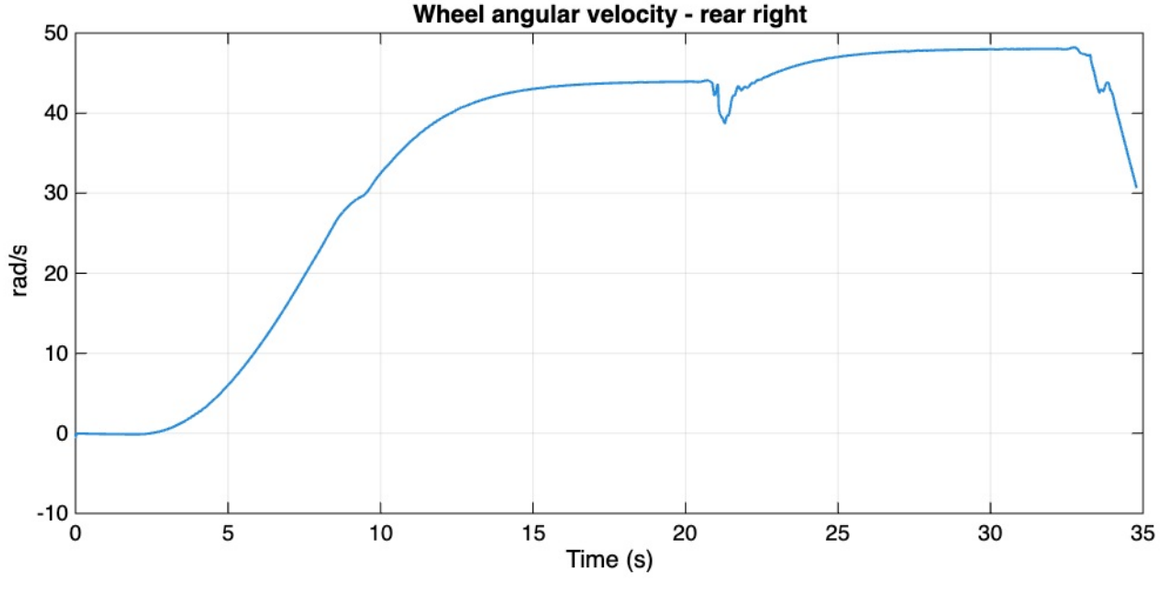
Simple signal plots for tuning - plot page 6

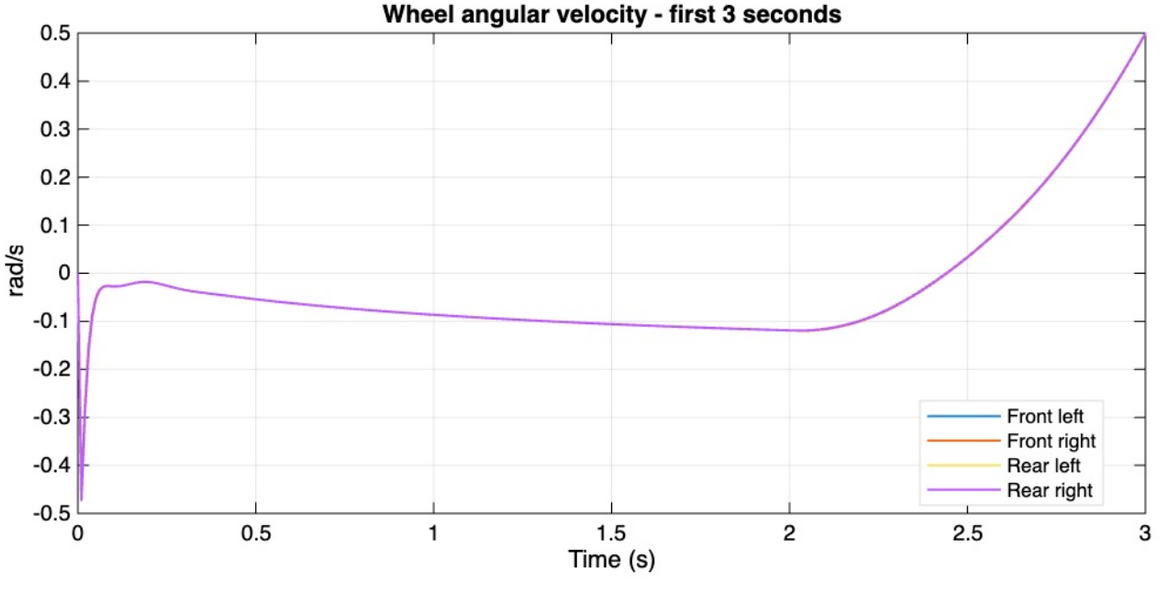


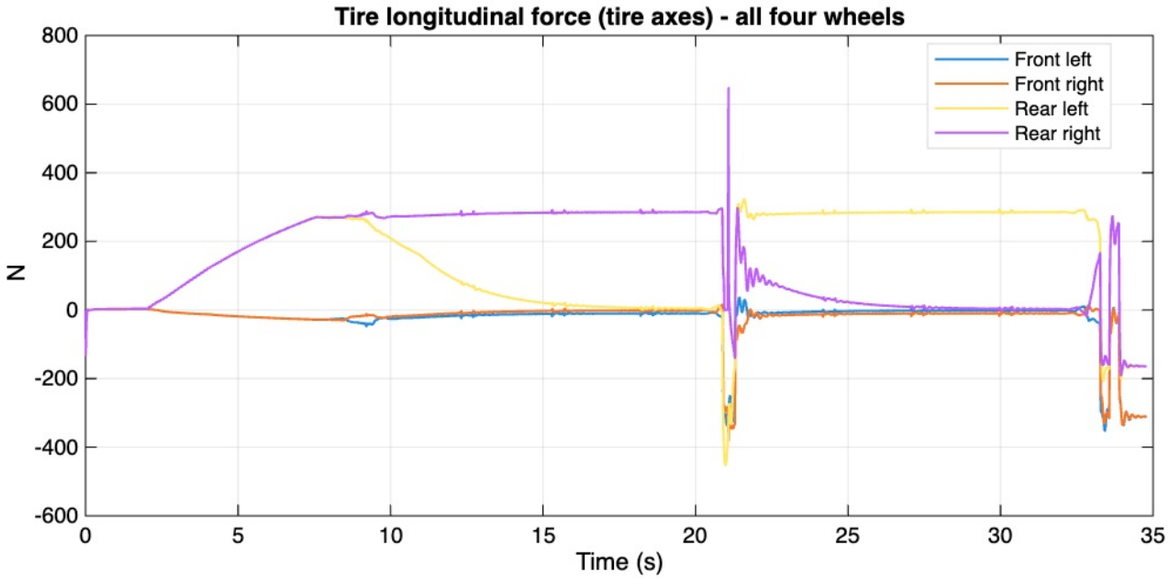



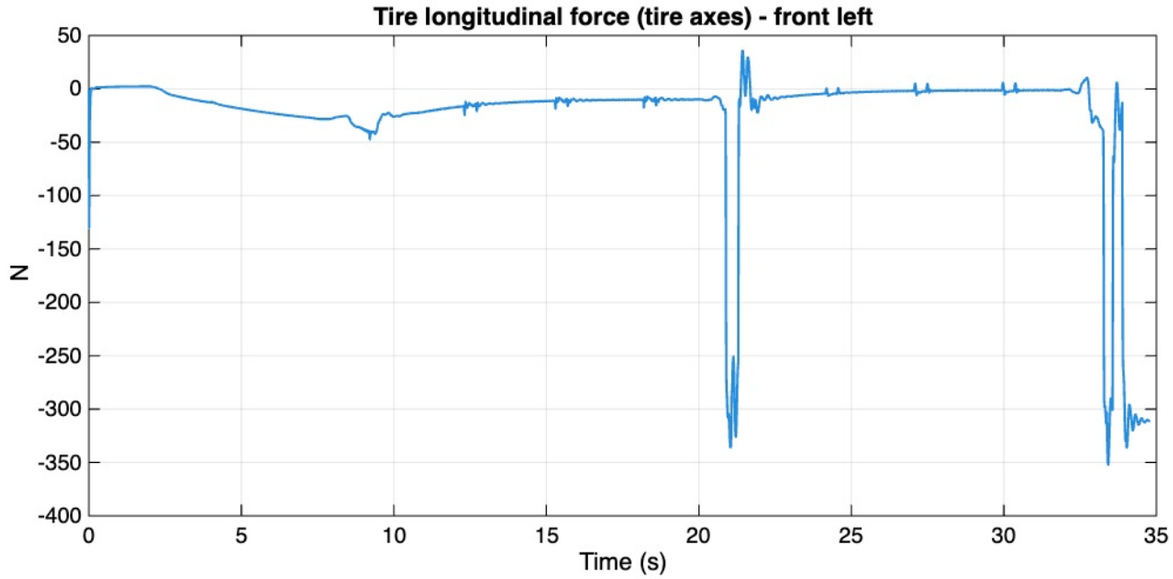



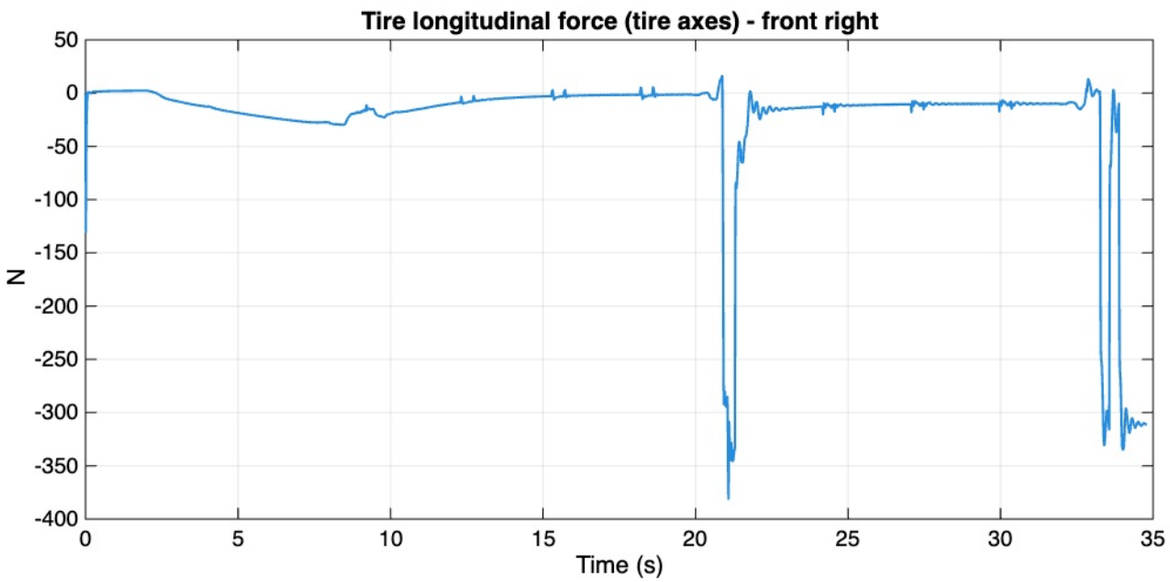



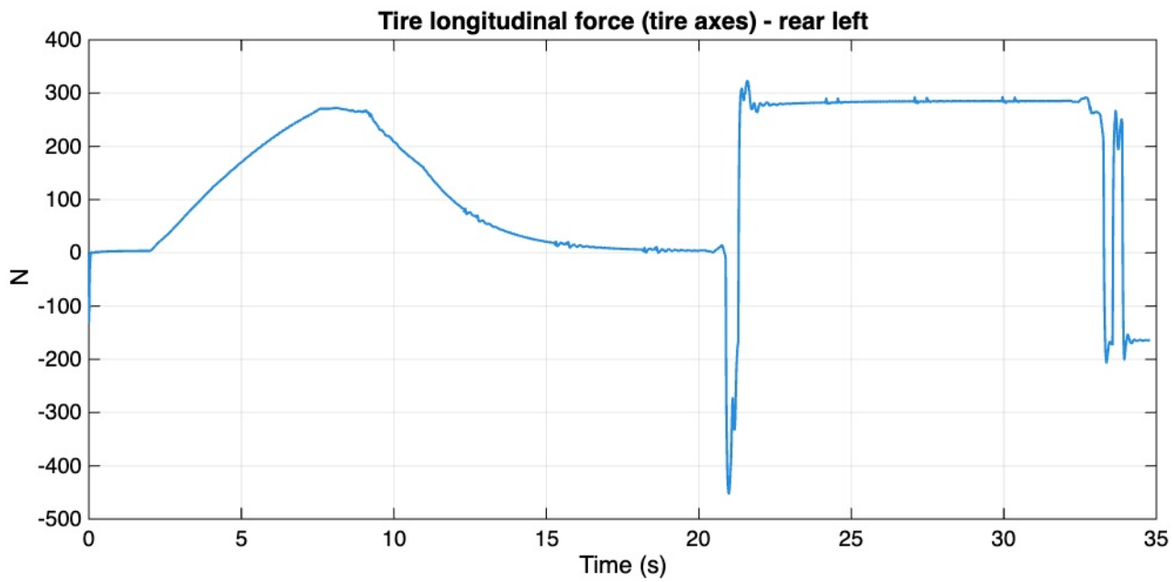



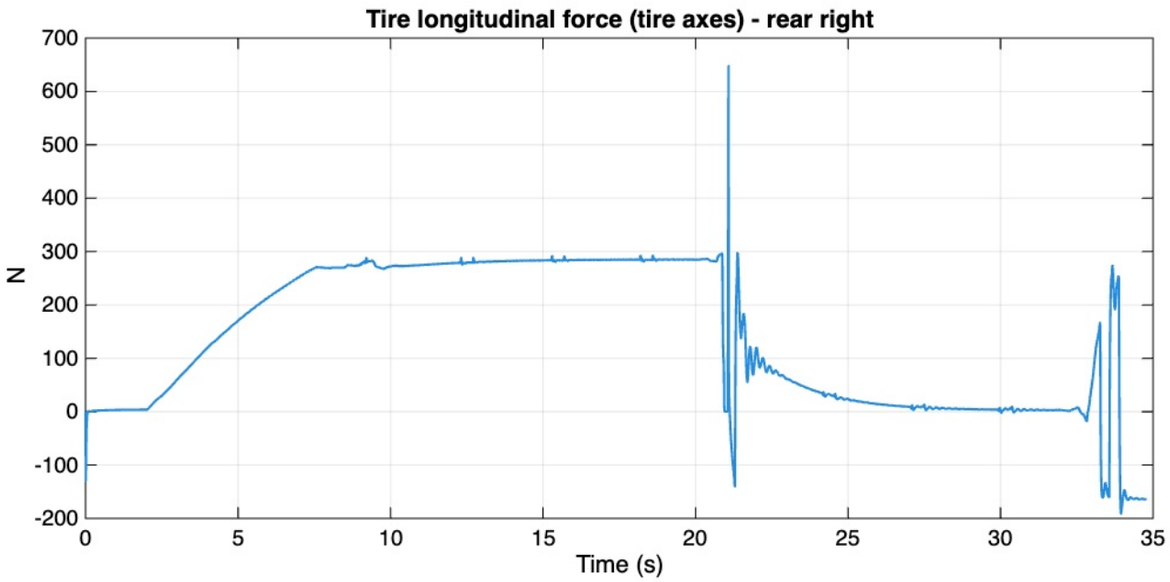
Simple signal plots for tuning - plot page 7

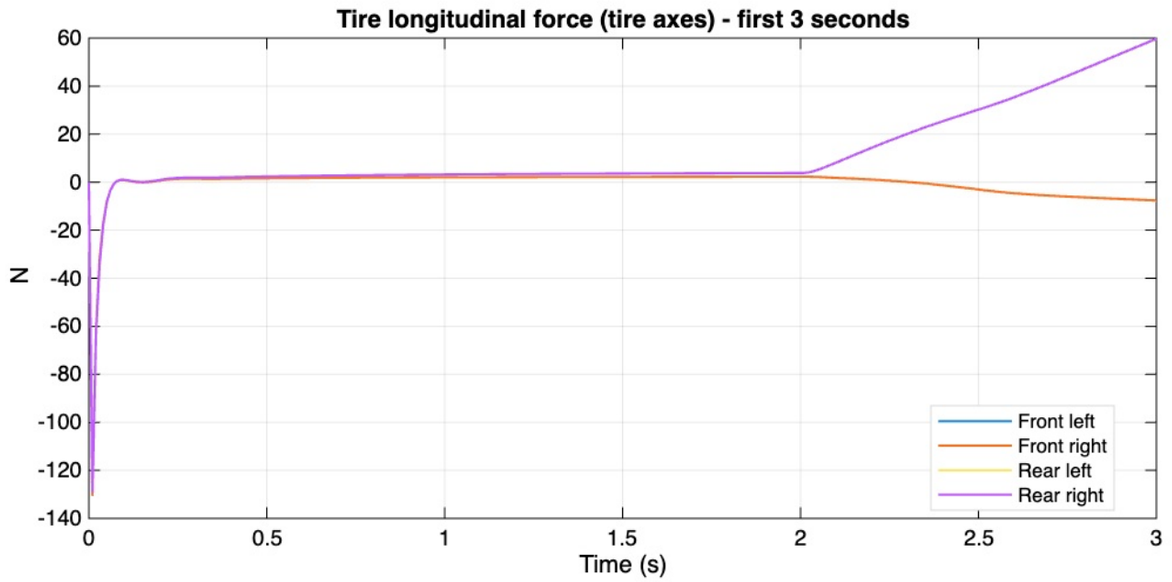


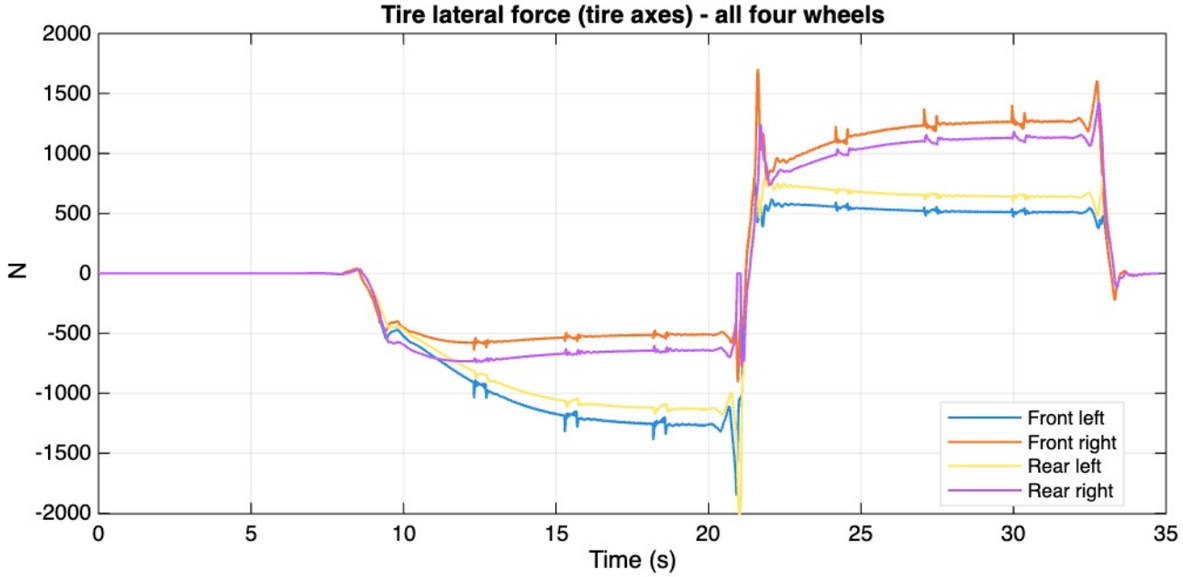



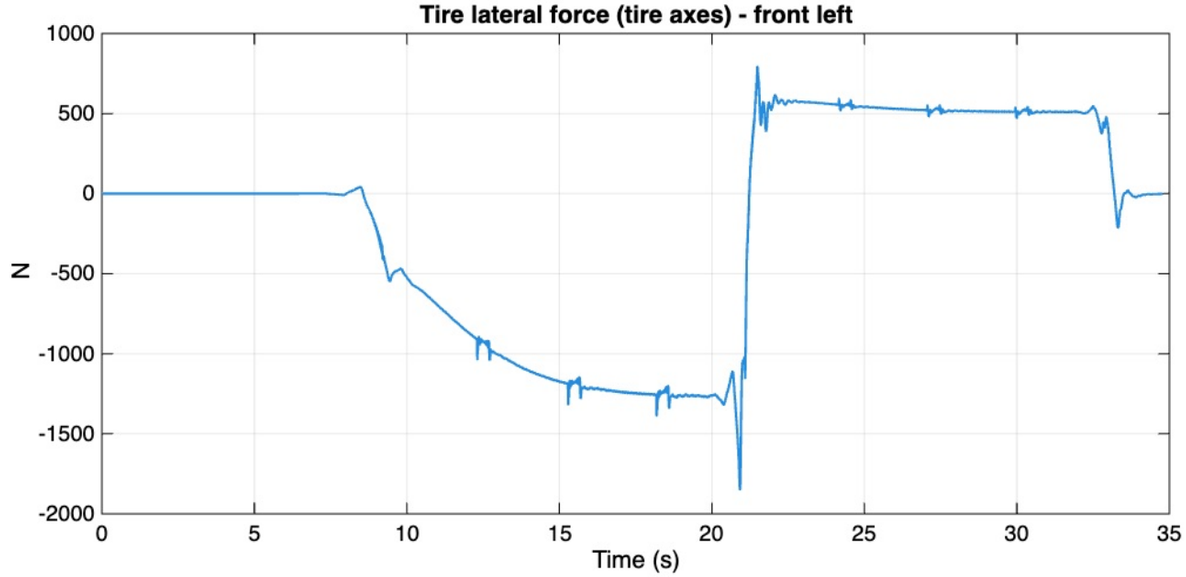



Simple signal plots for tuning - plot page 8

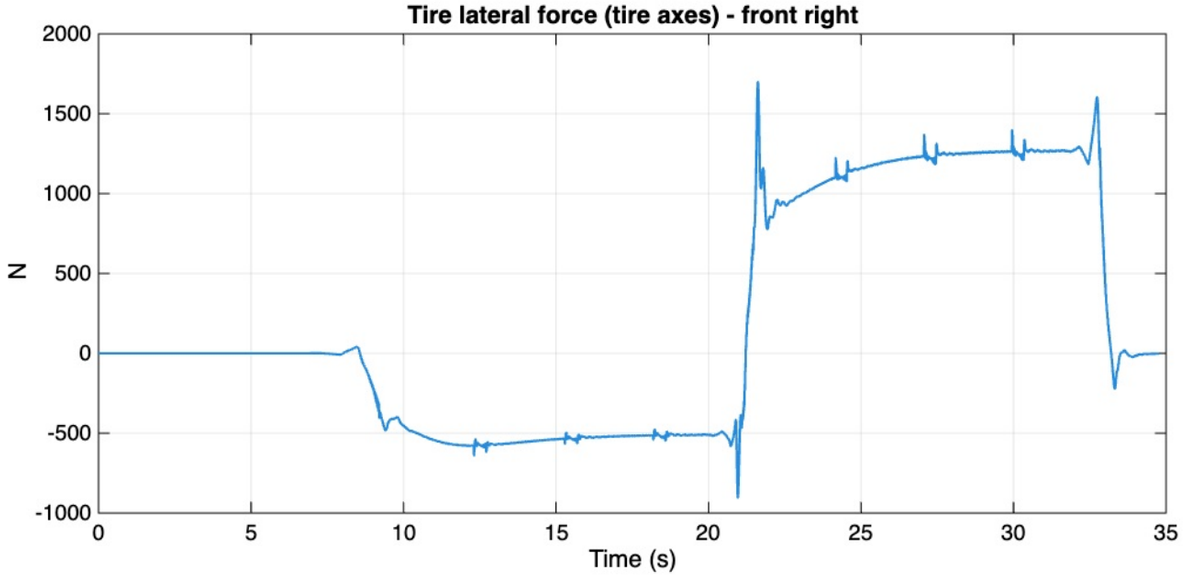
**Tire lateral force (tire axes) - all four wheels**



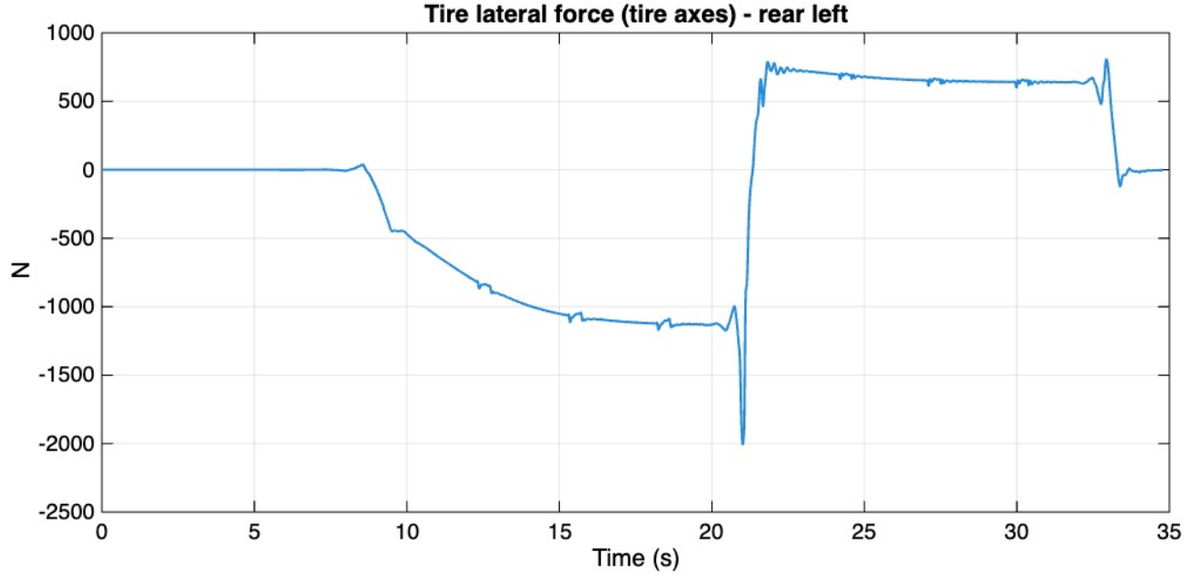

**Tire lateral force (tire axes) - front left**



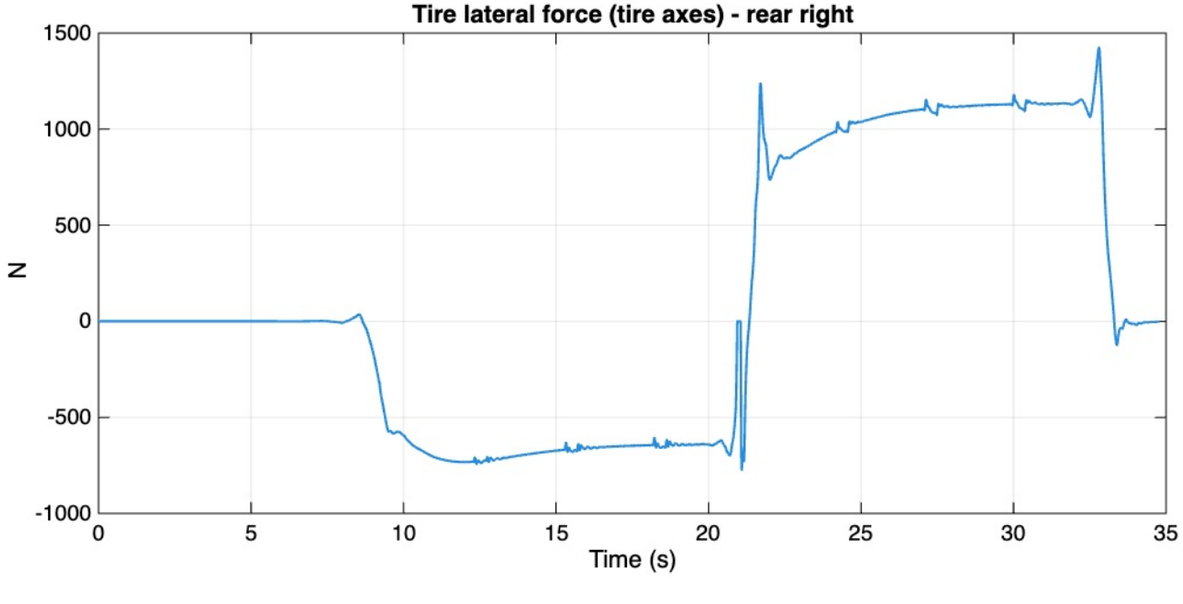

**Tire lateral force (tire axes) - front right**



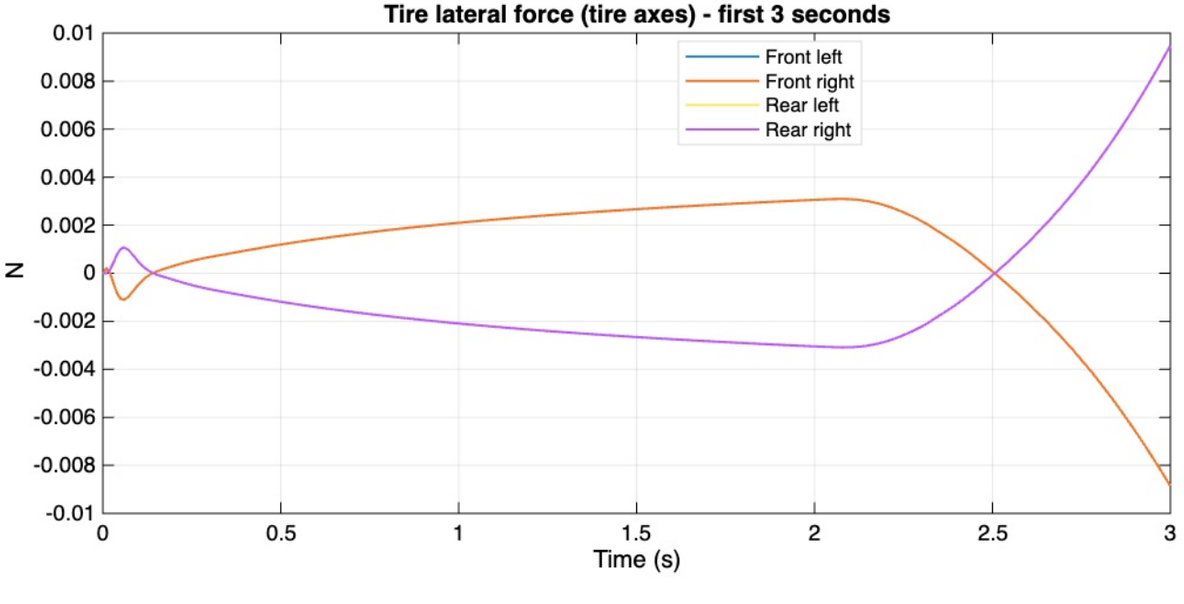

**Tire lateral force (tire axes) - rear left**



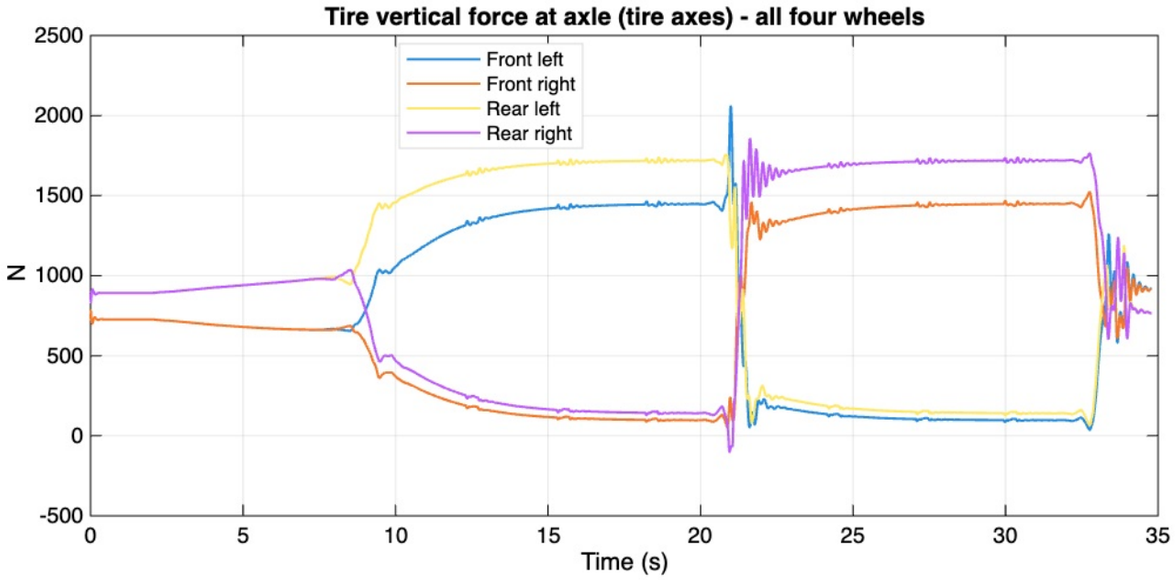

**Tire lateral force (tire axes) - rear right**

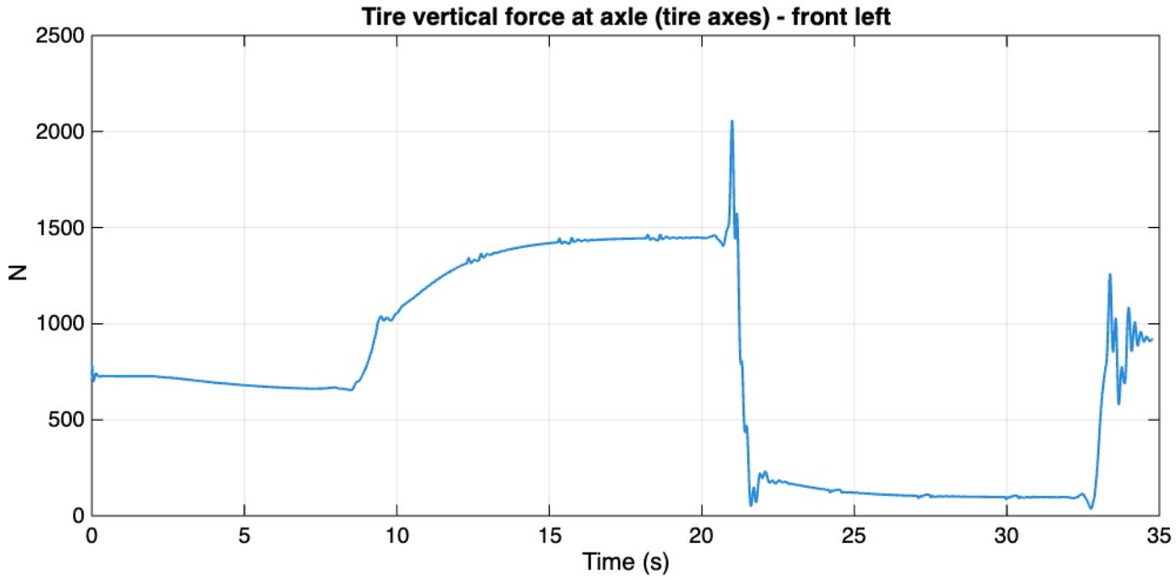


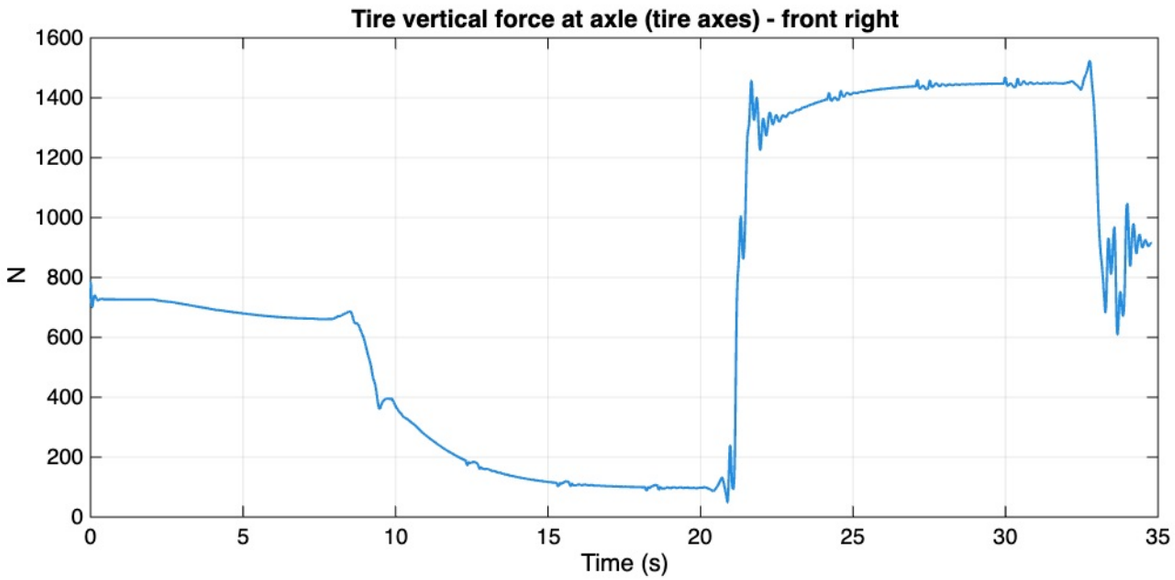

**Tire lateral force (tire axes) - first 3 seconds**

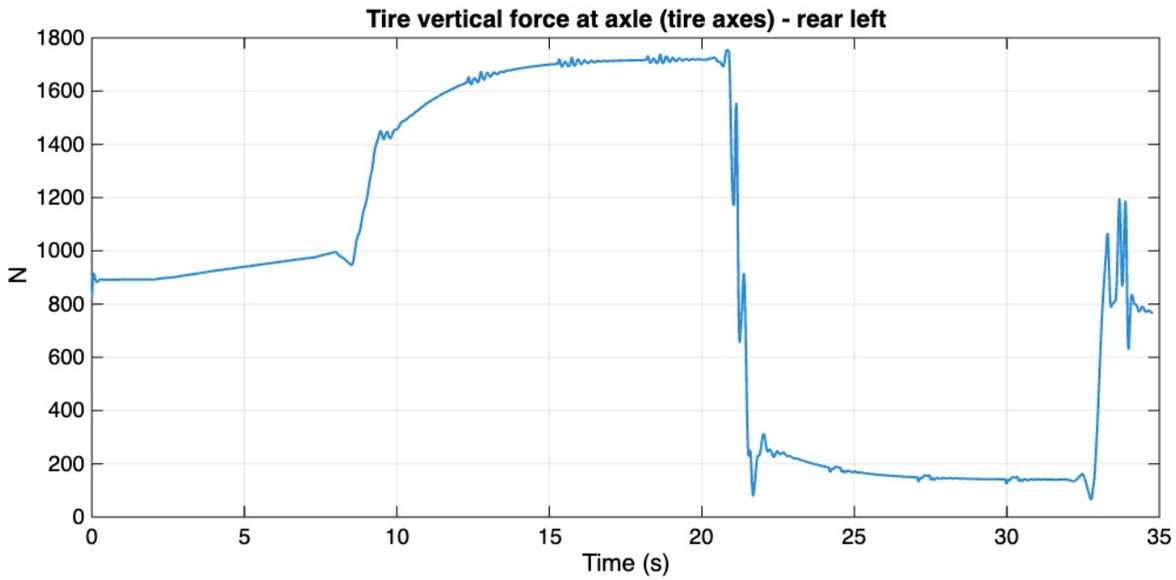


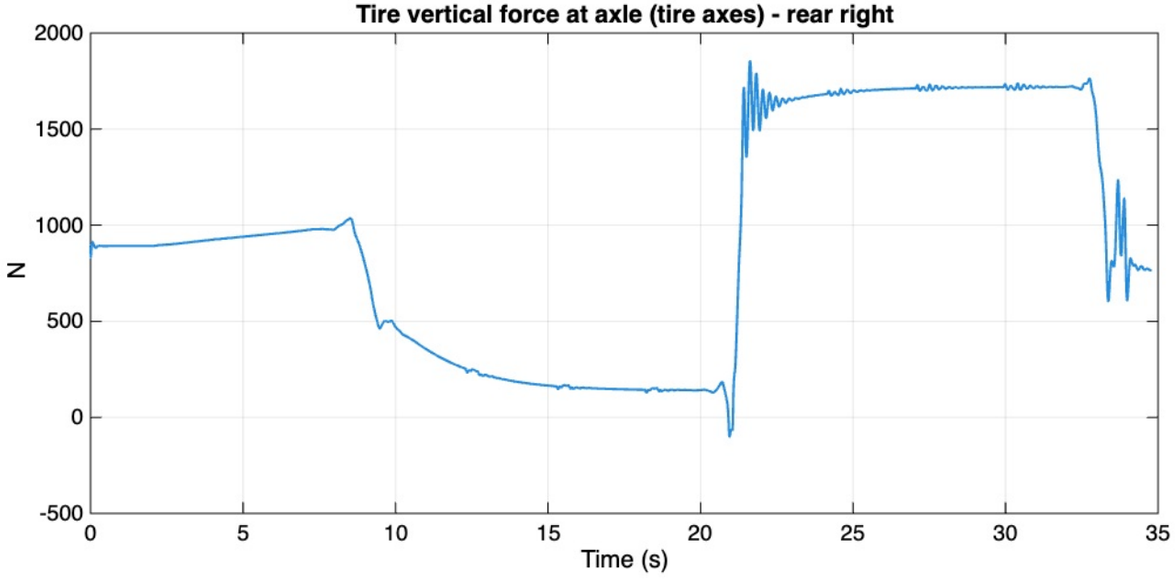
Simple signal plots for tuning - plot page 9

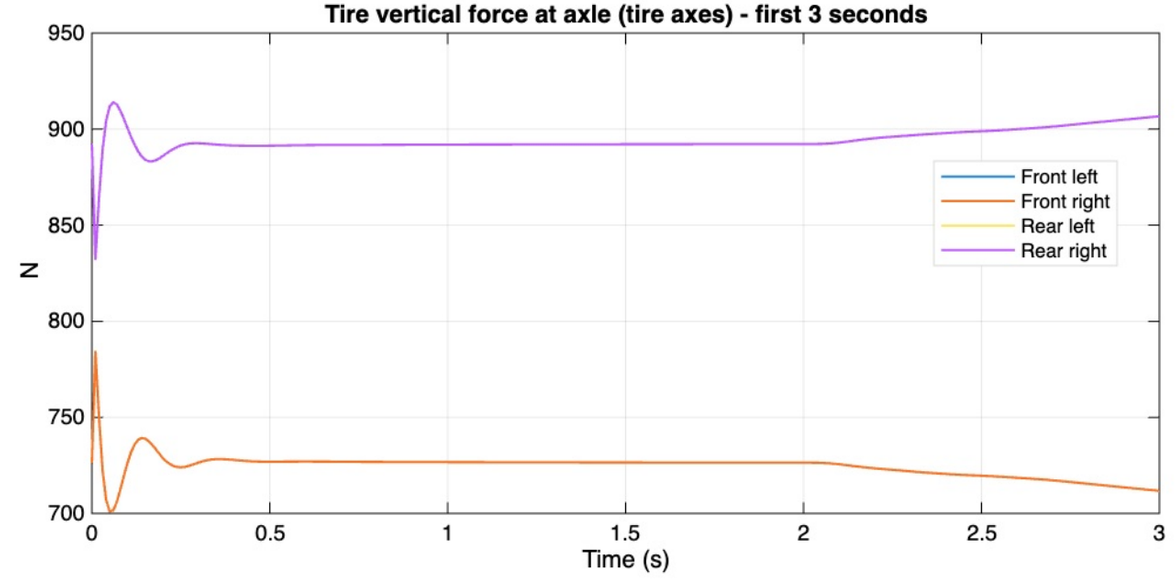


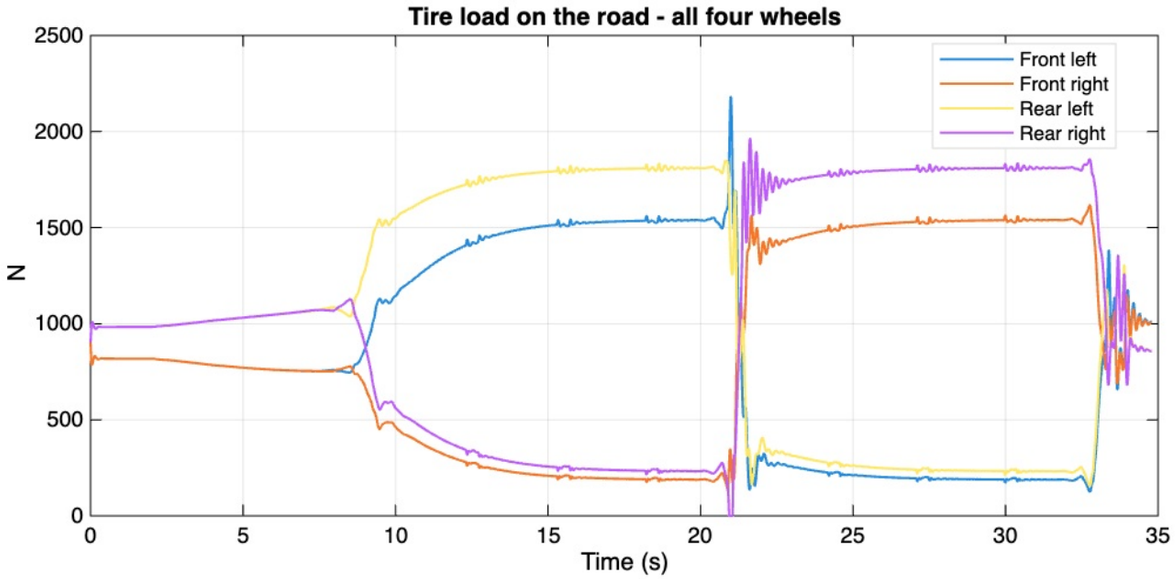



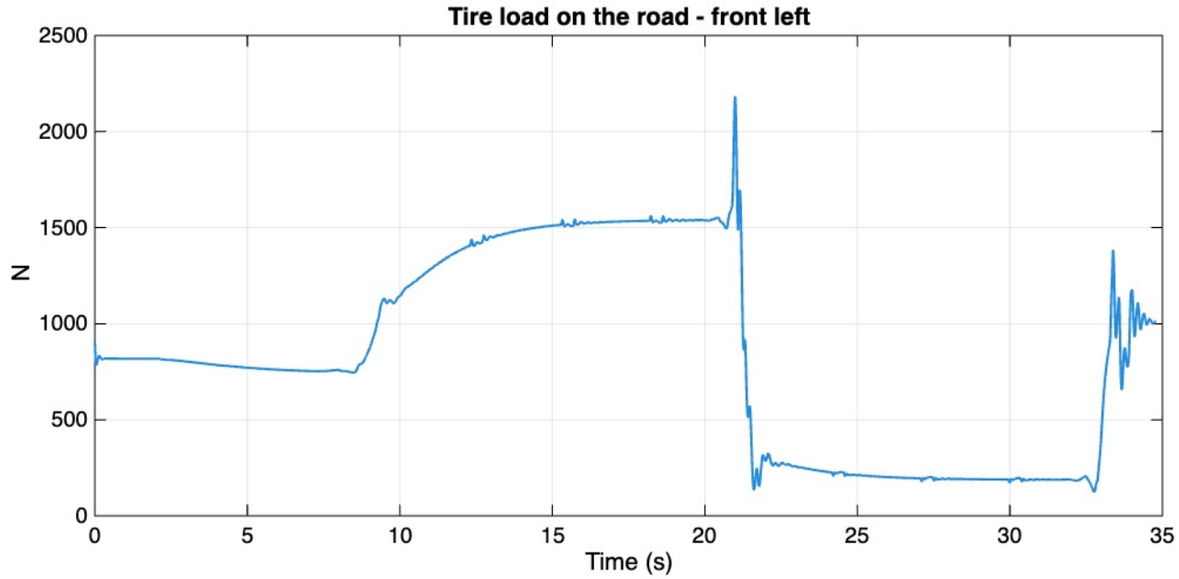



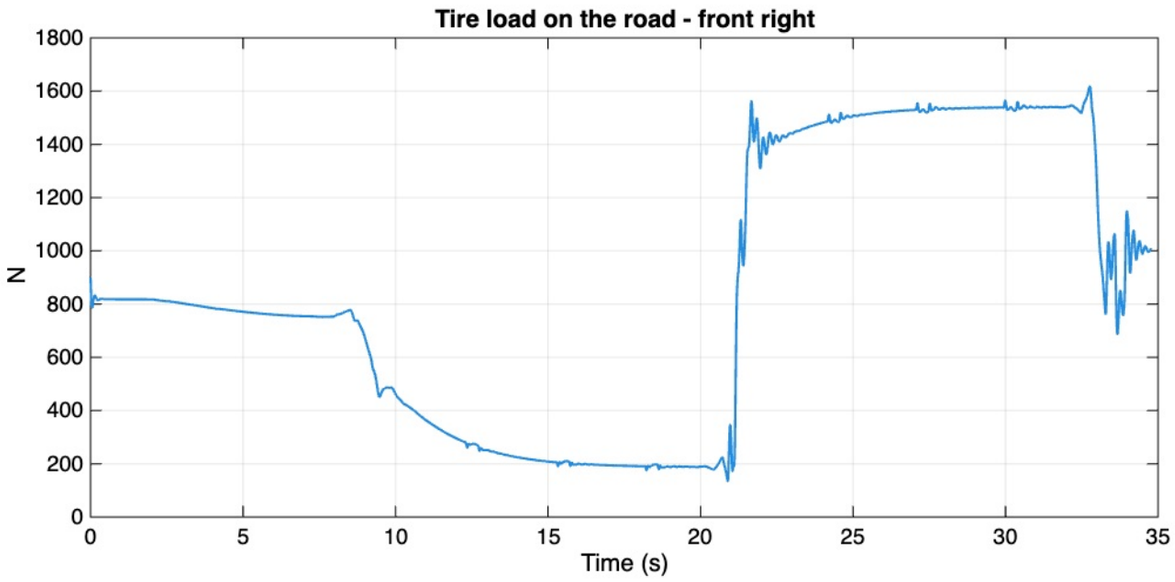



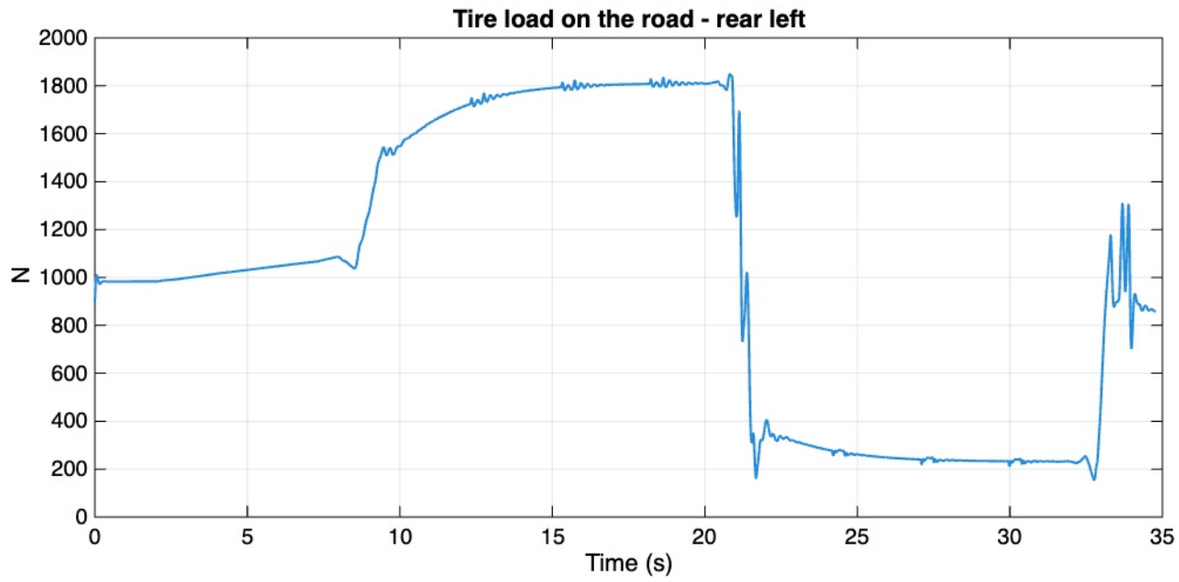



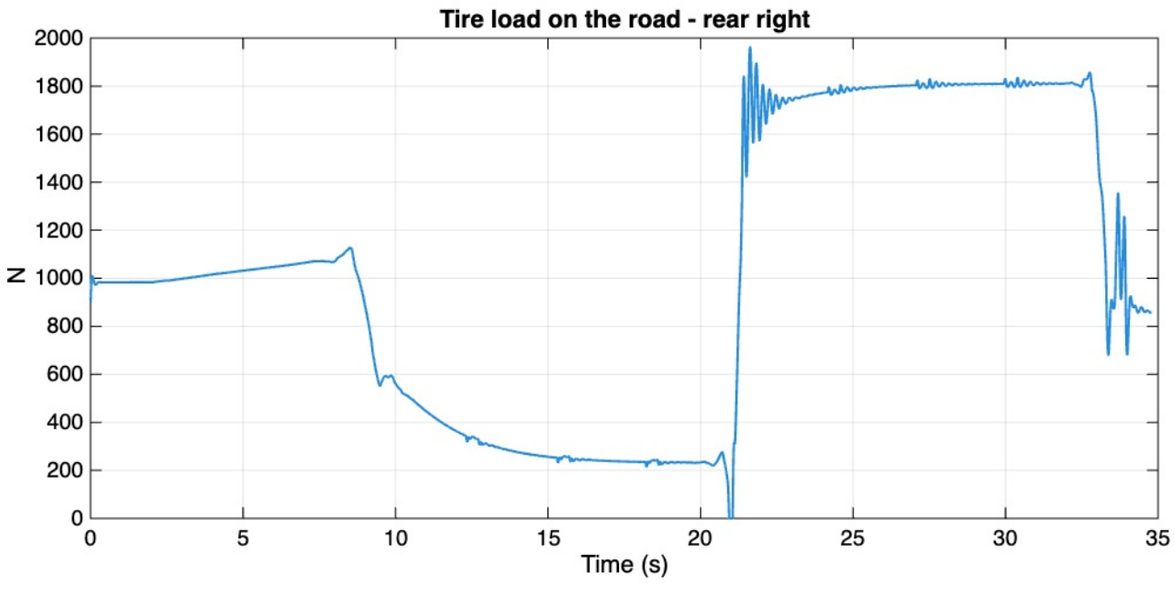
Simple signal plots for tuning - plot page 10

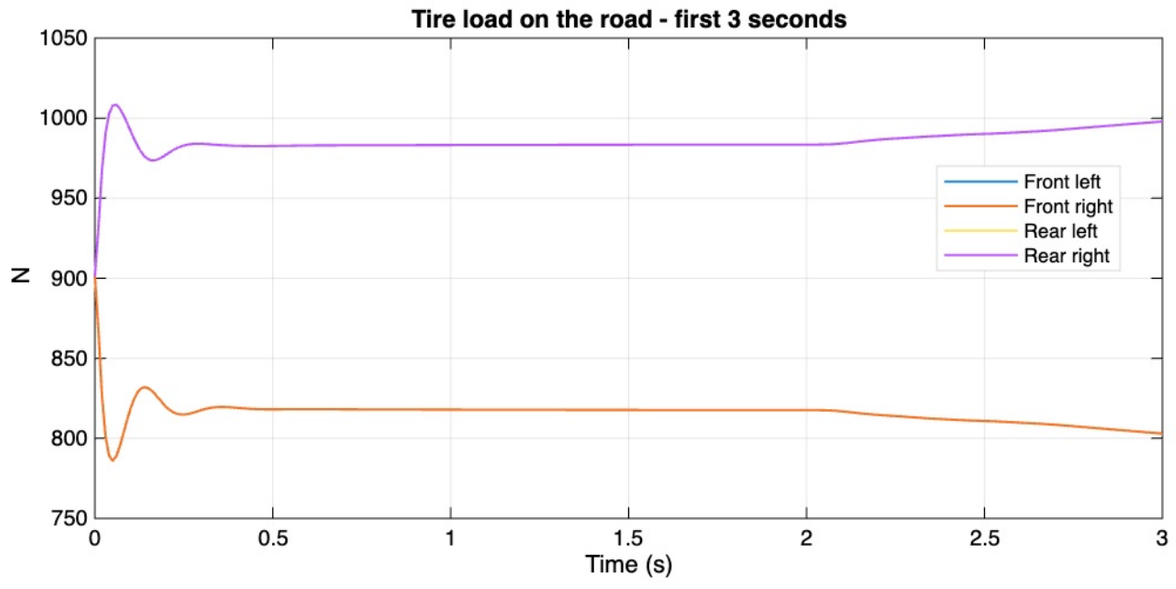


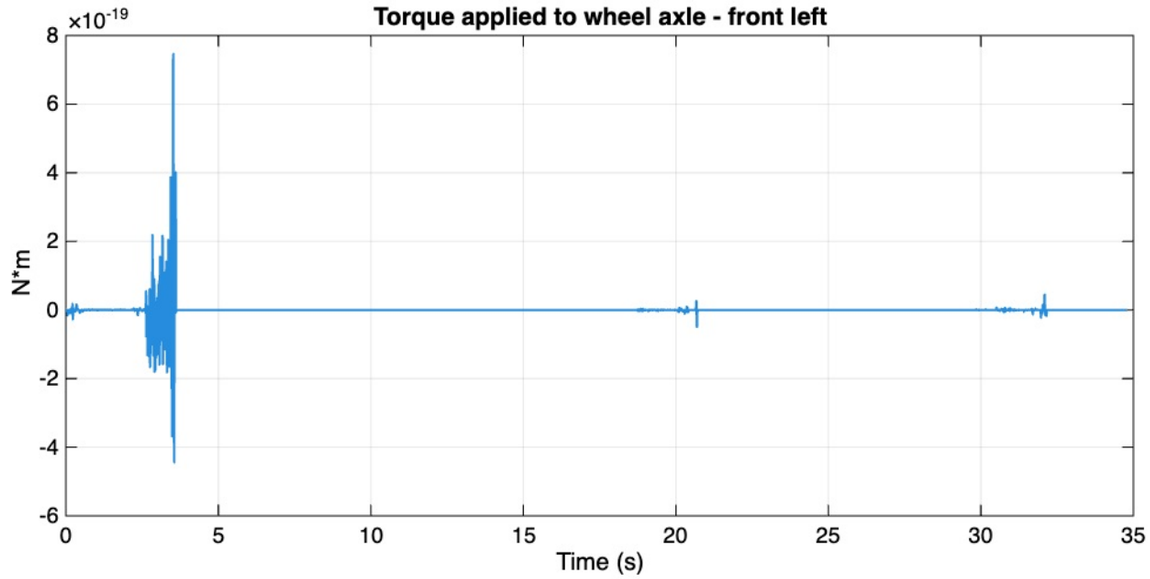



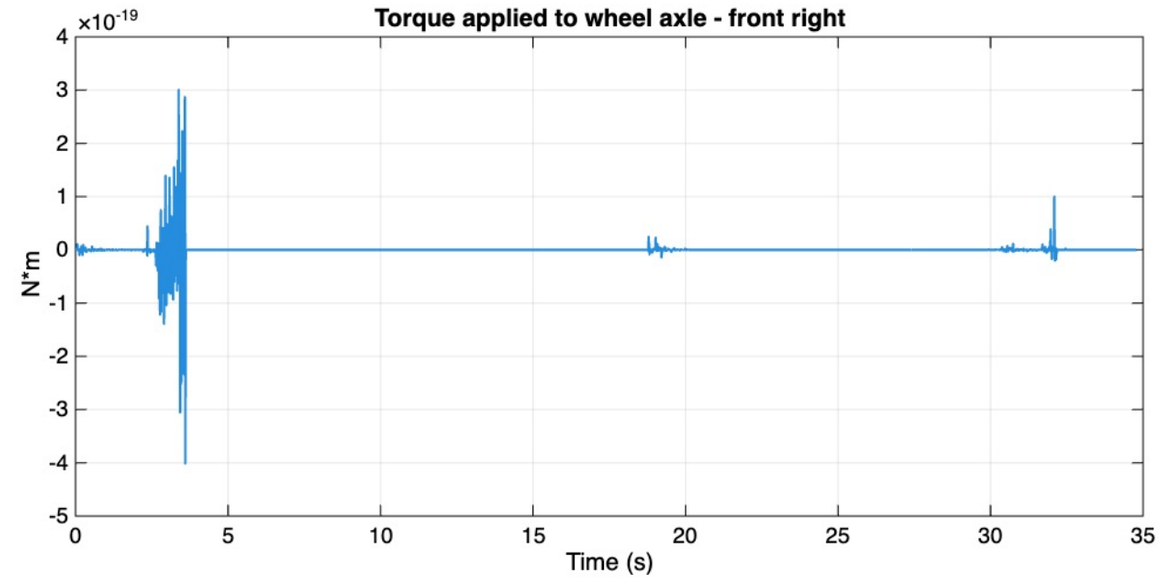



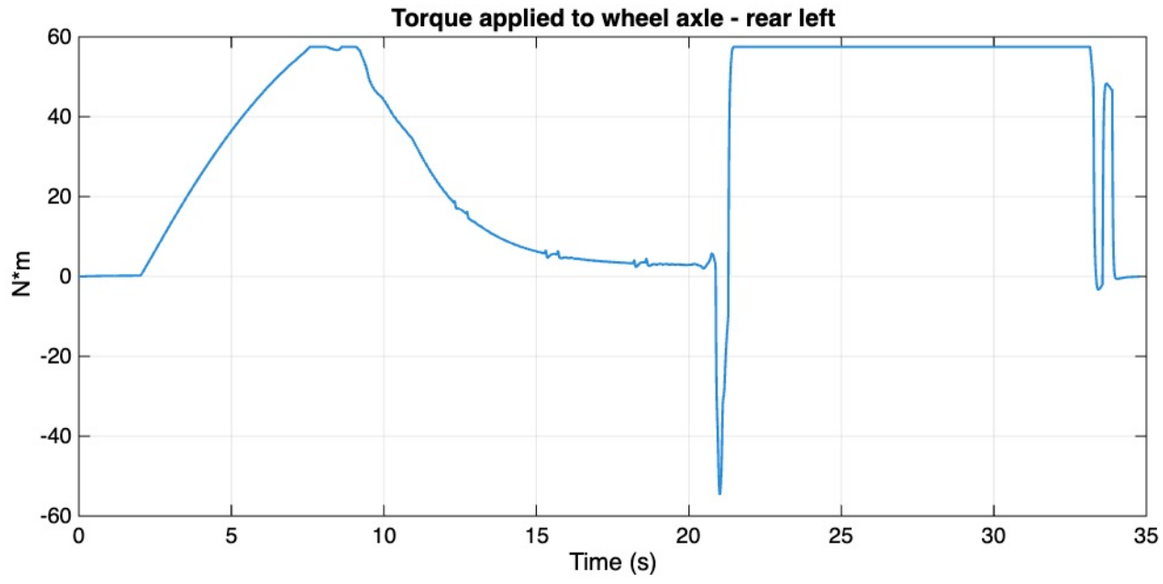



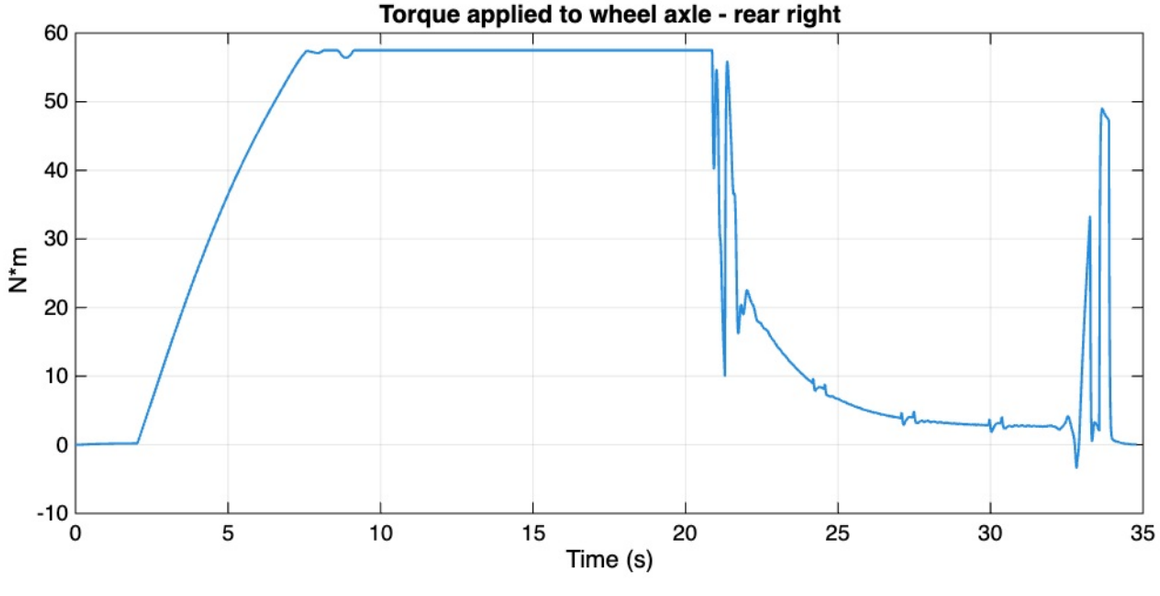



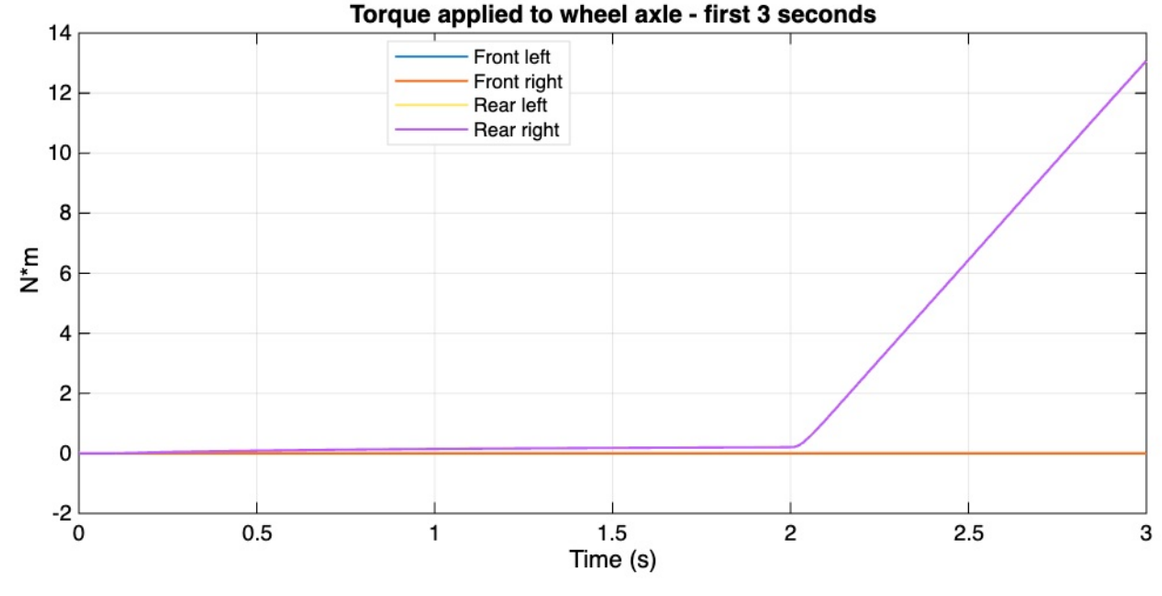

Torque applied to wheel axle - all four wheels
N*m
Time (s)
Front left
Front right
Rear left
Rear right


Torque applied to wheel axle - front left
×10^-19
N*m
Time (s)


Torque applied to wheel axle - front right
×10^-19
N*m
Time (s)

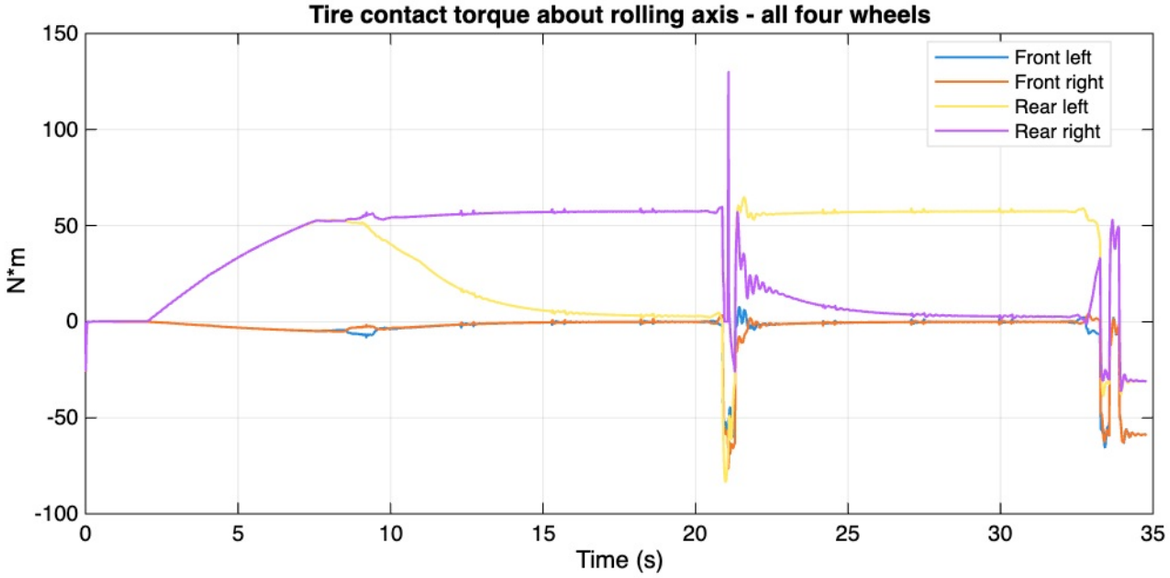

Torque applied to wheel axle - rear left
N*m
Time (s)


Torque applied to wheel axle - rear right
N*m
Time (s)

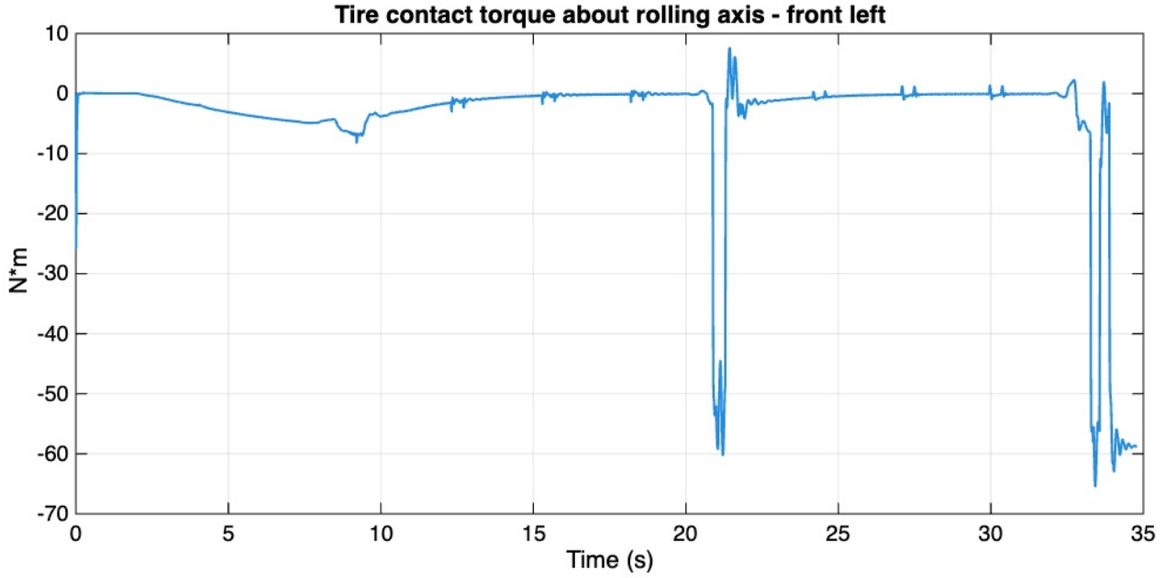

Torque applied to wheel axle - first 3 seconds
Front left
Front right
Rear left
Rear right
N*m
Time (s)

Simple signal plots for tuning - plot page 12

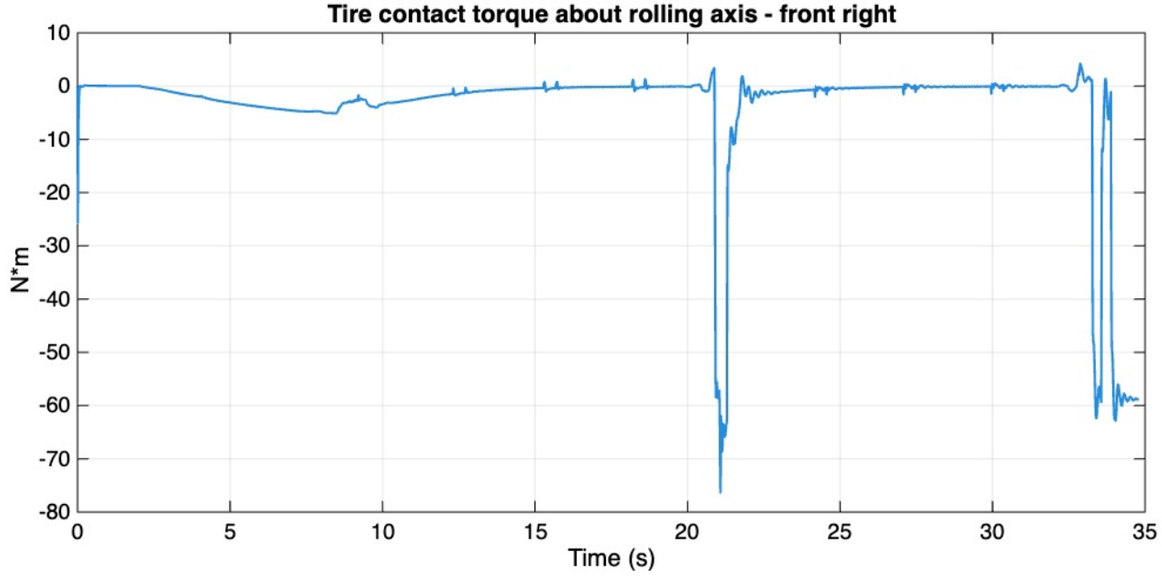
**Tire contact torque about rolling axis - all four wheels**



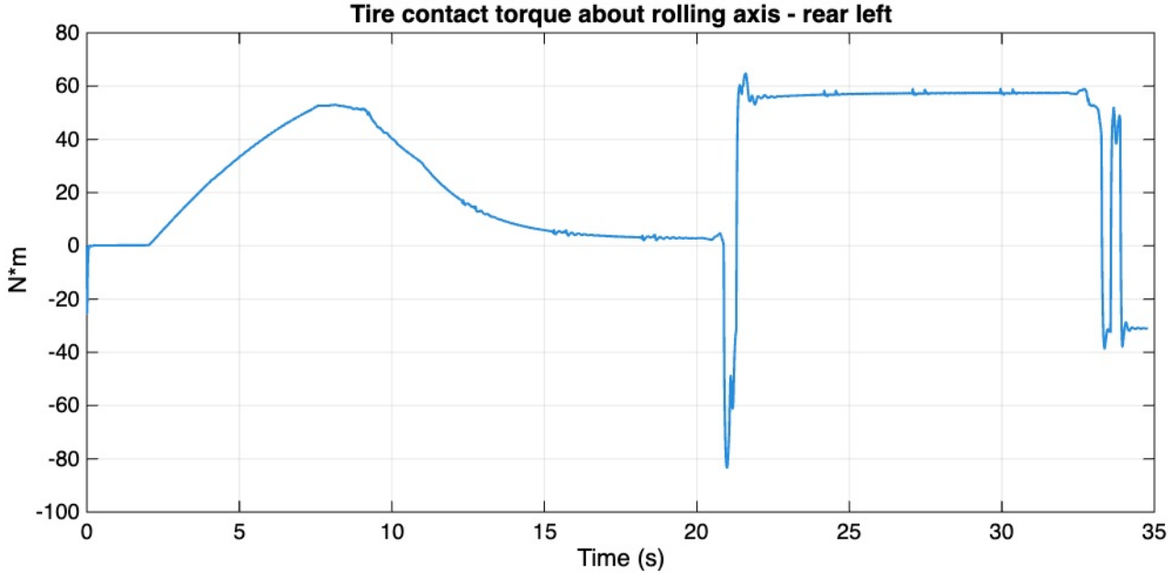

**Tire contact torque about rolling axis - front left**



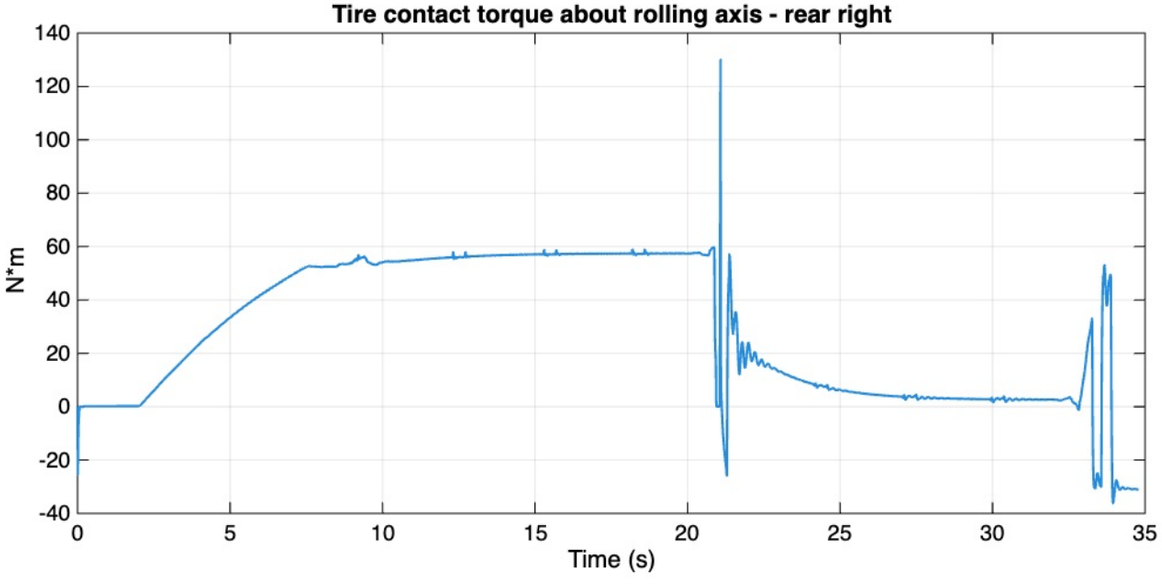

**Tire contact torque about rolling axis - front right**



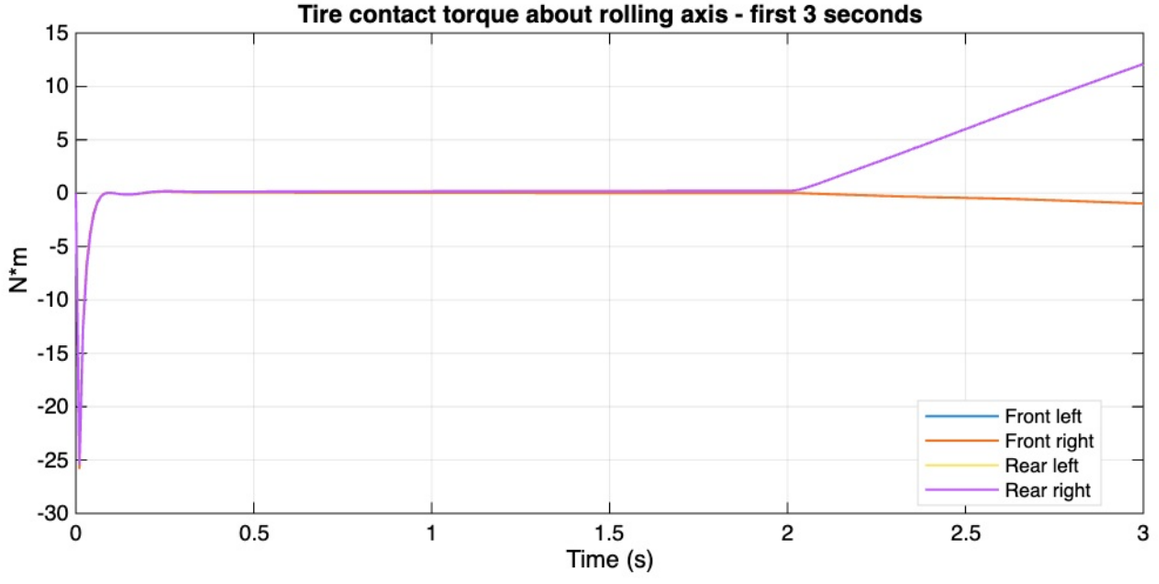

**Tire contact torque about rolling axis - rear left**



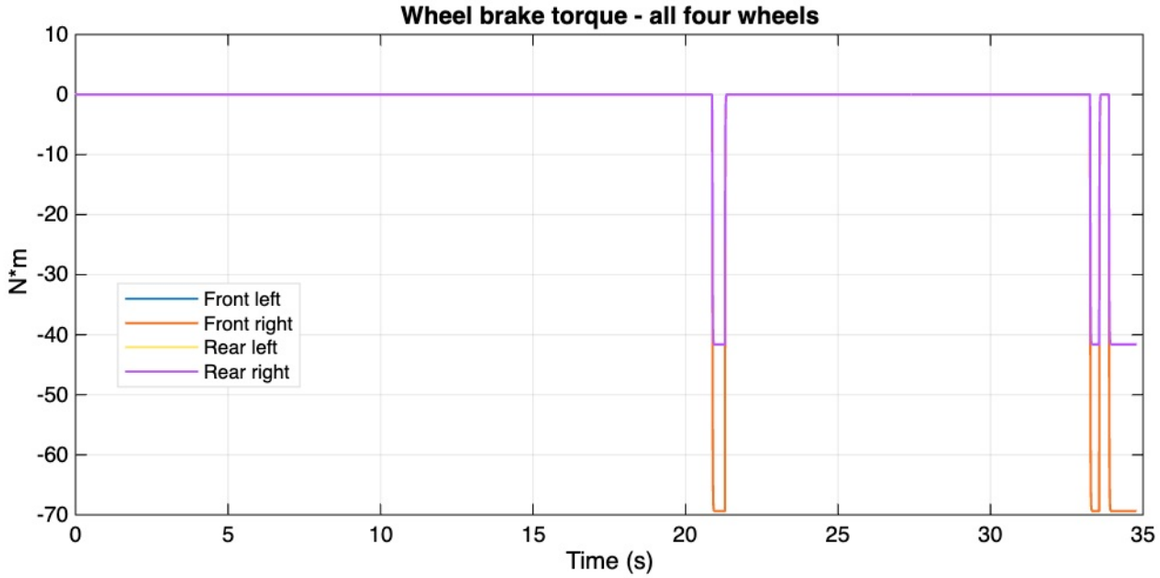

**Tire contact torque about rolling axis - rear right**

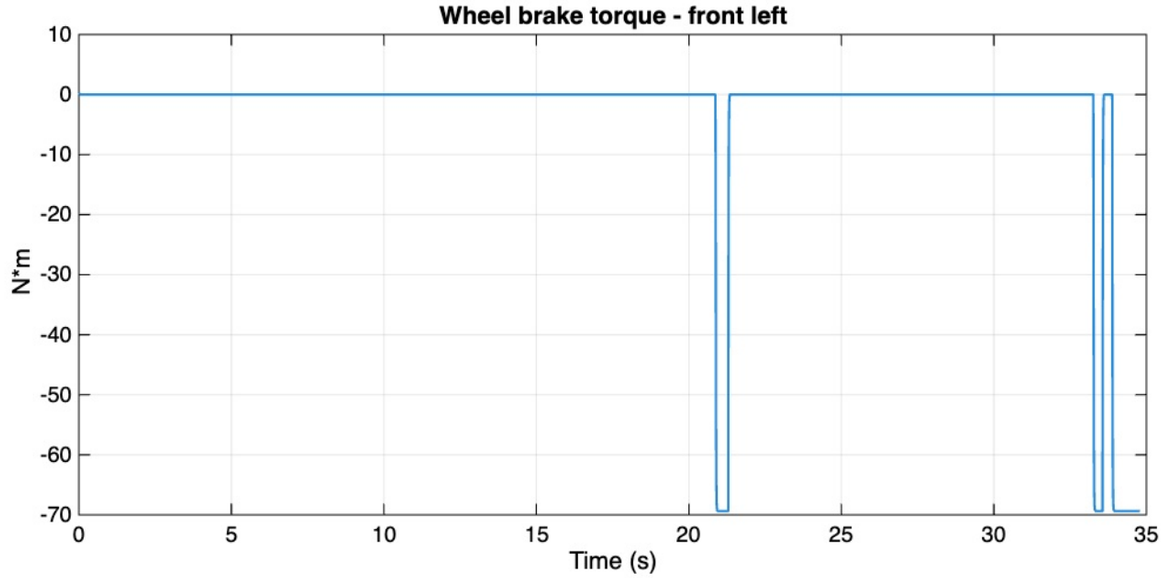


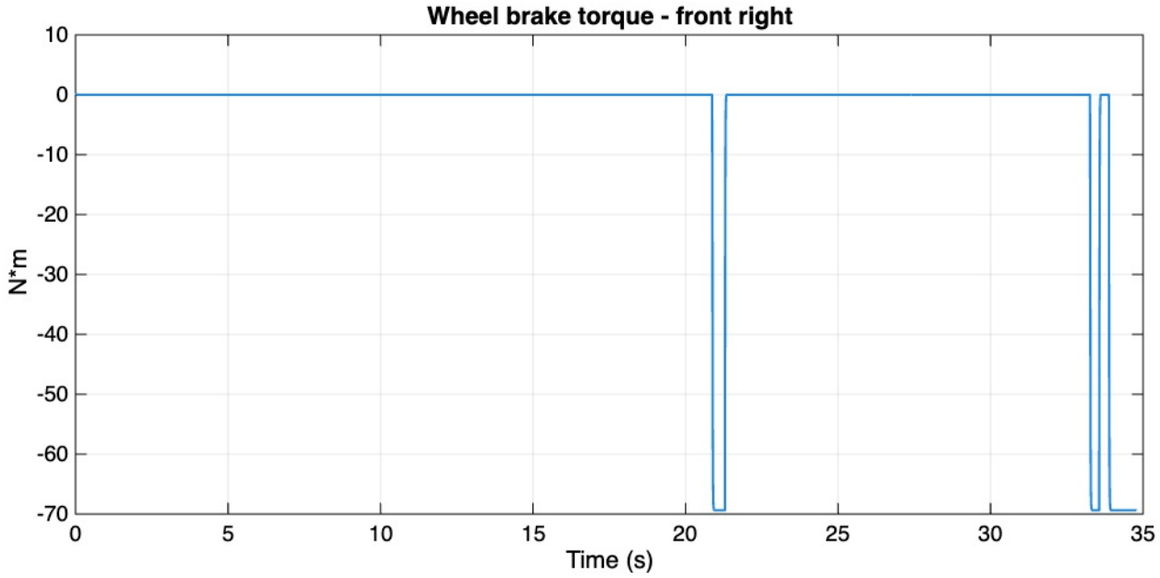

**Tire contact torque about rolling axis - first 3 seconds**

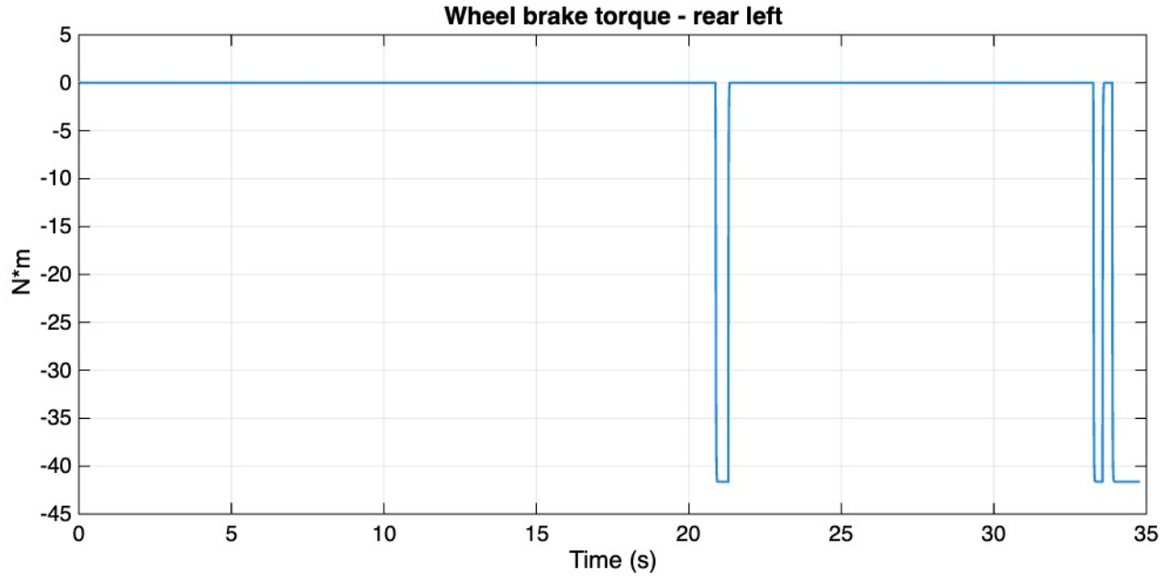


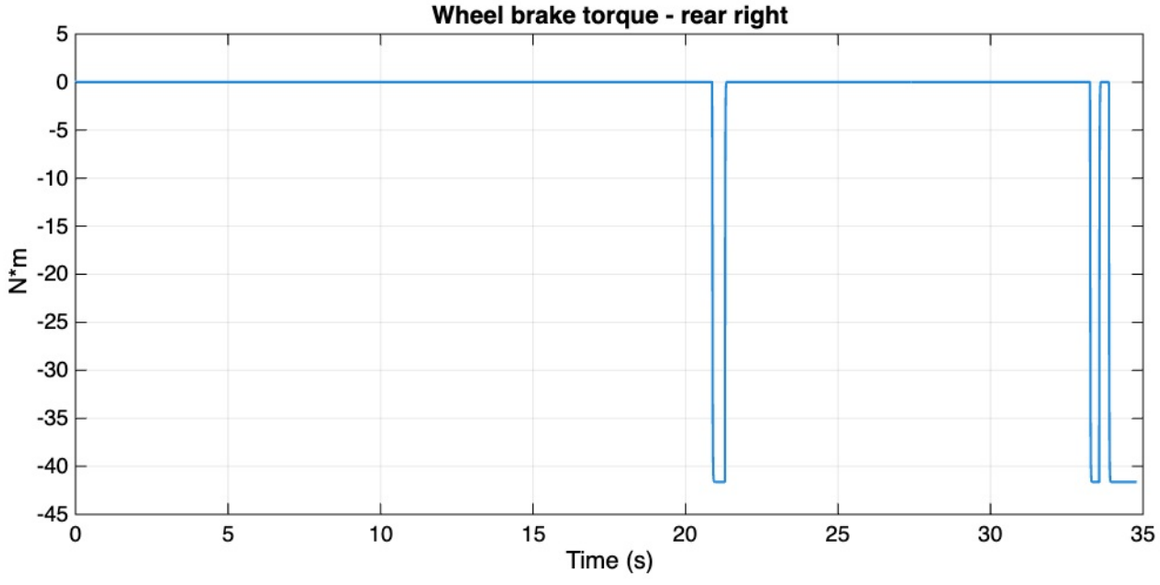
Simple signal plots for tuning - plot page 13

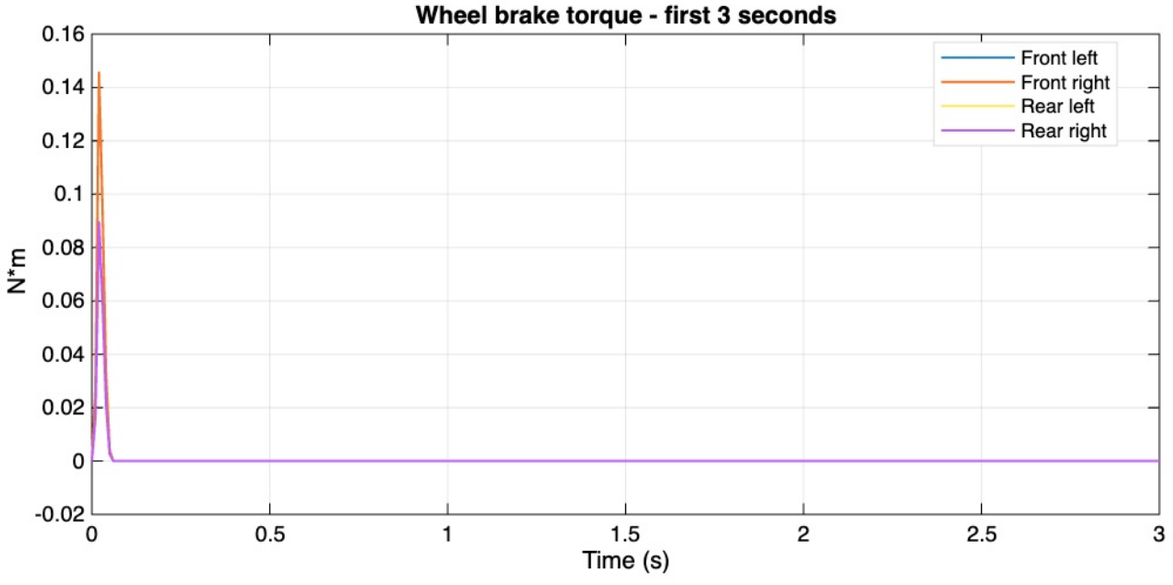


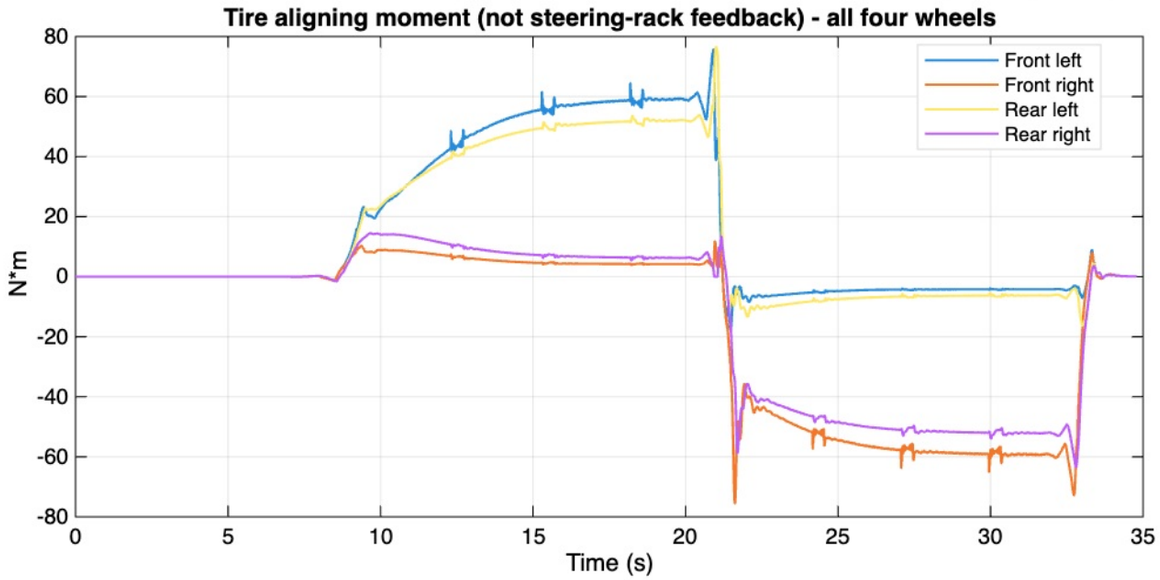



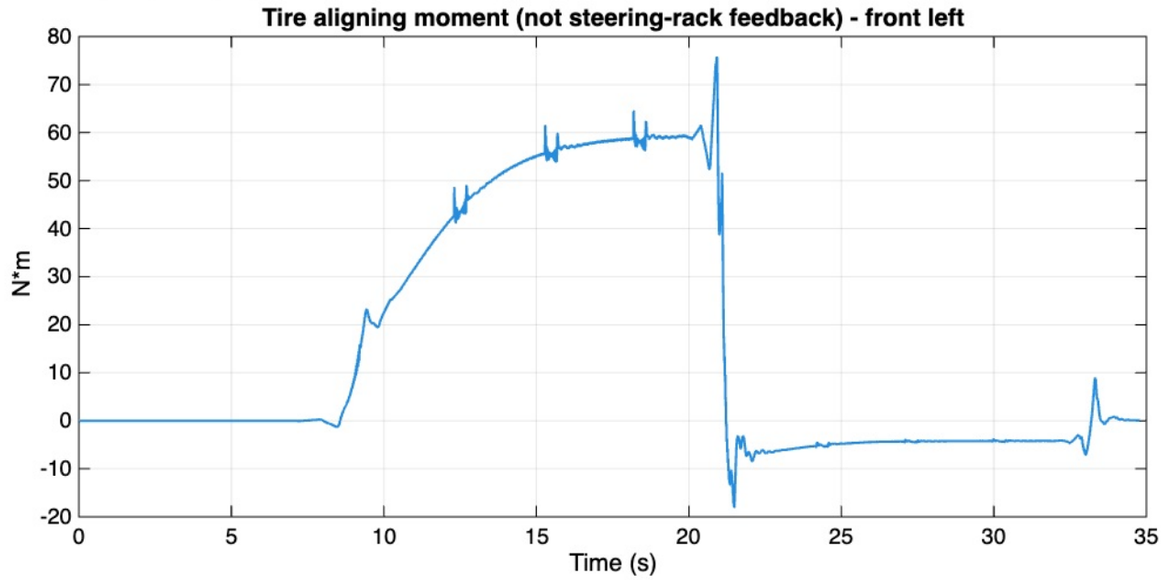



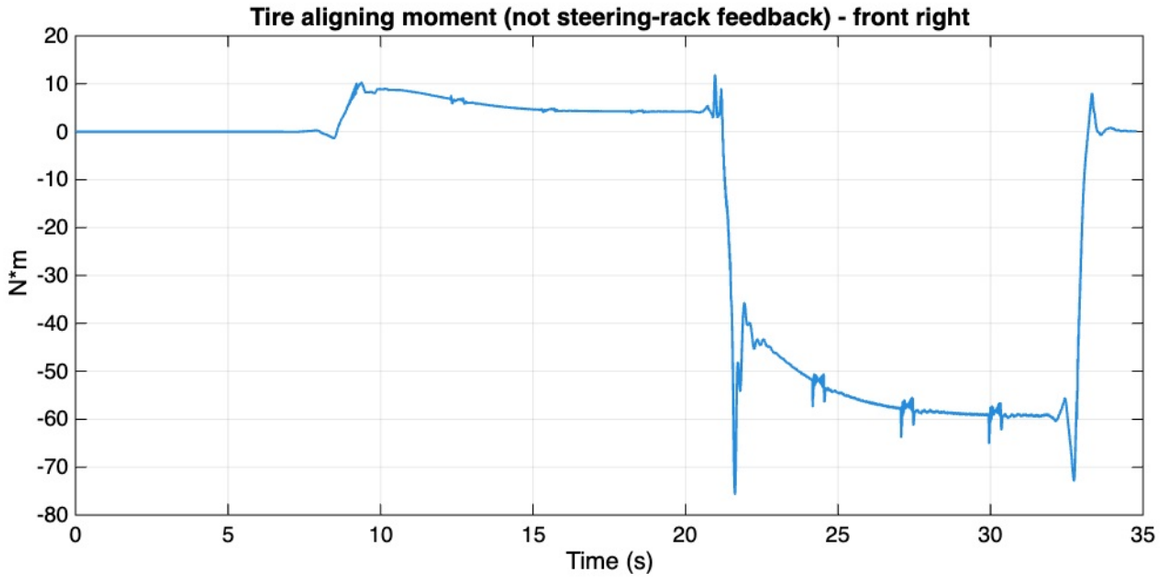



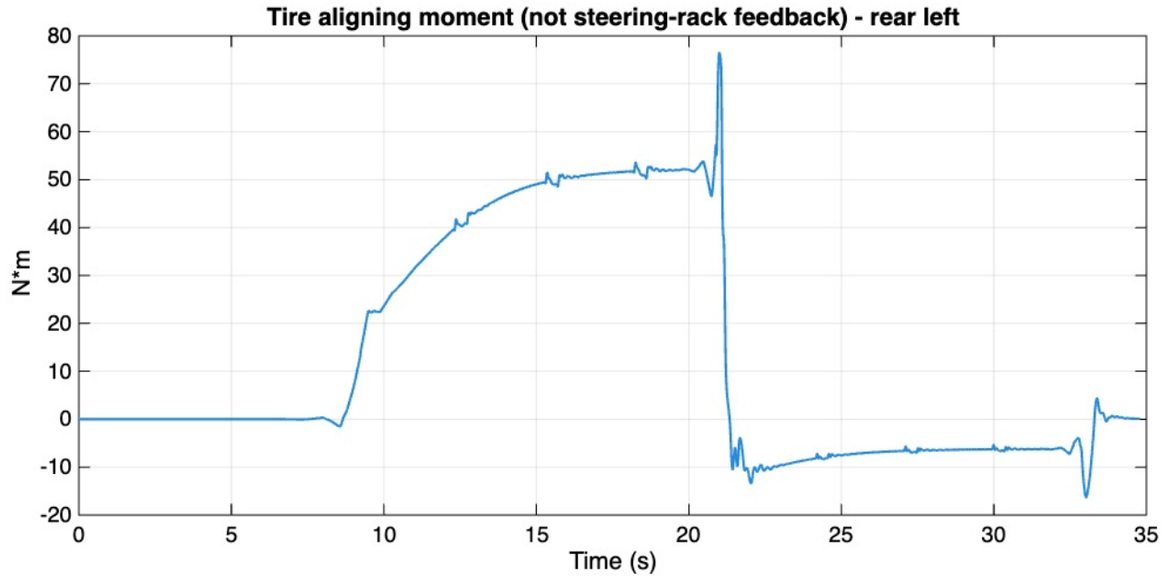



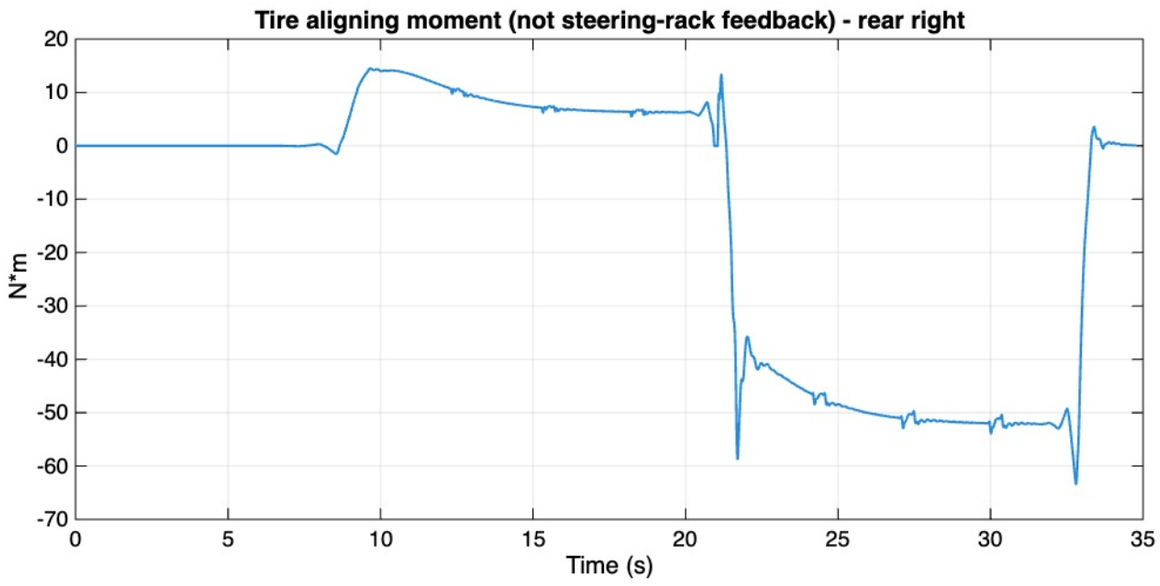
# Simple signal plots for tuning - plot page 14

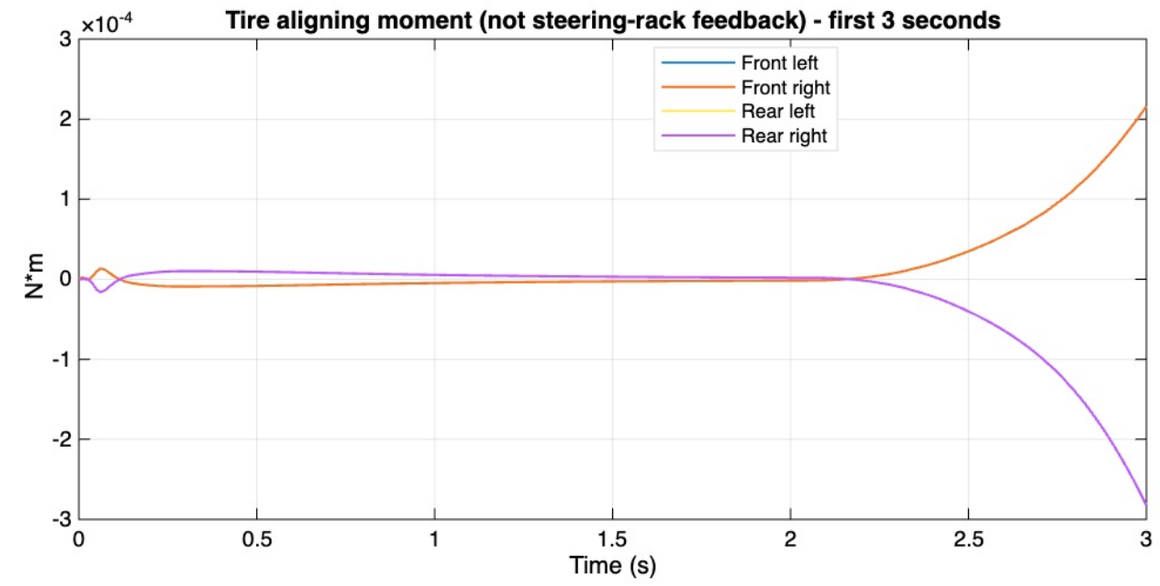


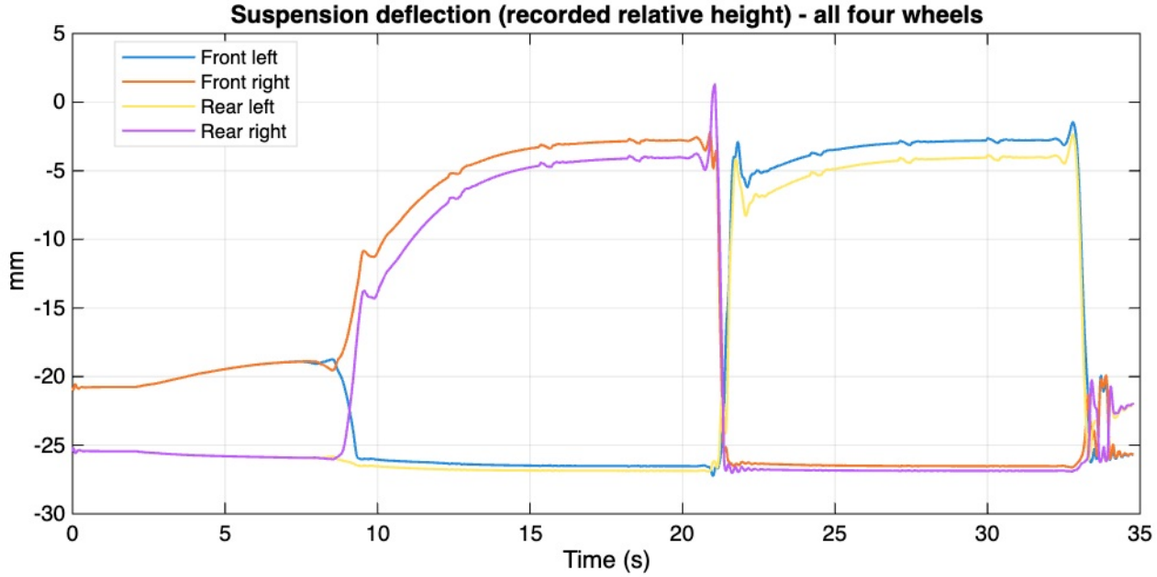



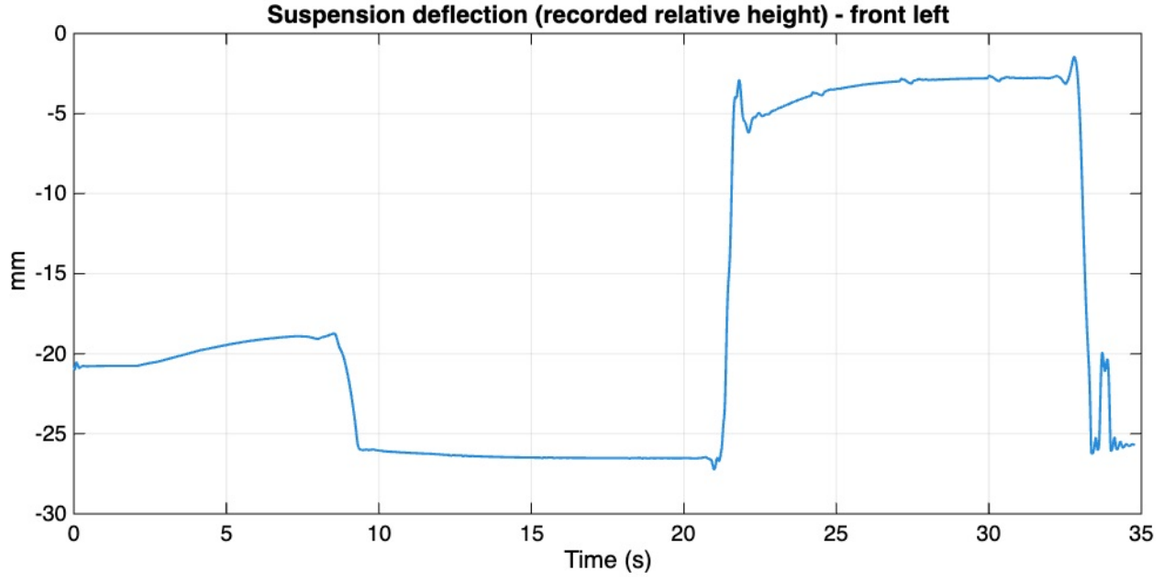



Simple signal plots for tuning - plot page 15

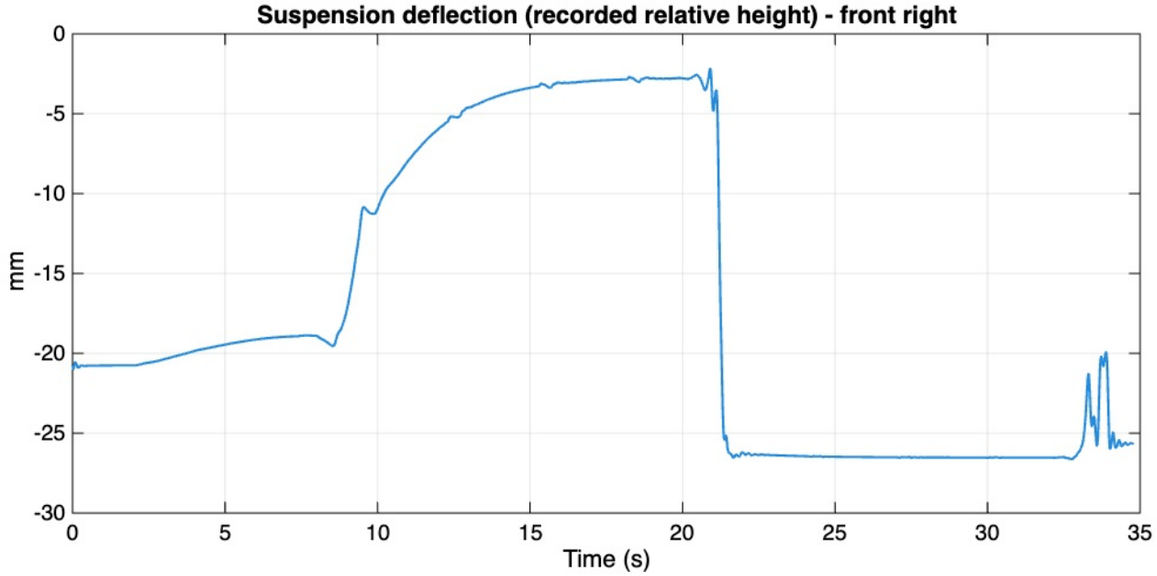
**Suspension deflection (recorded relative height) - all four wheels**



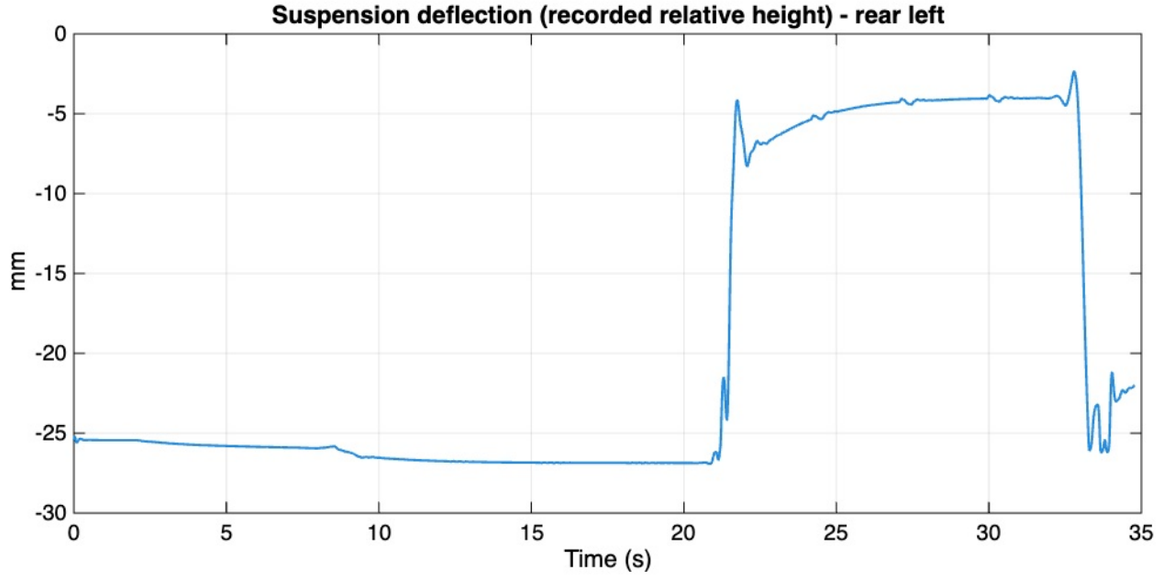

**Suspension deflection (recorded relative height) - front left**



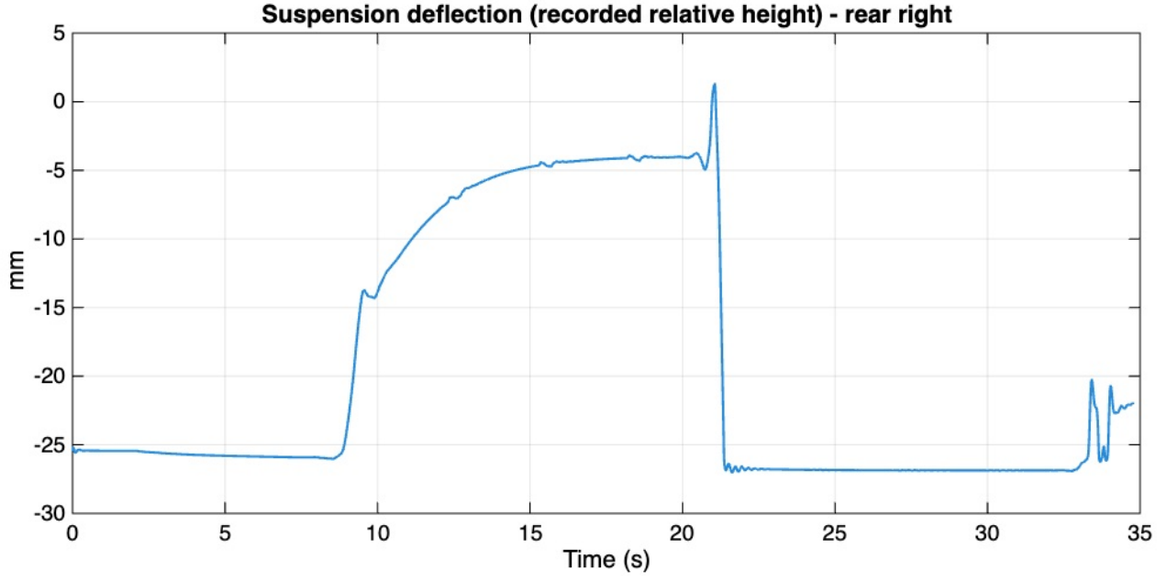

**Suspension deflection (recorded relative height) - front right**



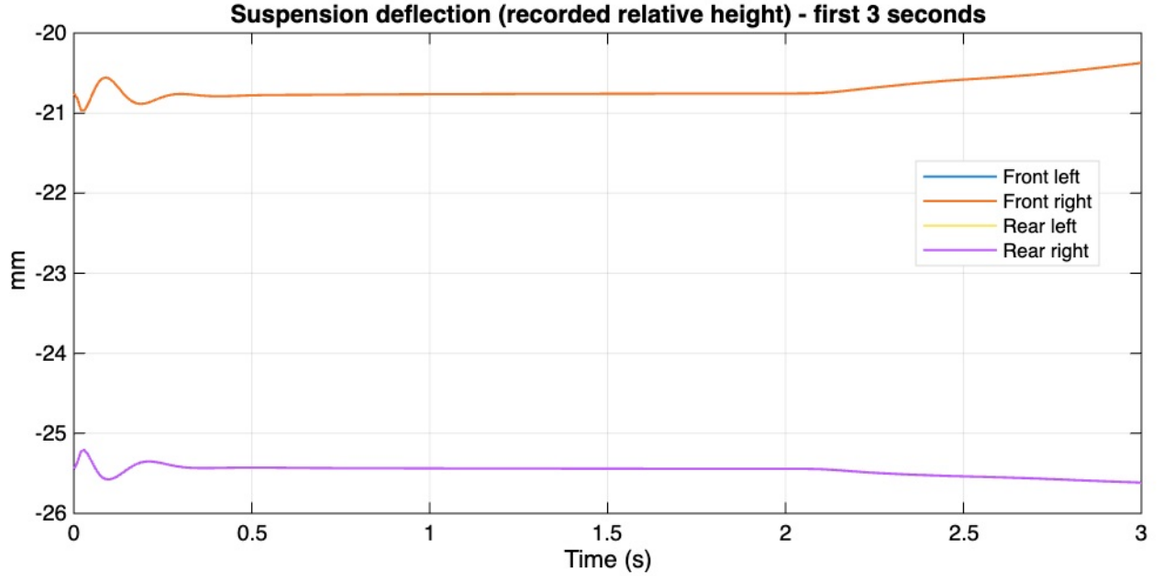

**Suspension deflection (recorded relative height) - rear left**




**Suspension deflection (recorded relative height) - rear right**

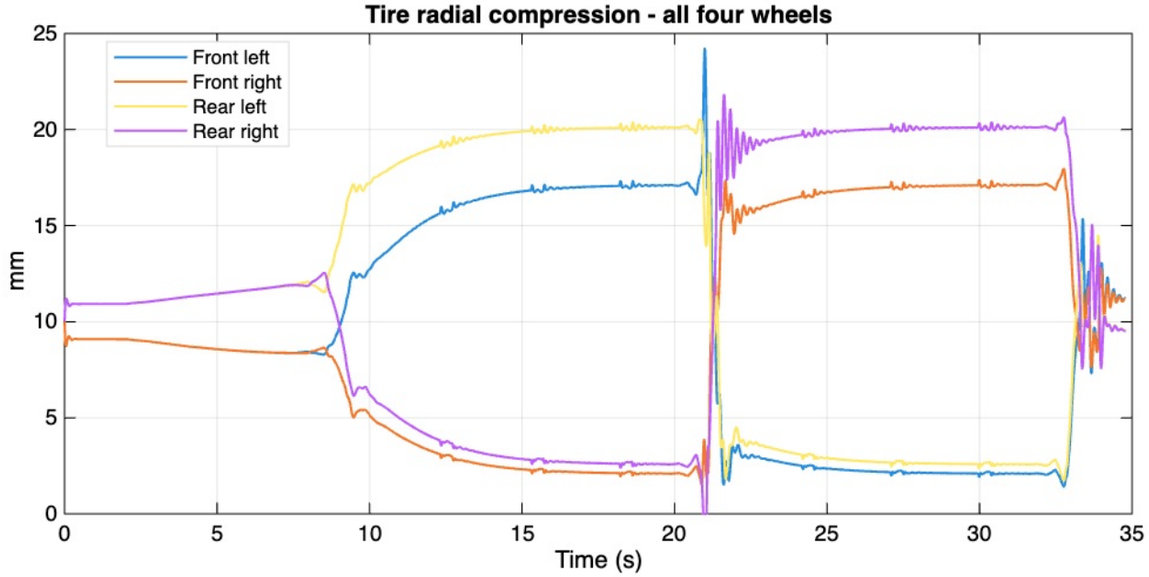



**Suspension deflection (recorded relative height) - first 3 seconds**

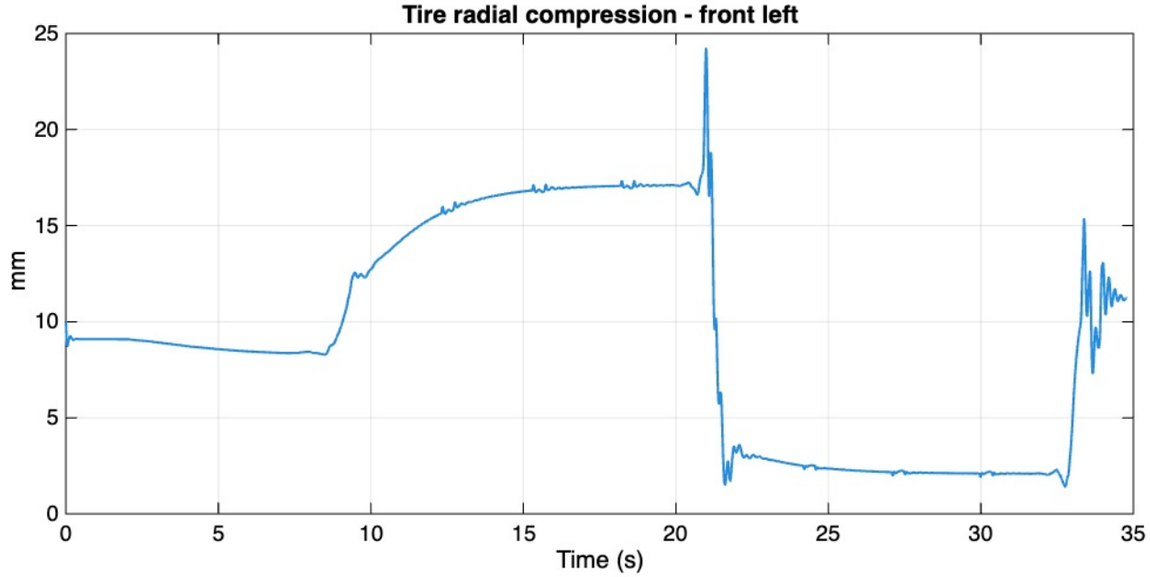


Simple signal plots for tuning - plot page 16

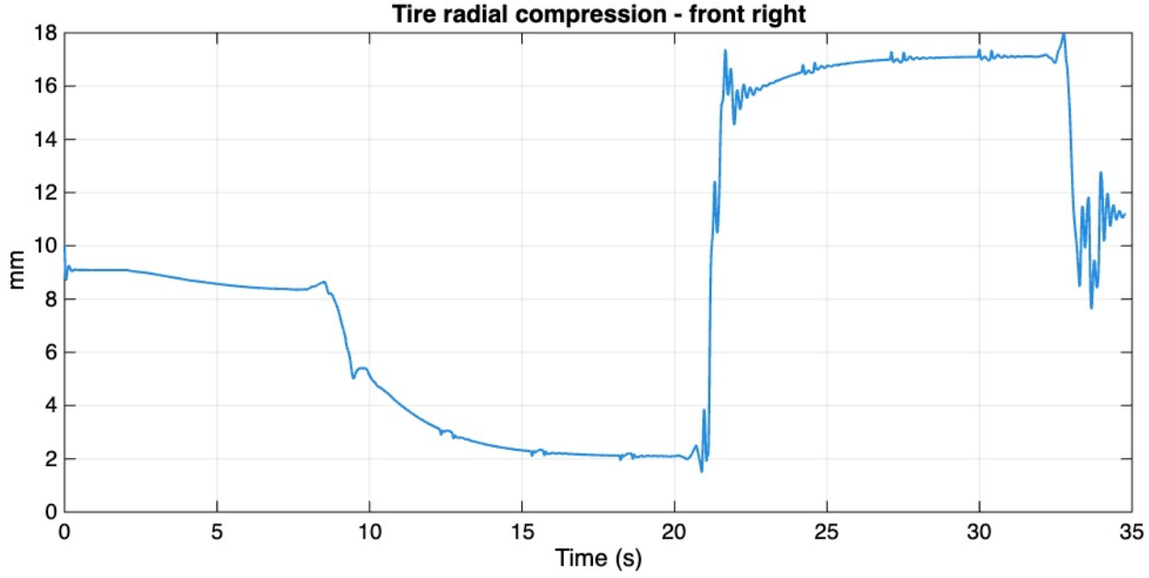
**Tire radial compression - all four wheels**



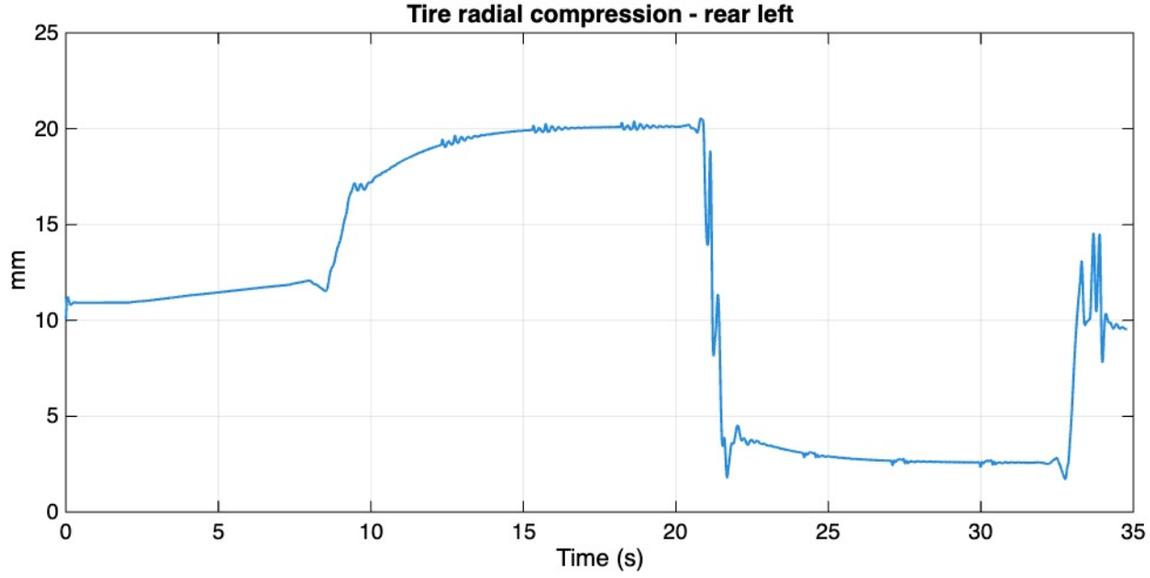

**Tire radial compression - front left**



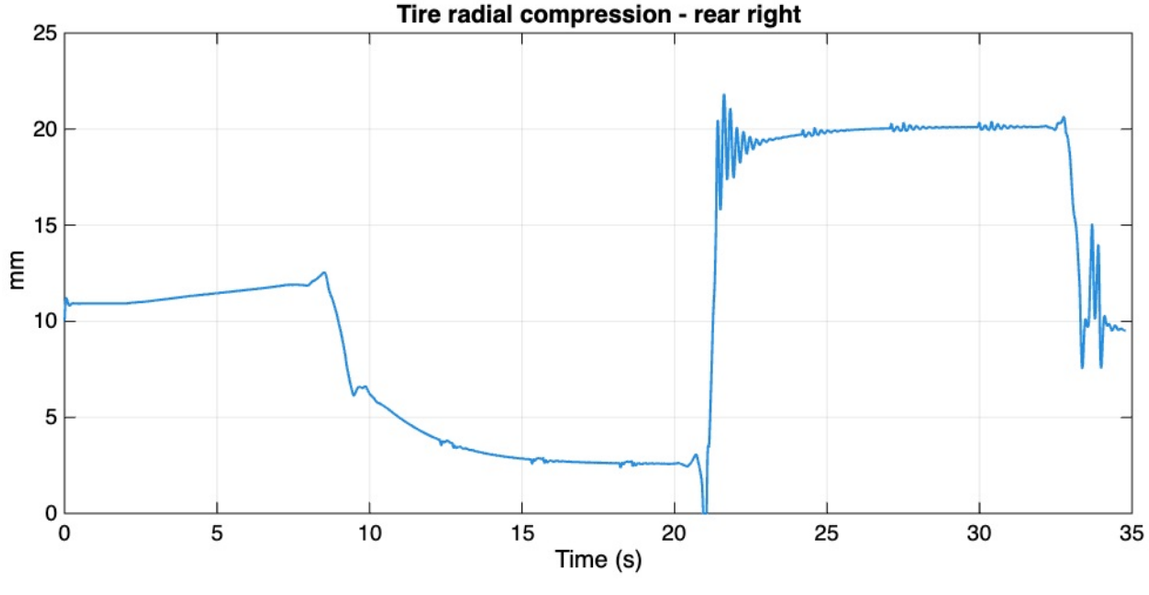

**Tire radial compression - front right**



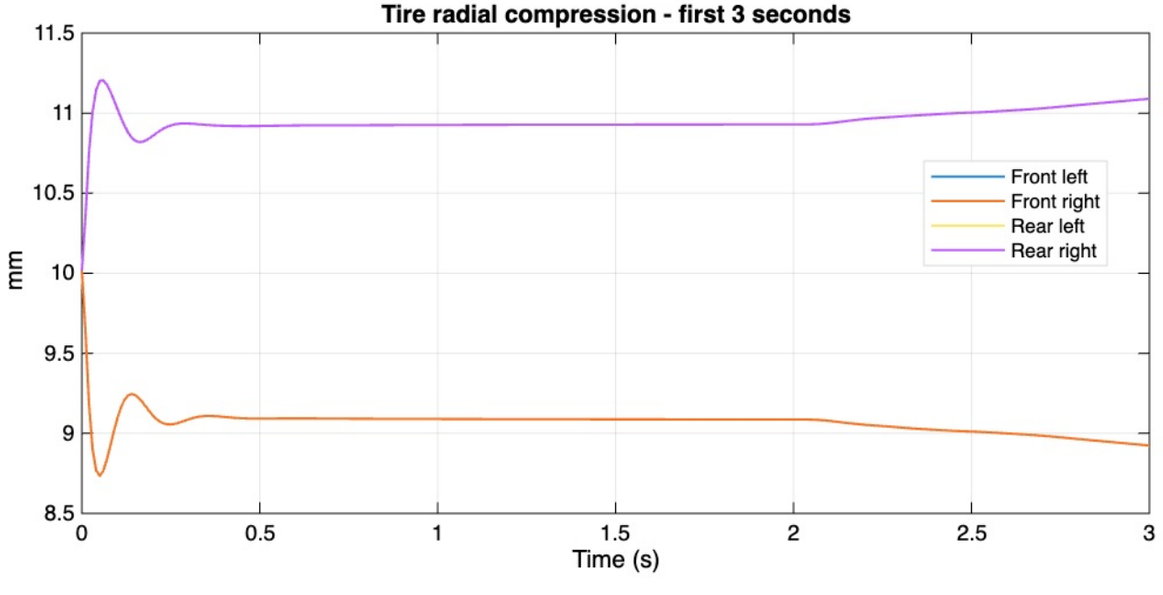

**Tire radial compression - rear left**




**Tire radial compression - rear right**

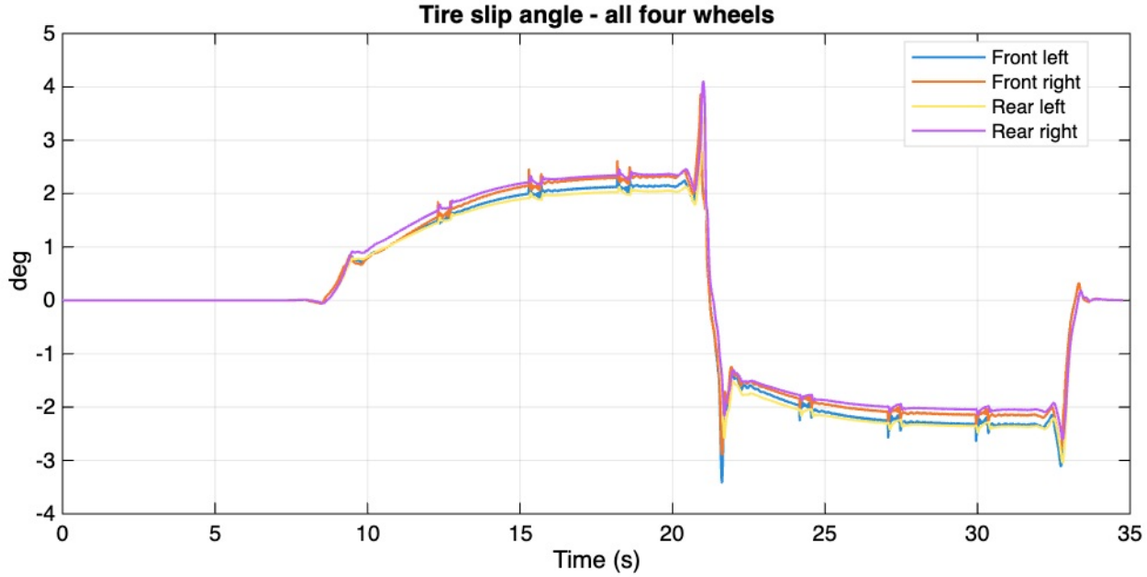



**Tire radial compression - first 3 seconds**

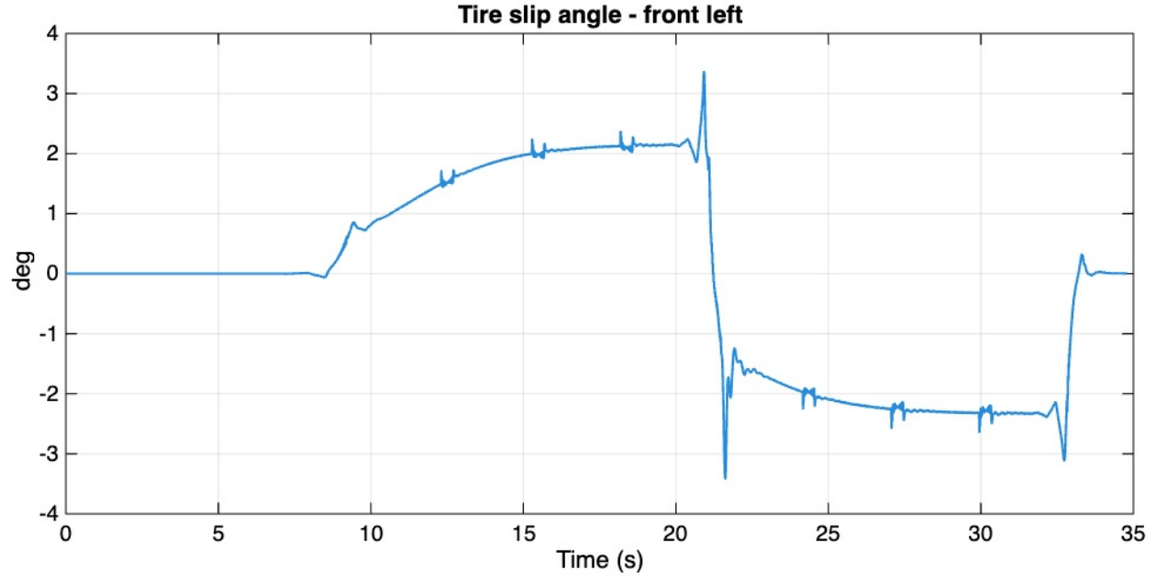


Simple signal plots for tuning - plot page 17

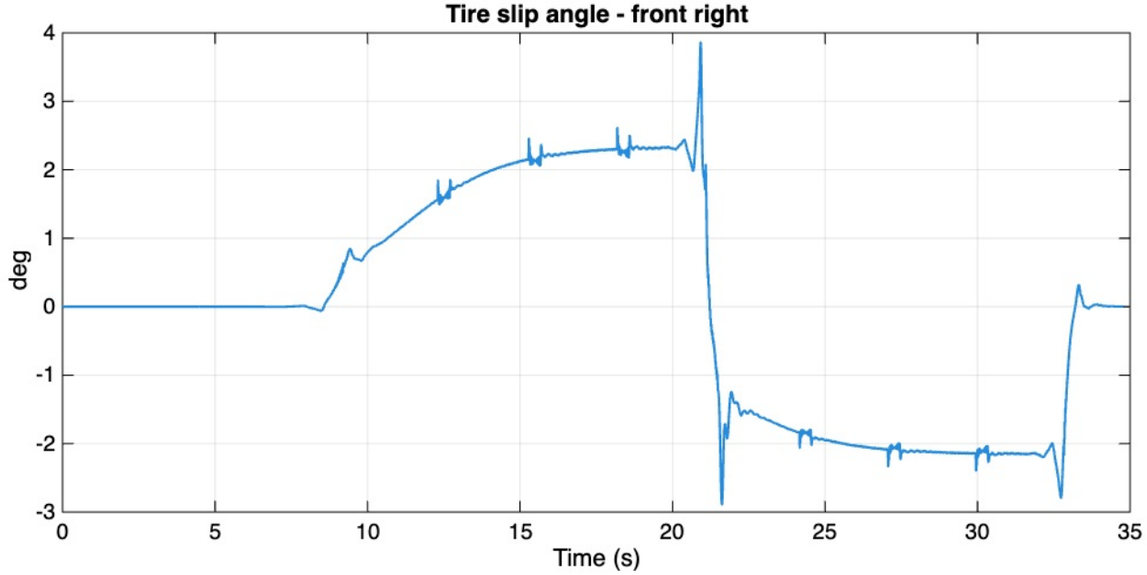
**Tire slip angle - all four wheels**



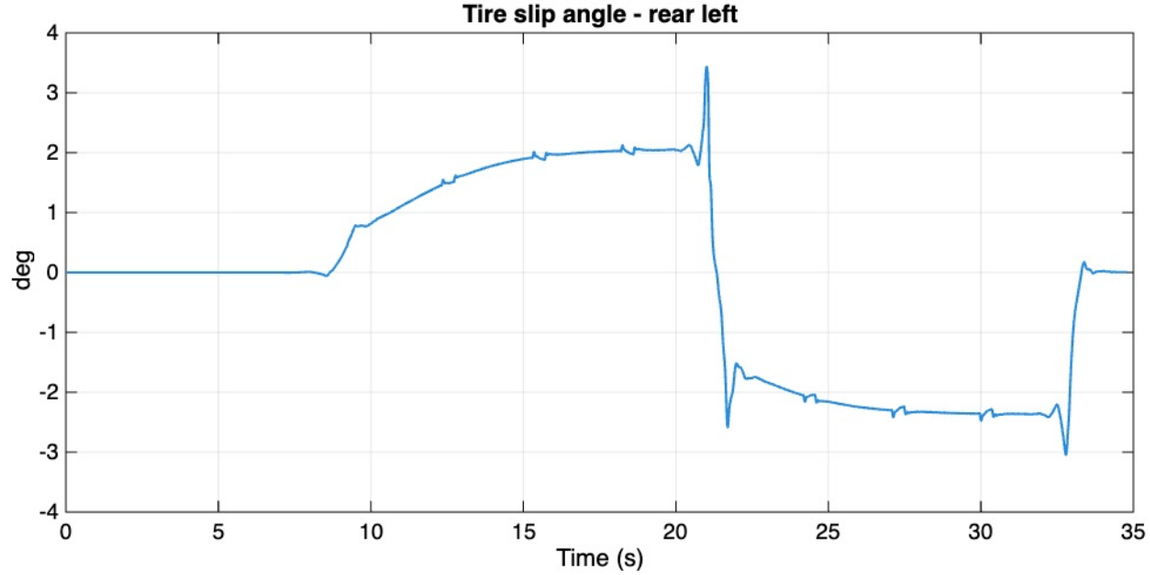

**Tire slip angle - front left**



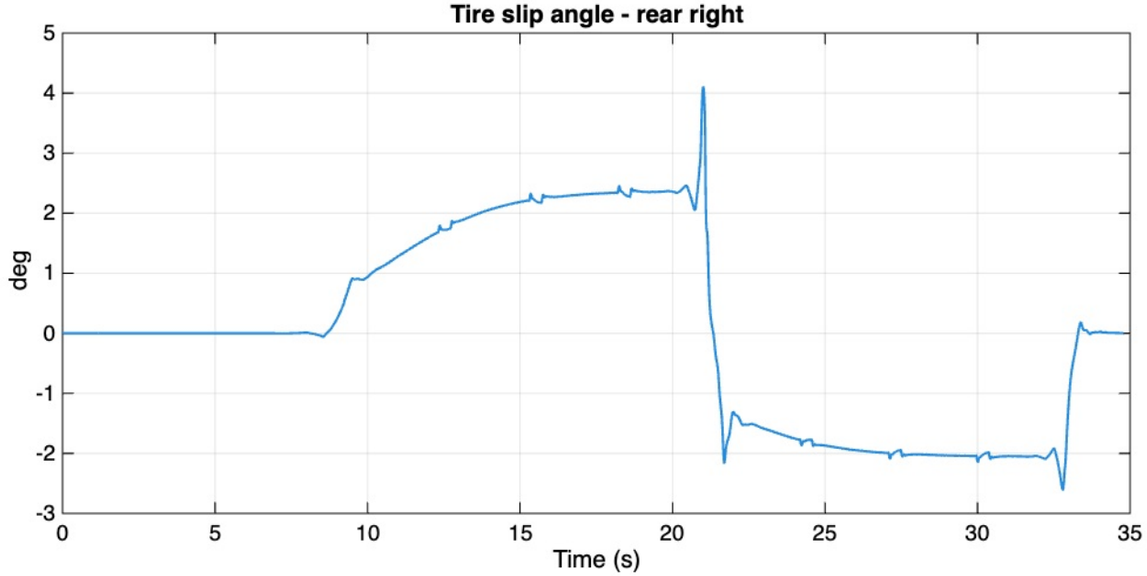

**Tire slip angle - front right**



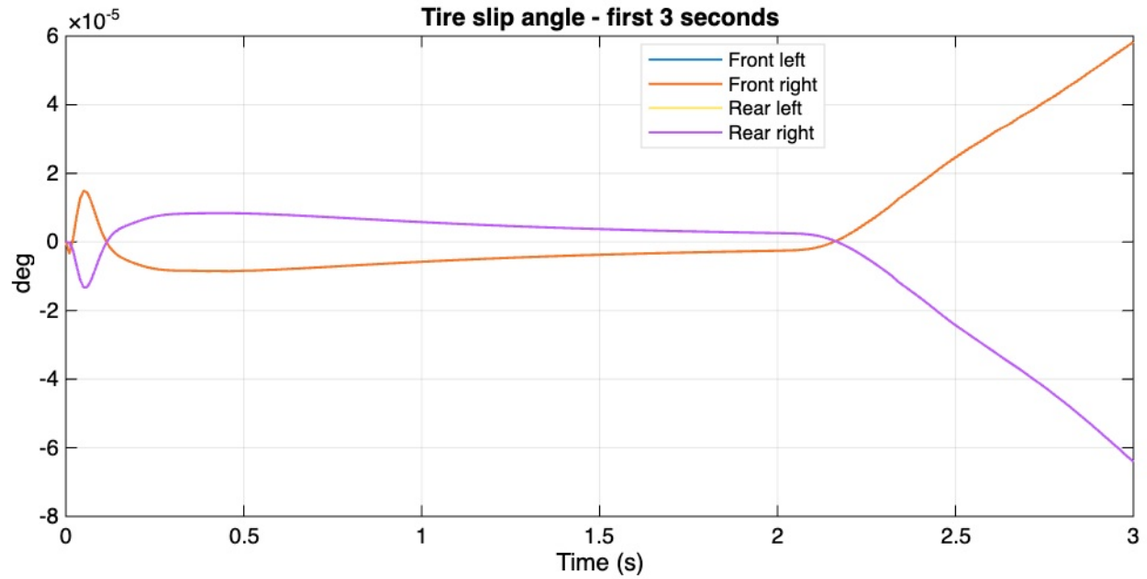

**Tire slip angle - rear left**



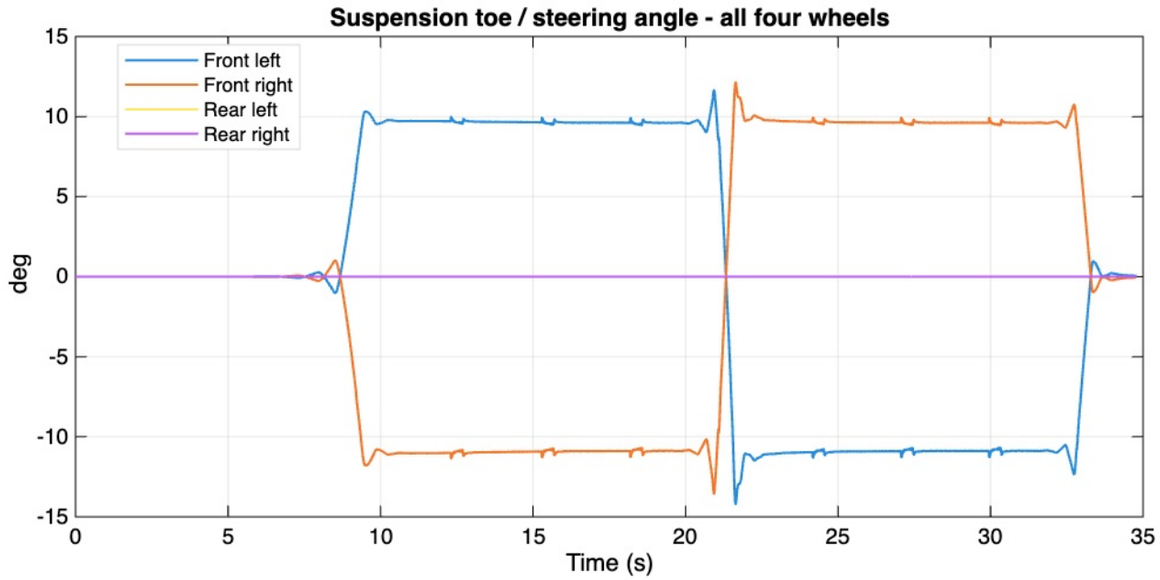

**Tire slip angle - rear right**

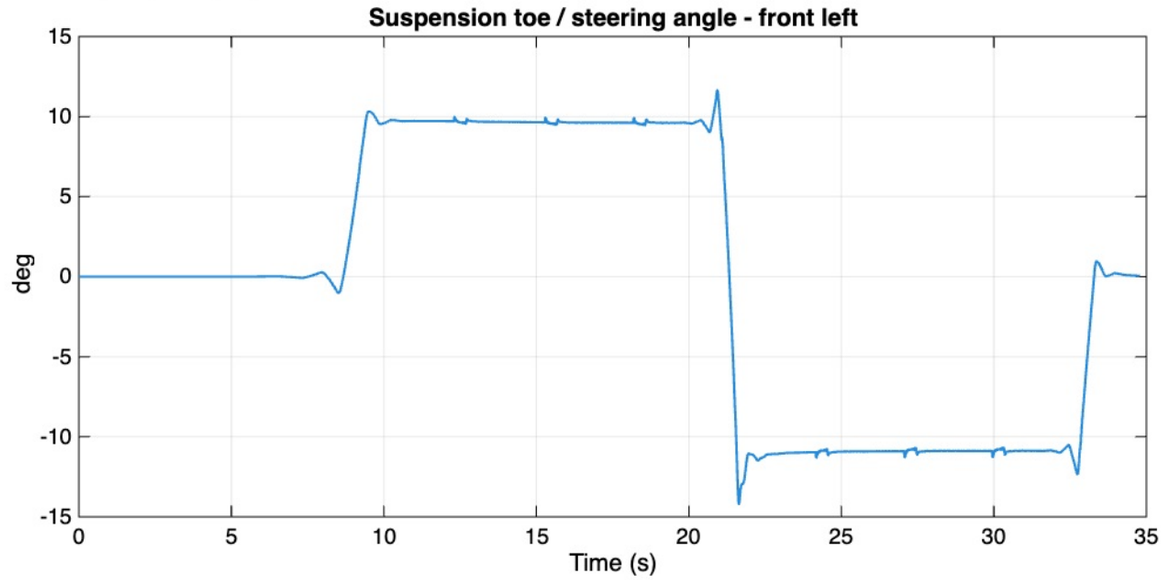


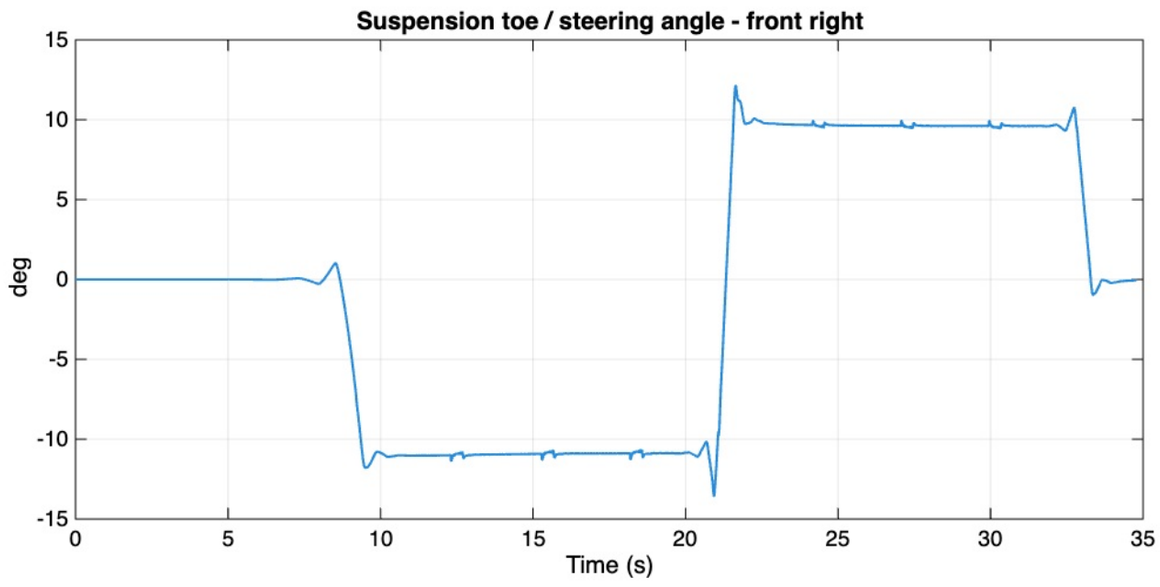

**Tire slip angle - first 3 seconds**

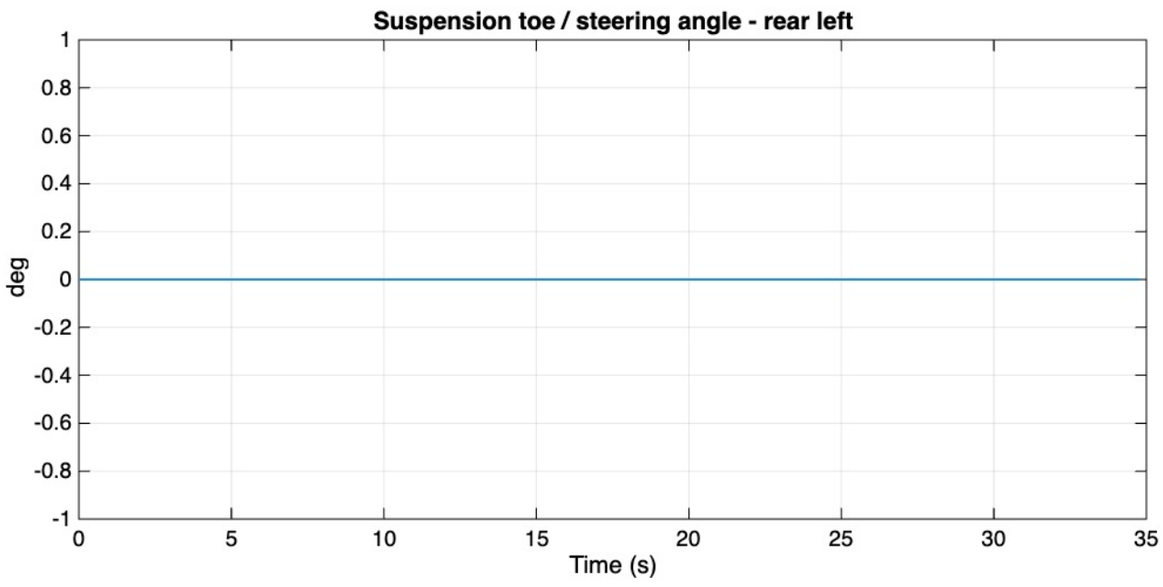


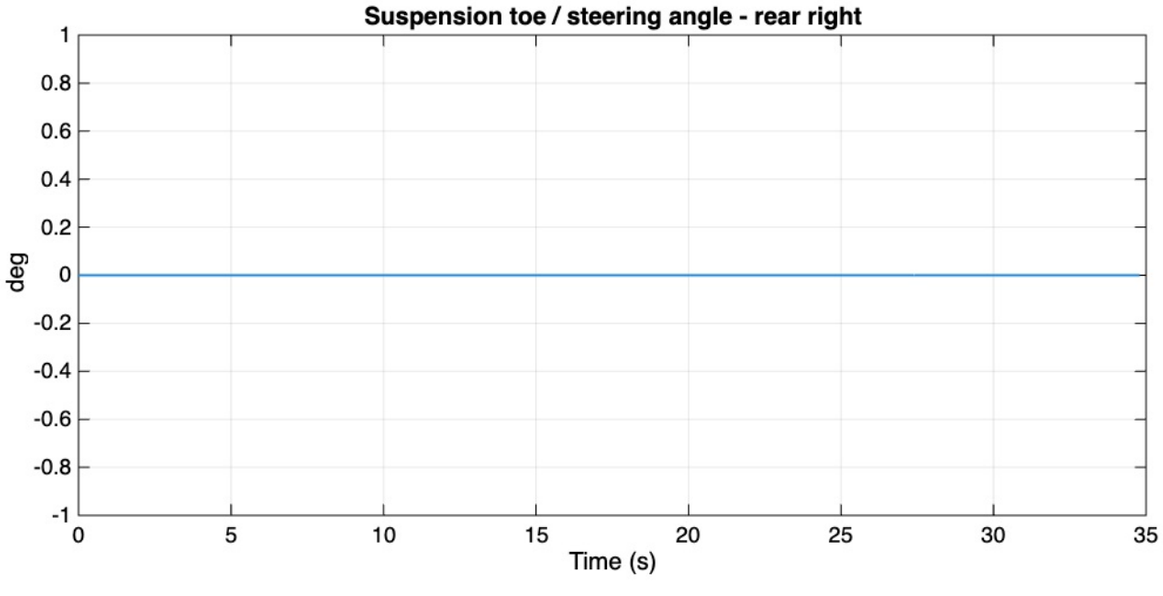
Simple signal plots for tuning - plot page 18

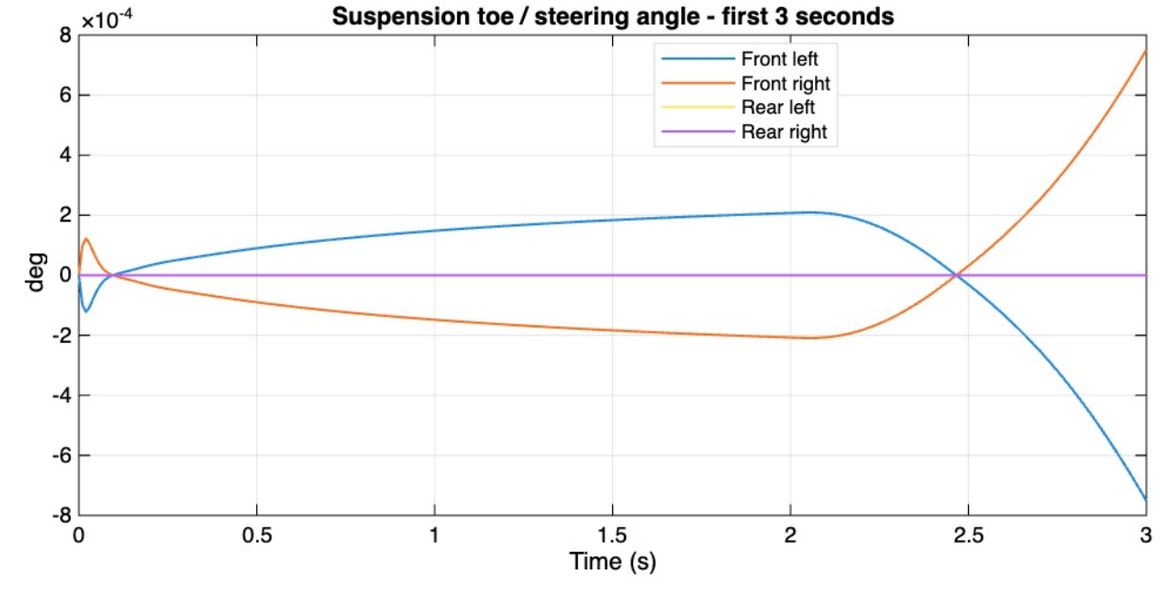


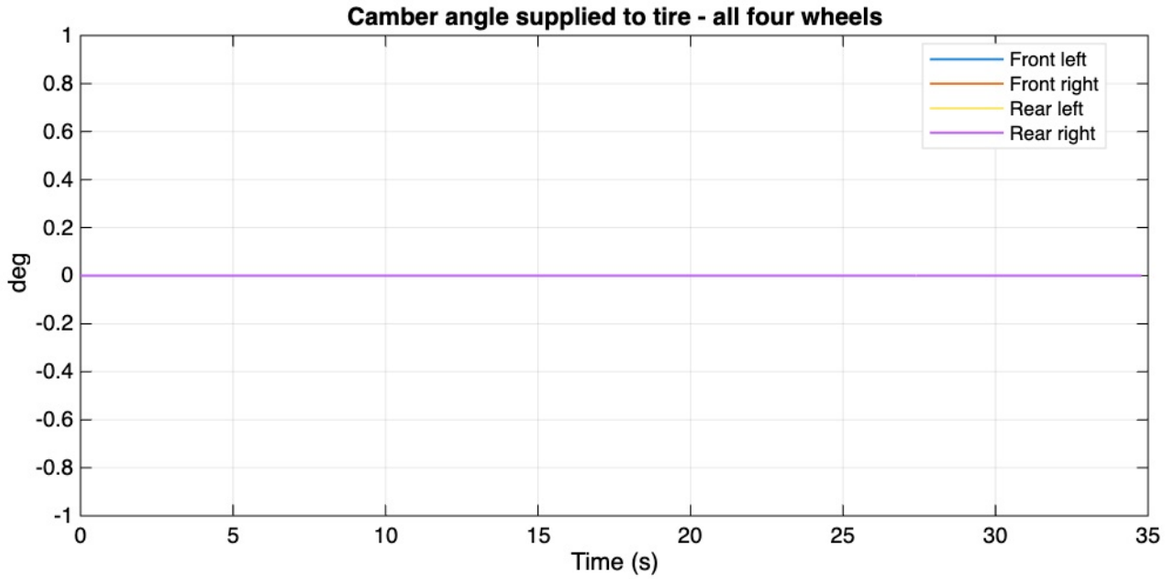



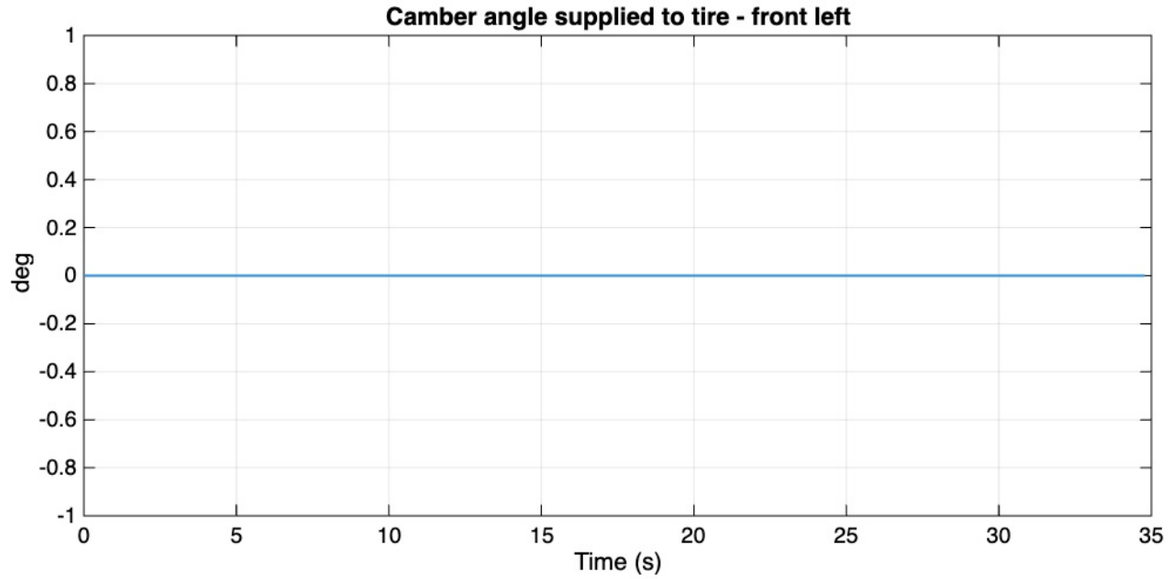



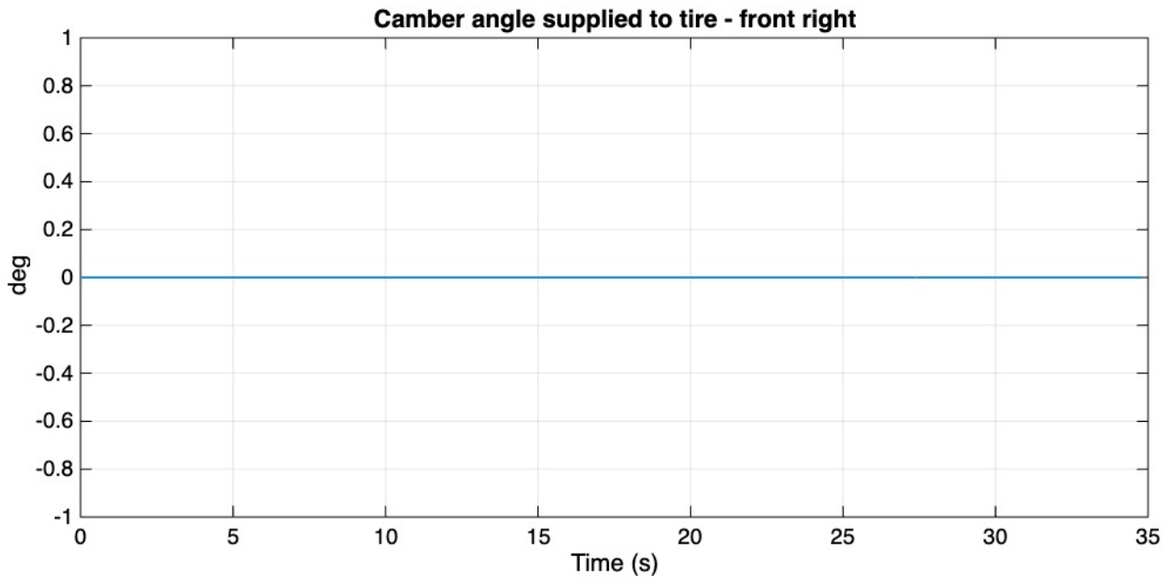



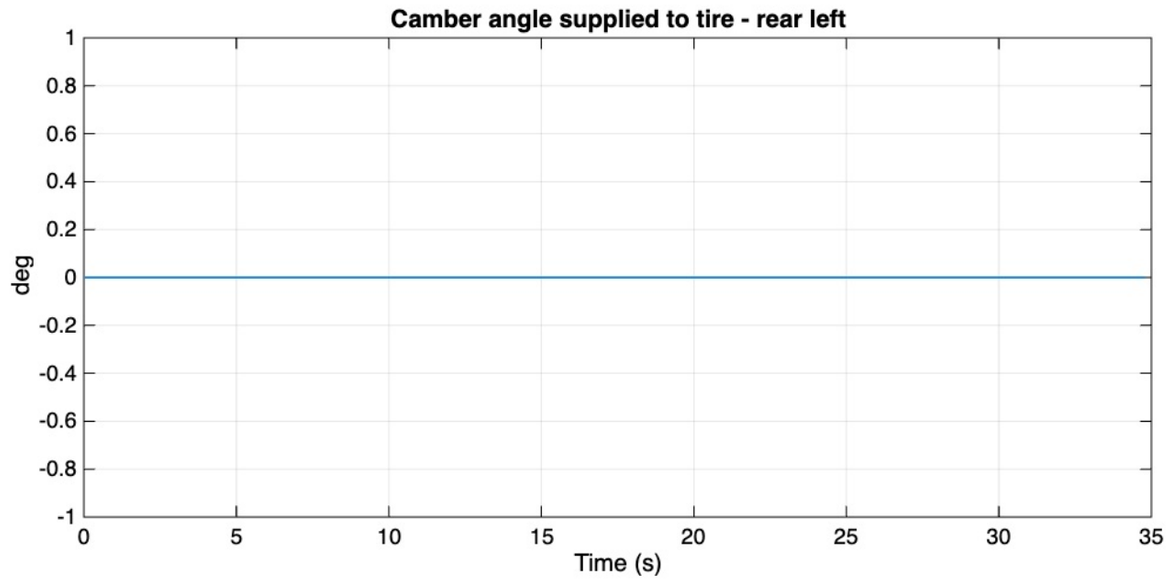



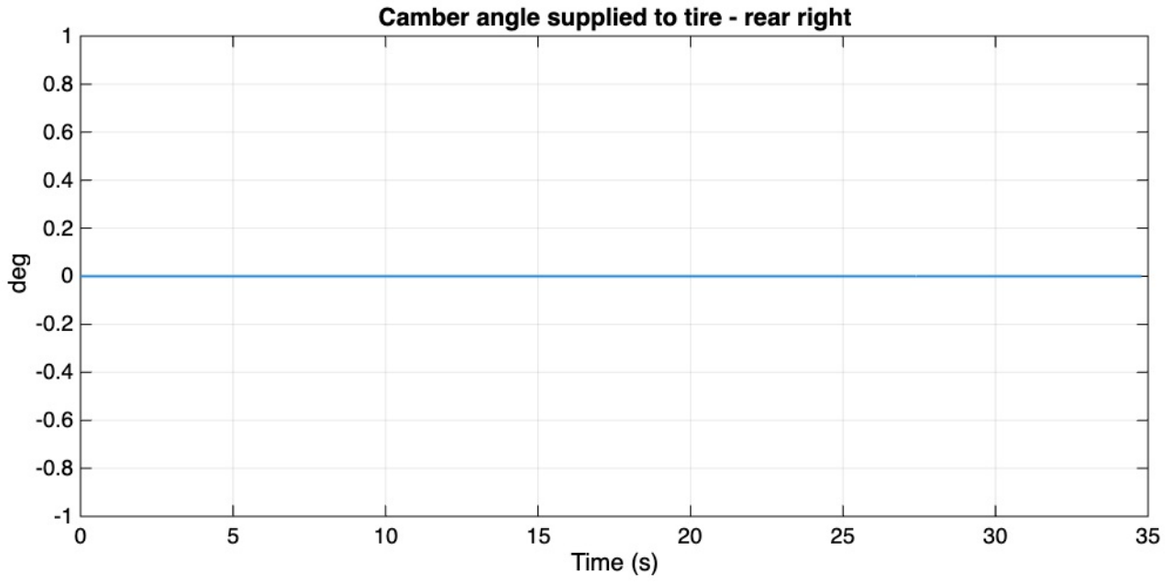



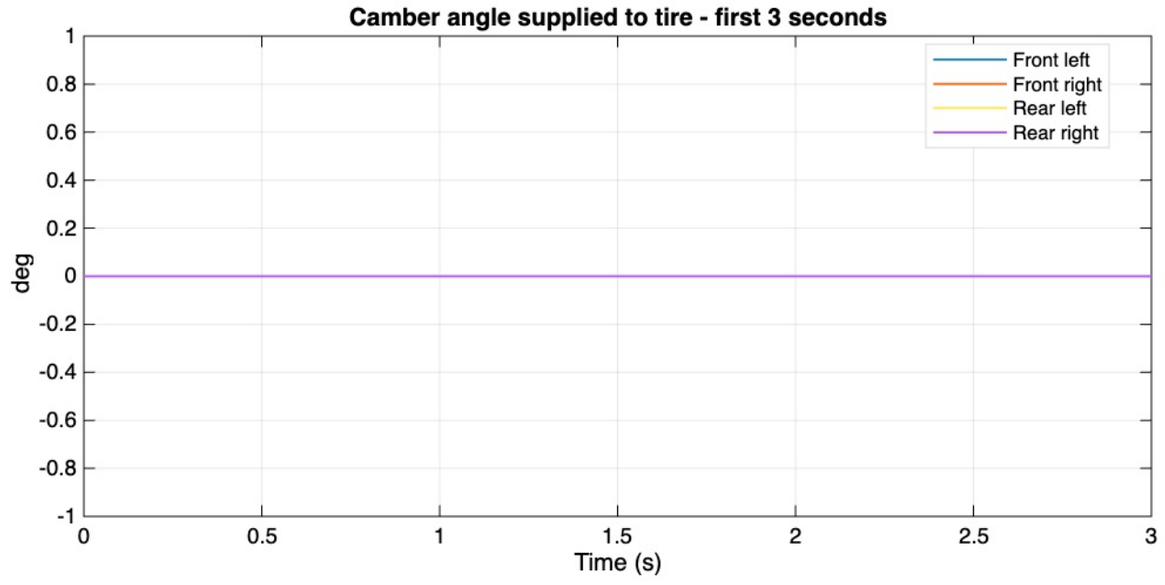

Camber angle supplied to tire - all four wheels
Front left
Front right
Rear left
Rear right
deg
Time (s)


Camber angle supplied to tire - front left
deg
Time (s)

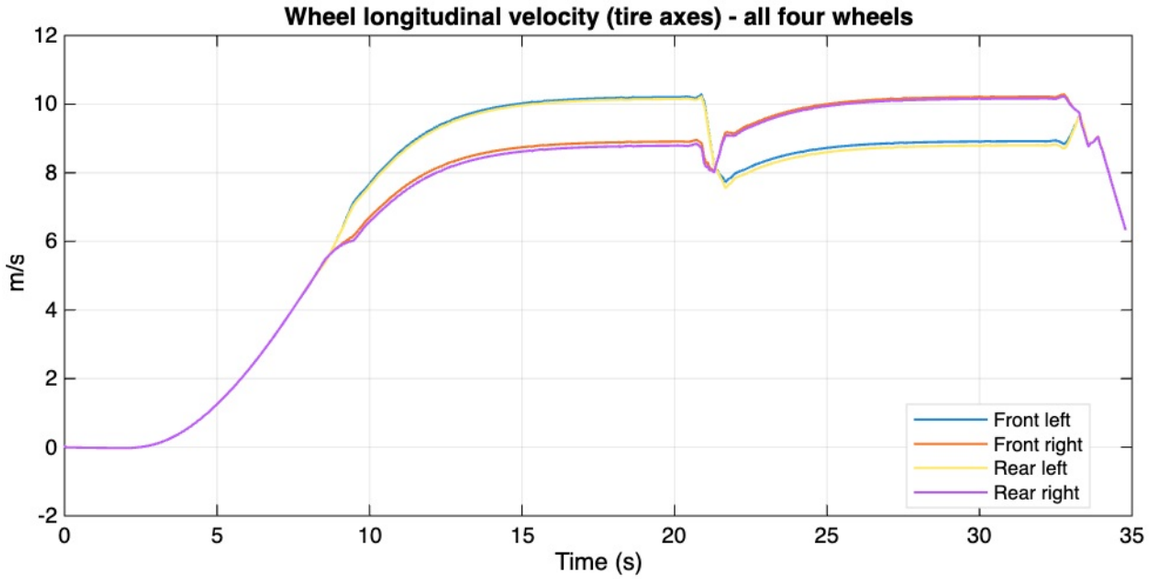

Camber angle supplied to tire - front right
deg
Time (s)

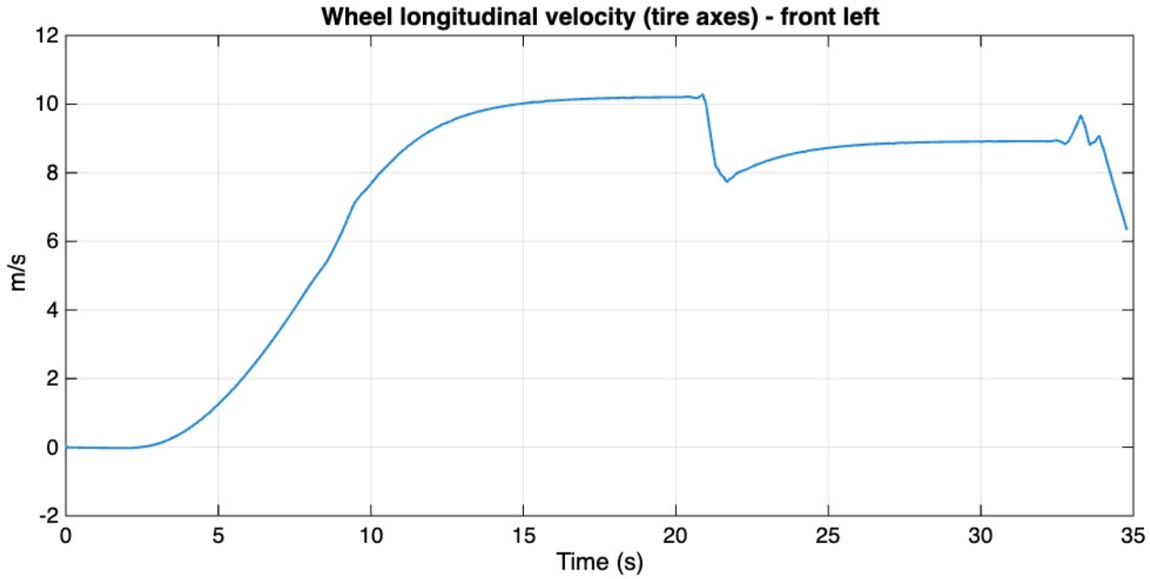

Camber angle supplied to tire - rear left
deg
Time (s)

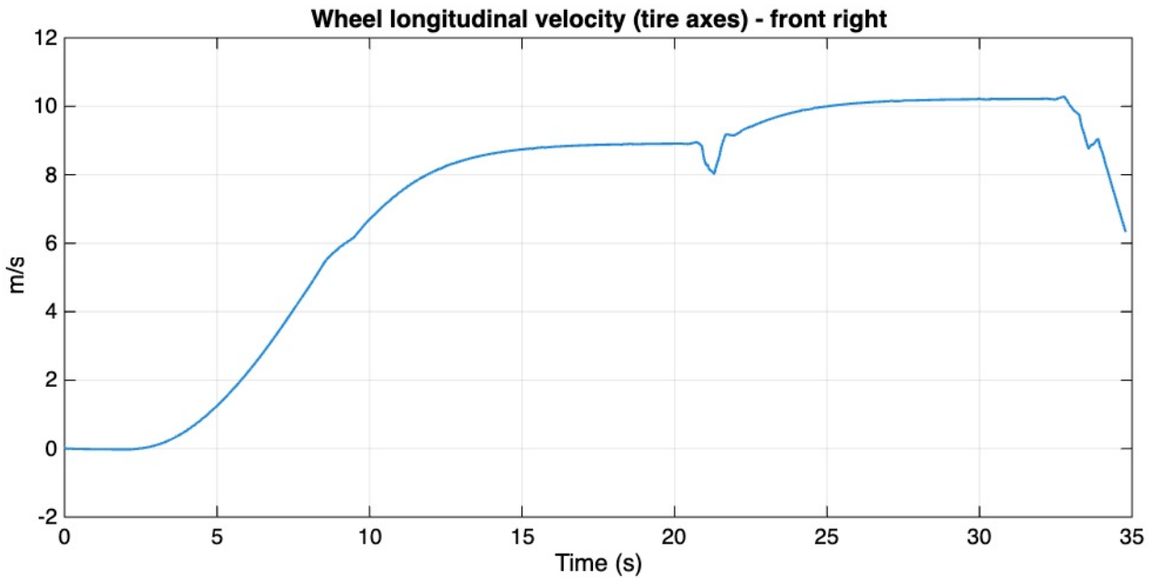

Camber angle supplied to tire - rear right
deg
Time (s)


Camber angle supplied to tire - first 3 seconds
Front left
Front right
Rear left
Rear right
deg
Time (s)

Simple signal plots for tuning - plot page 20

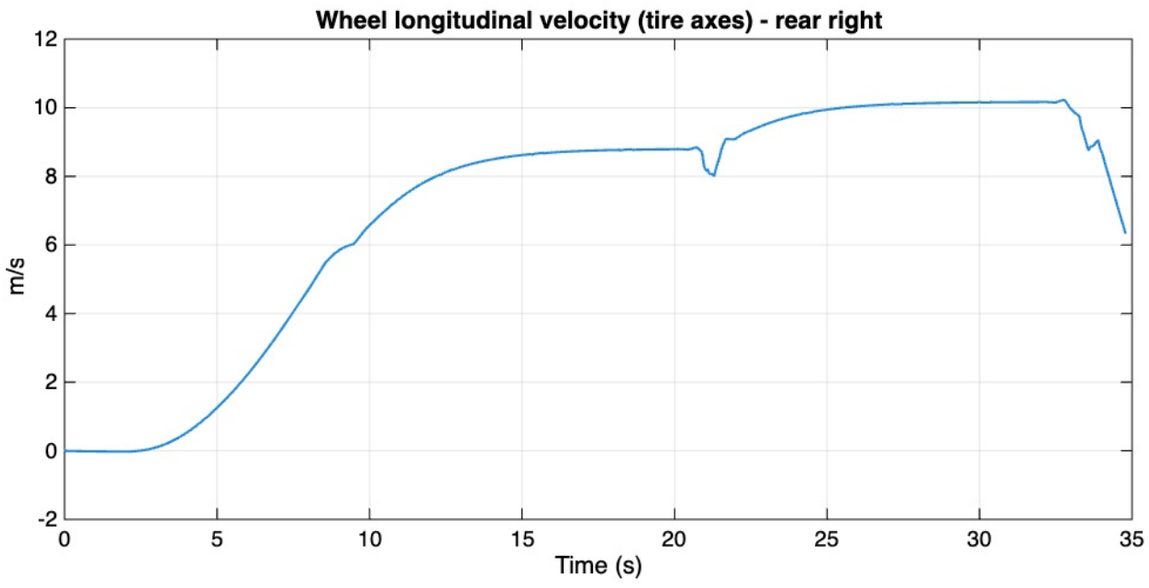


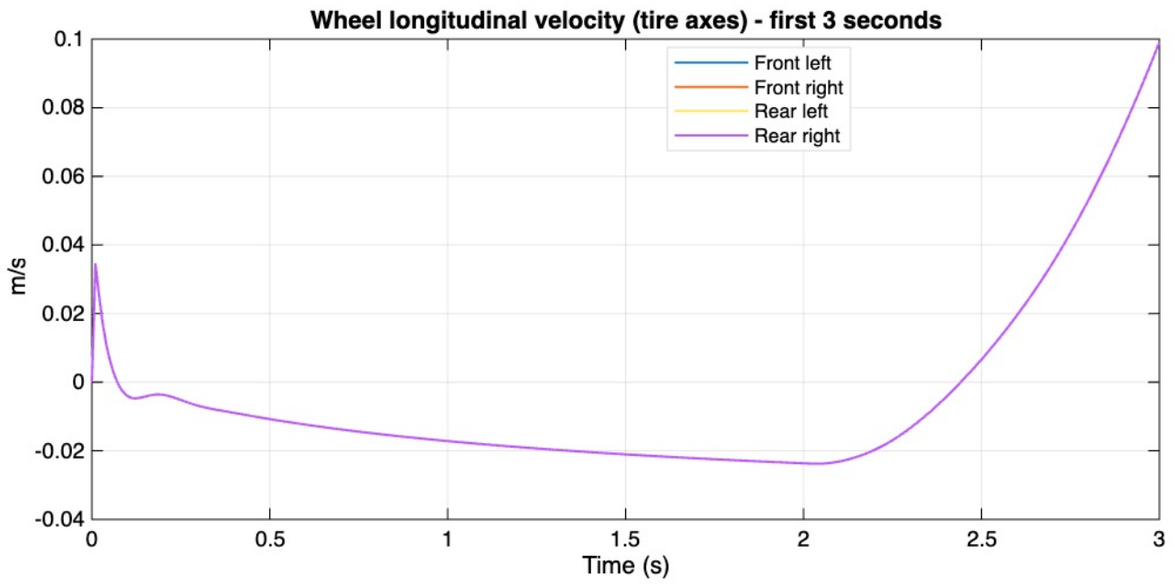



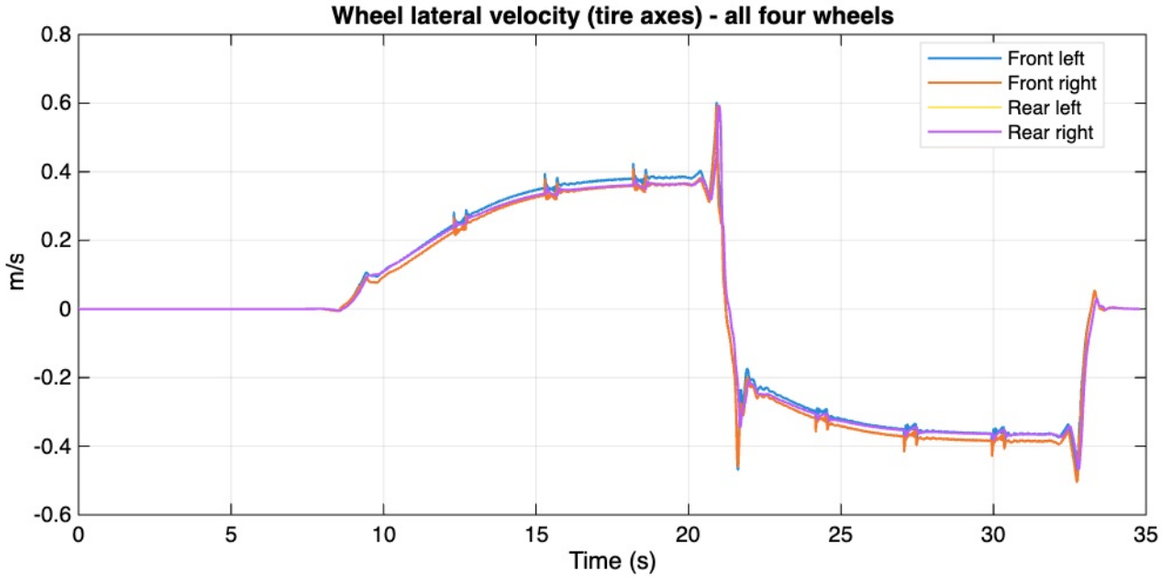



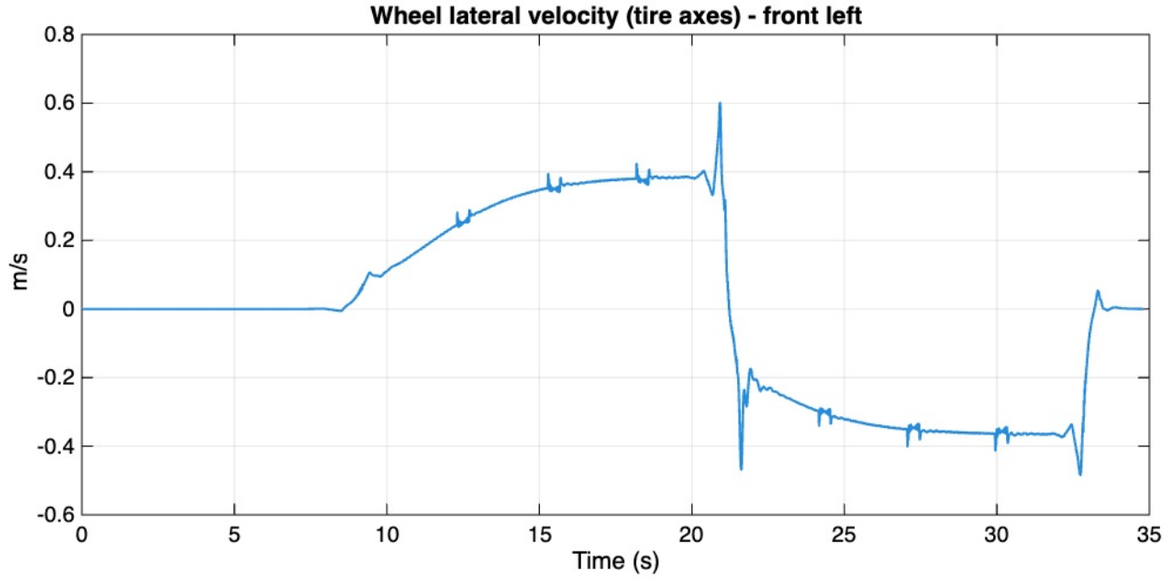


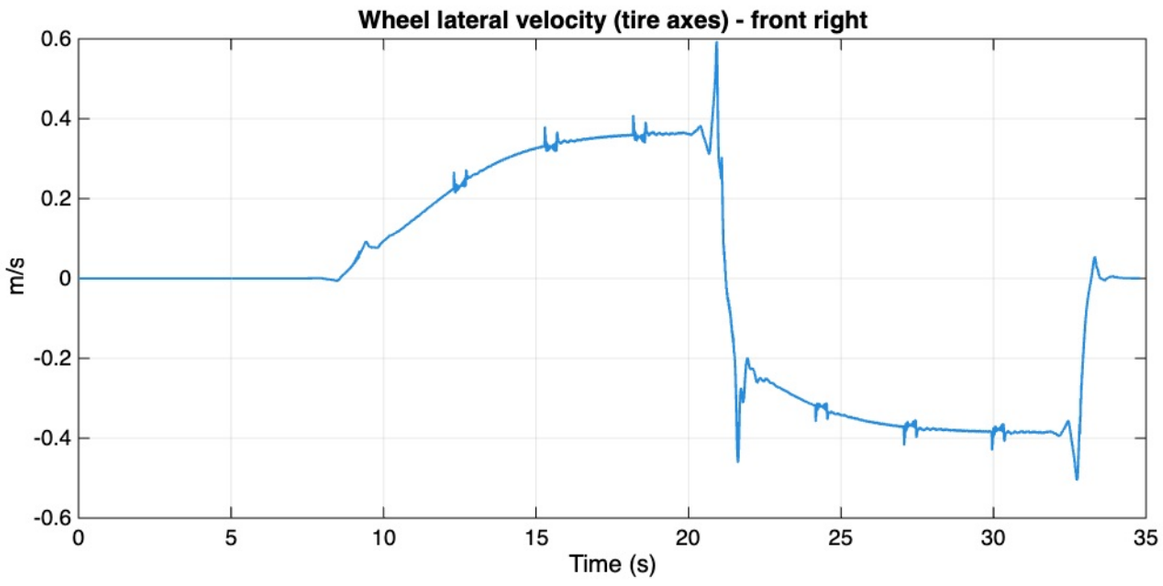


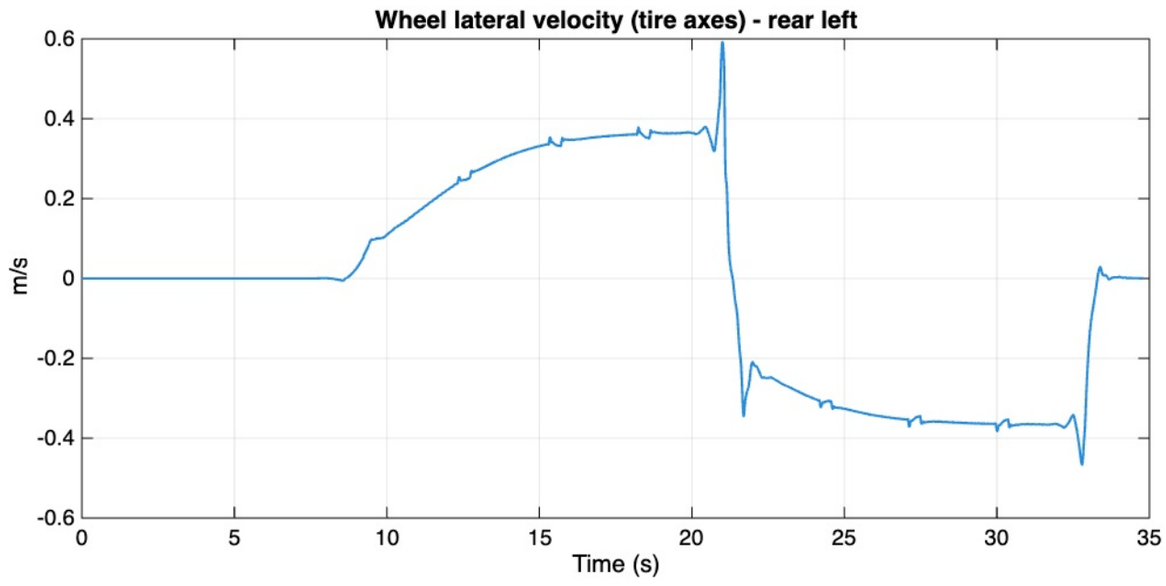


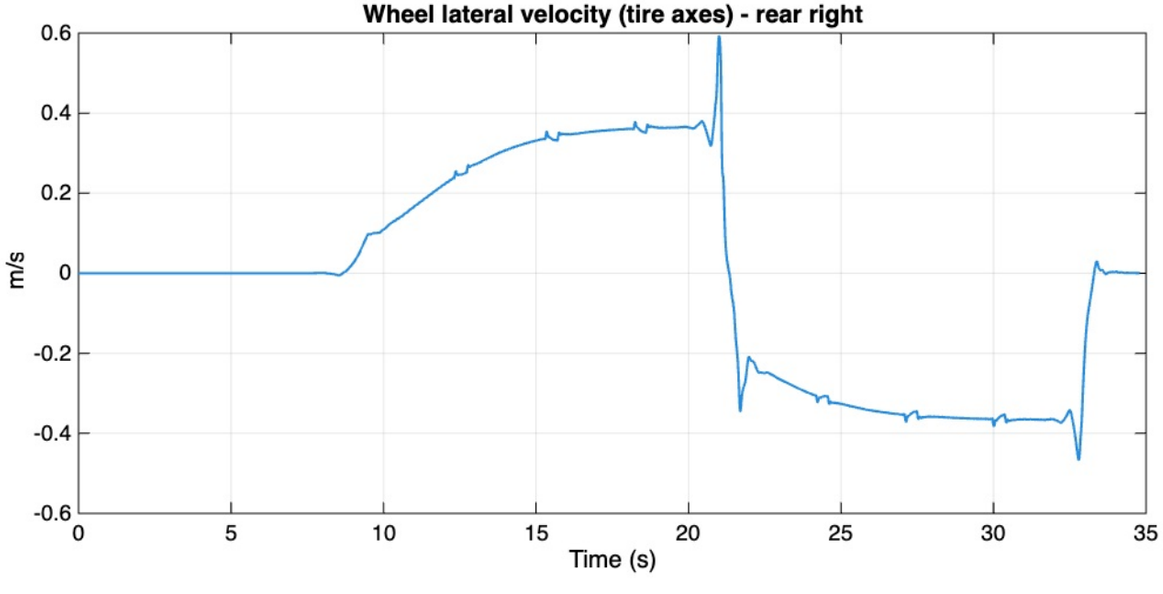




Wheel lateral velocity (tire axes) - all four wheels
Front left
Front right
Rear left
Rear right
m/s
Time (s)


Wheel lateral velocity (tire axes) - front left
m/s
Time (s)

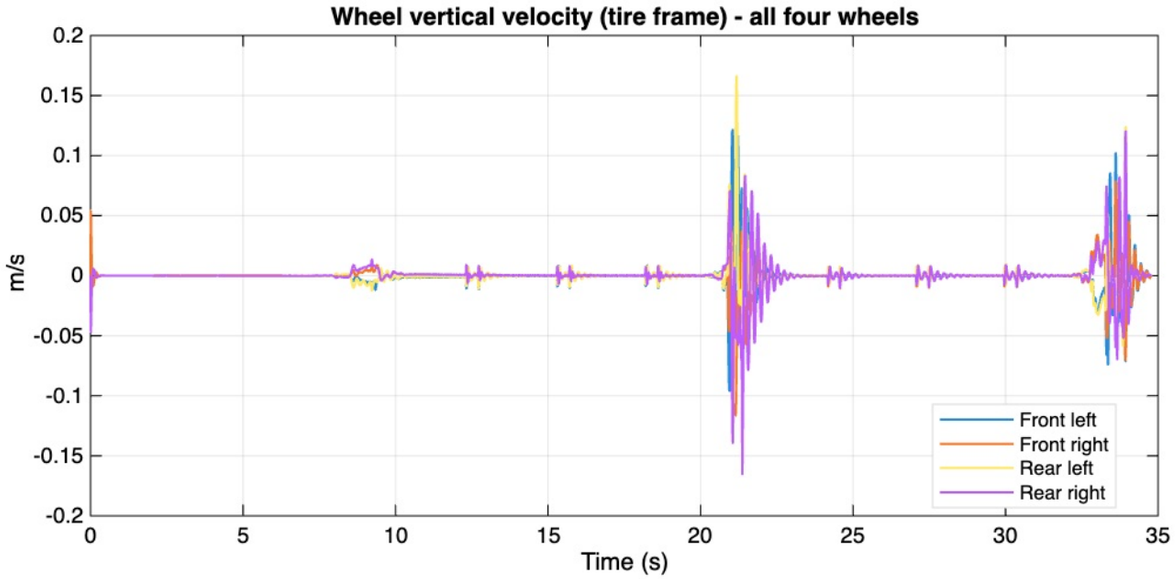

Wheel lateral velocity (tire axes) - front right
m/s
Time (s)

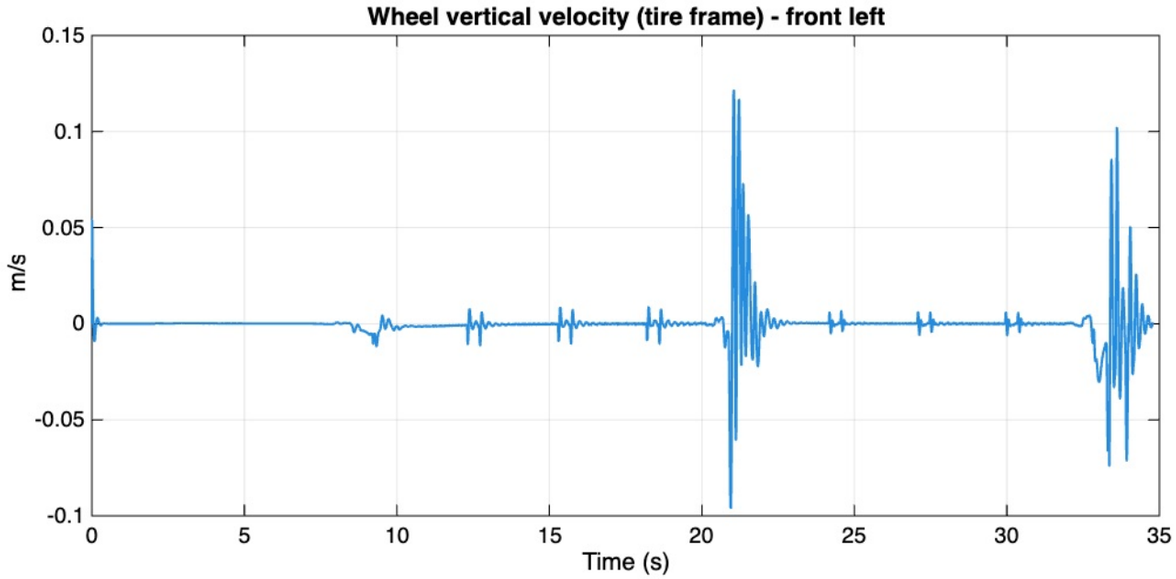

Wheel lateral velocity (tire axes) - rear left
m/s
Time (s)

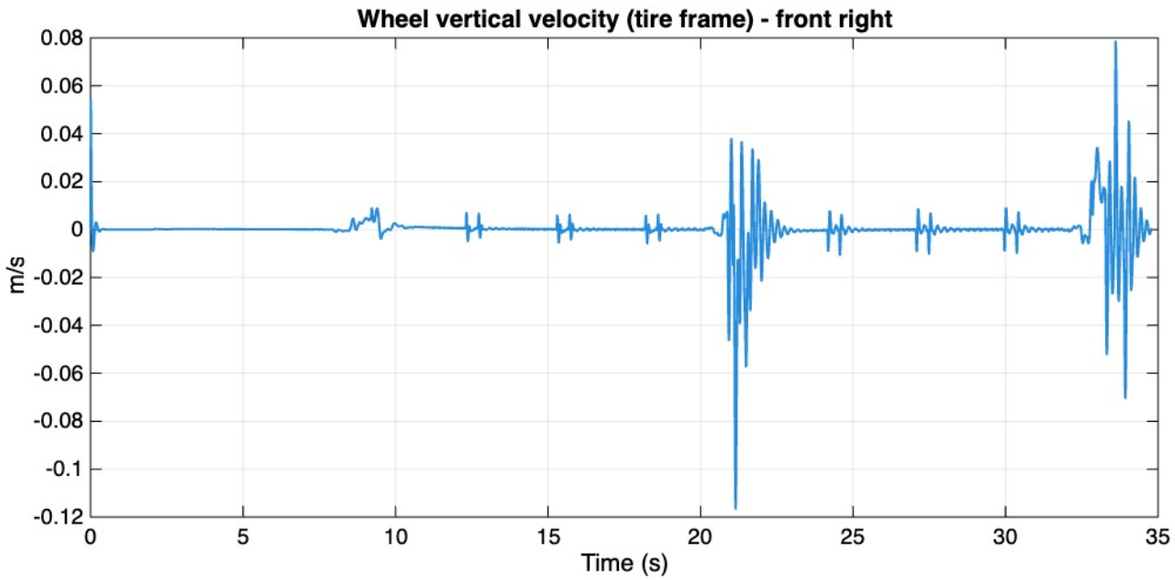

Wheel lateral velocity (tire axes) - rear right
m/s
Time (s)

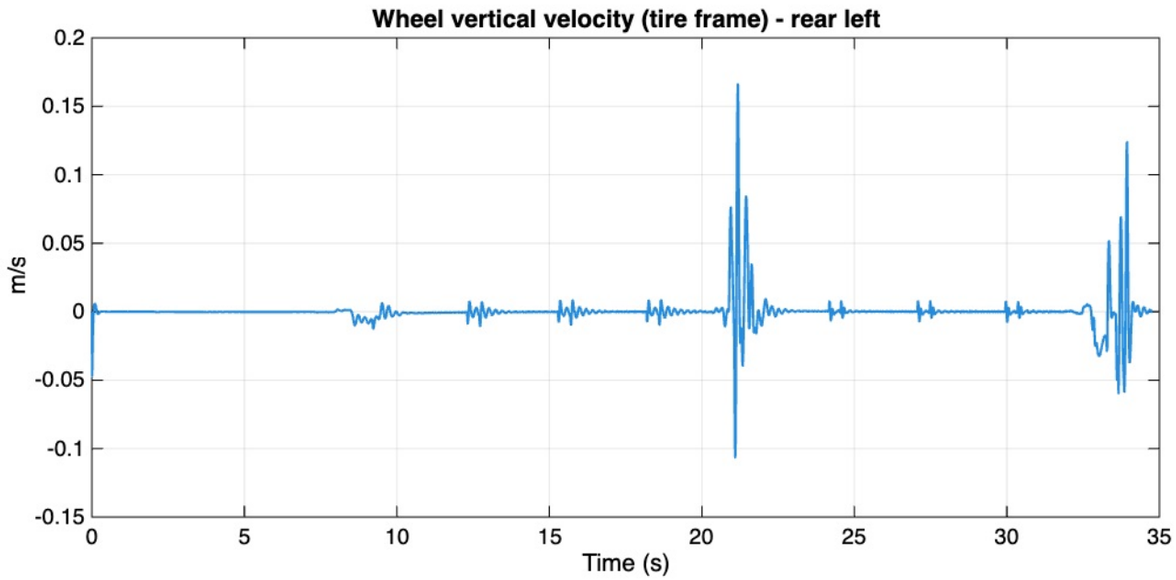

Wheel lateral velocity (tire axes) - first 3 seconds
×10-6
Front left
Front right
Rear left
Rear right
m/s
Time (s)

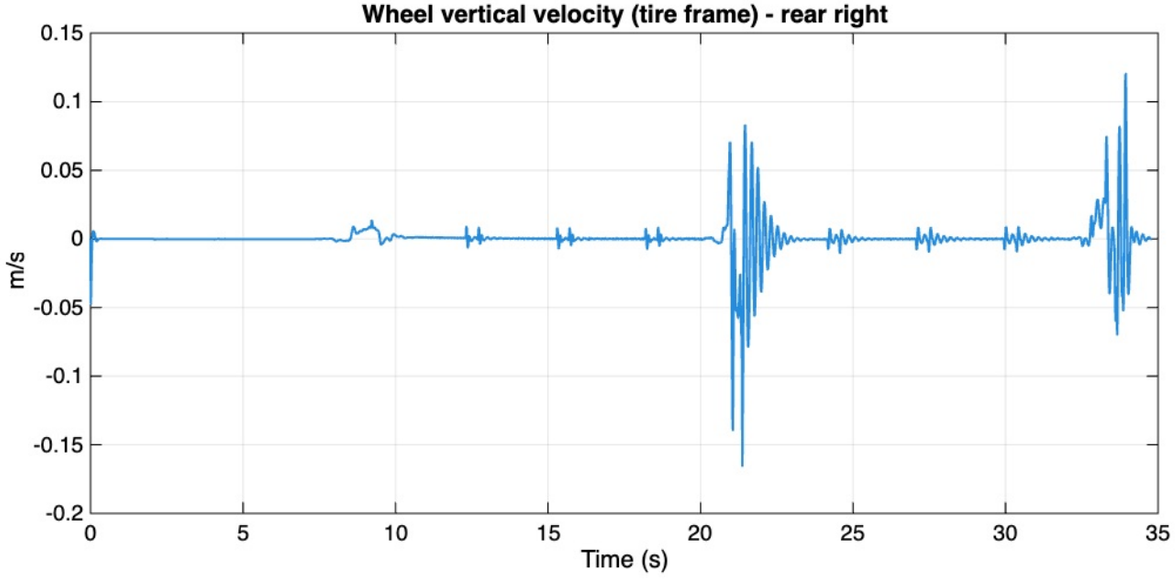
Simple signal plots for tuning - plot page 22

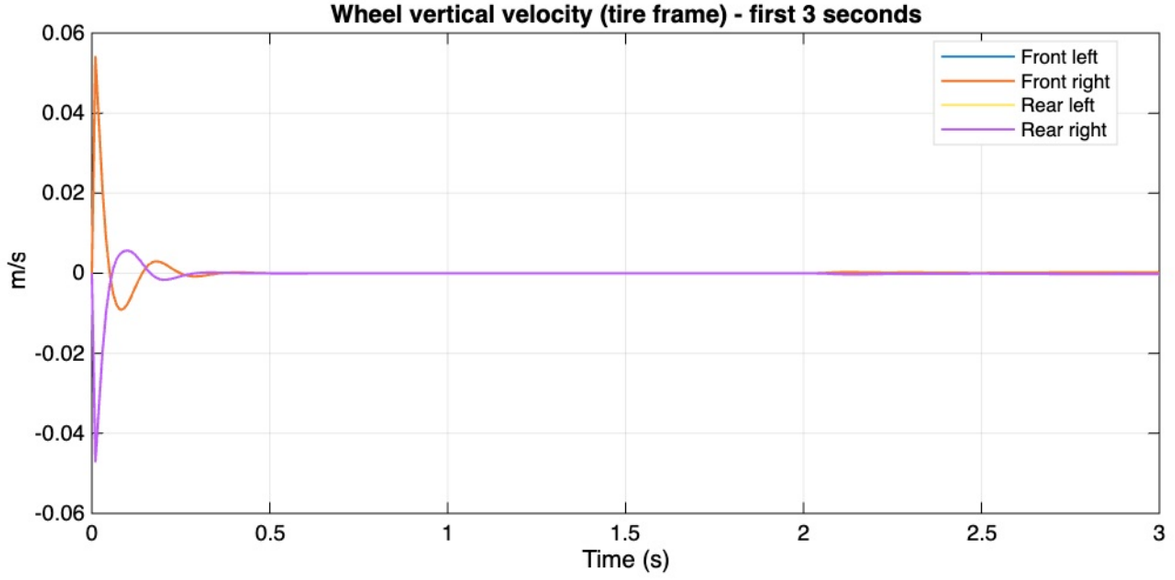


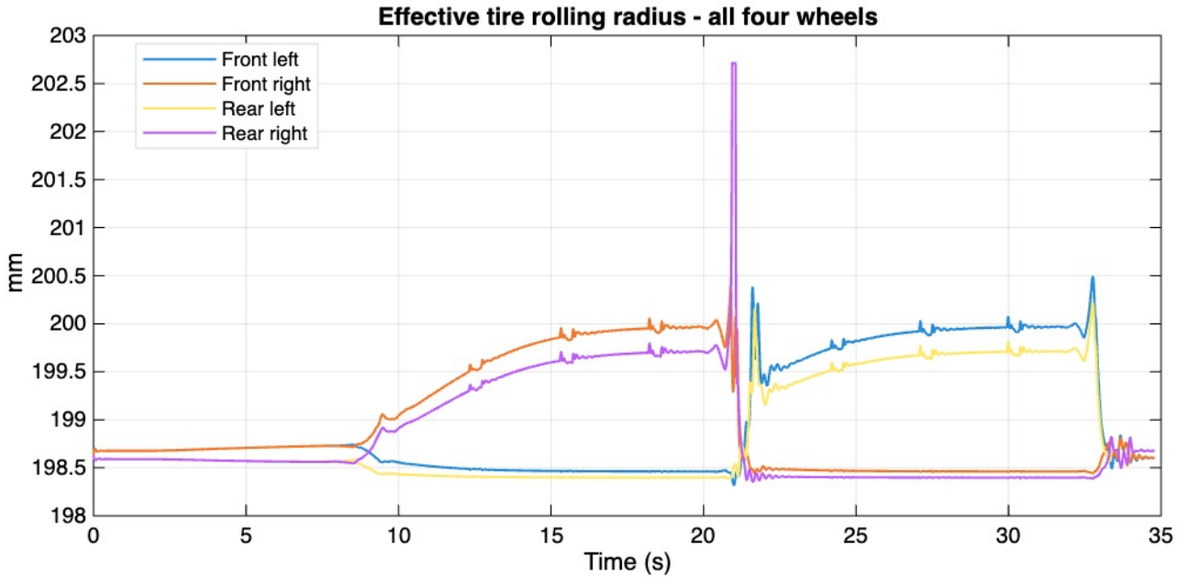



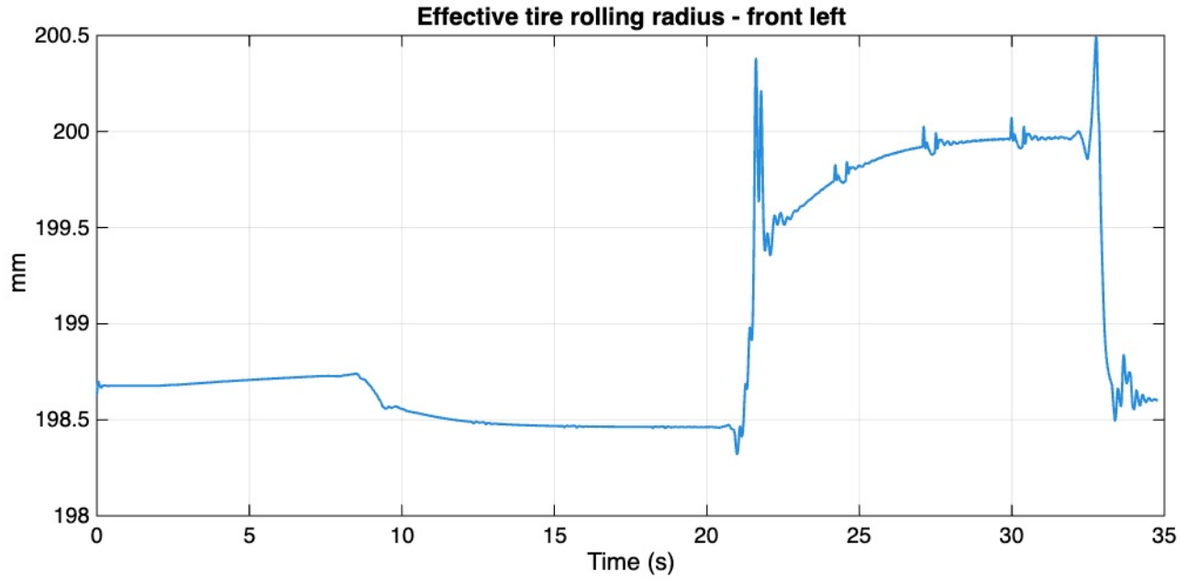



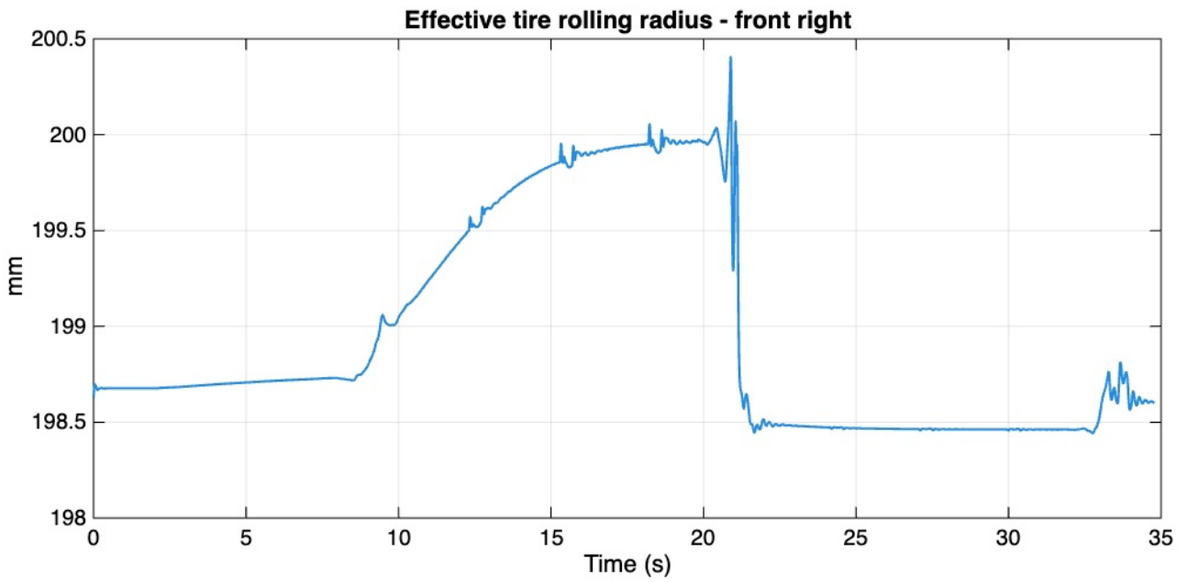



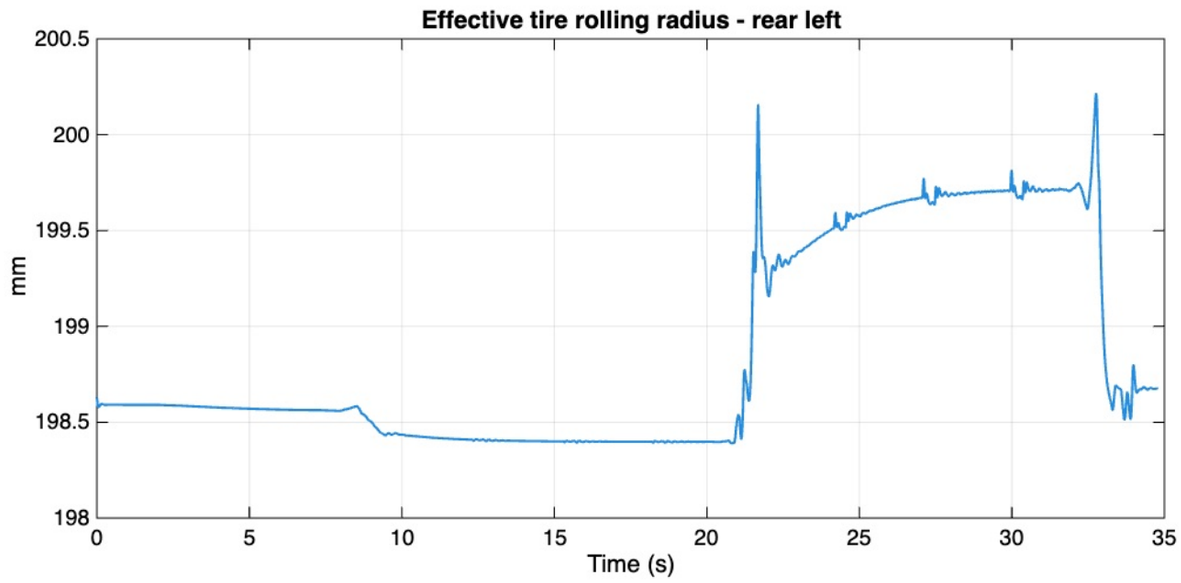



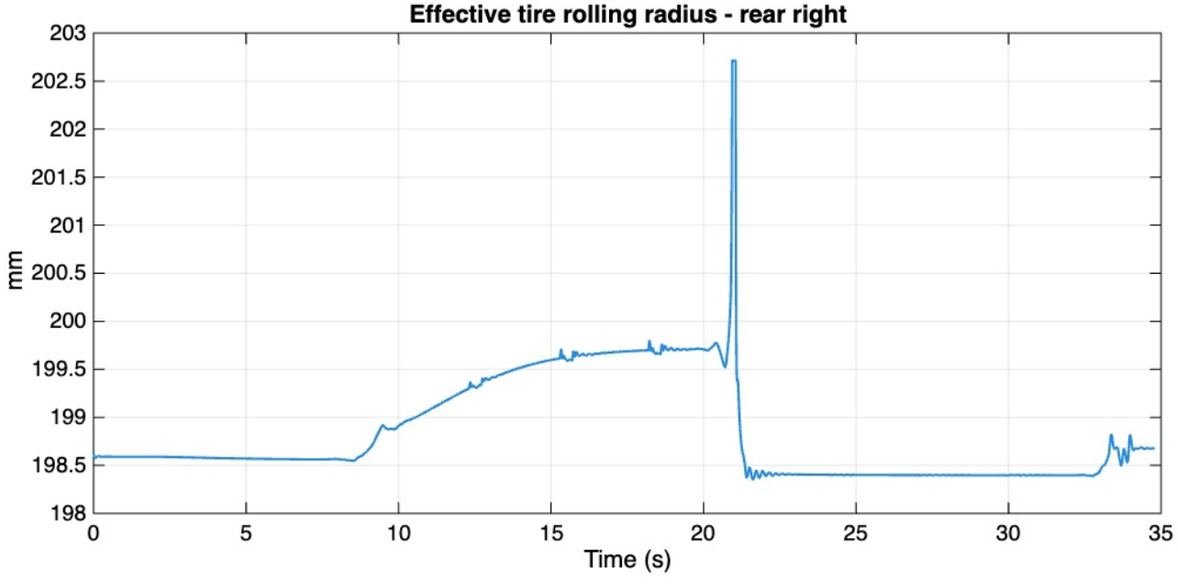



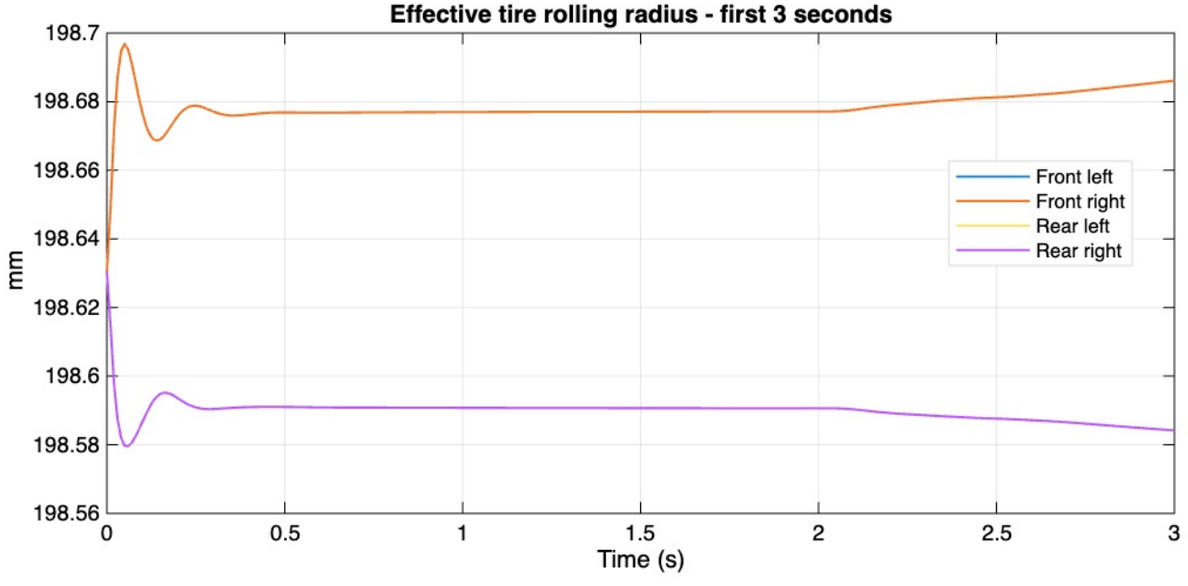

Effective tire rolling radius - all four wheels
Front left
Front right
Rear left
Rear right
mm
Time (s)


Effective tire rolling radius - front left
mm
Time (s)


Effective tire rolling radius - front right
mm
Time (s)

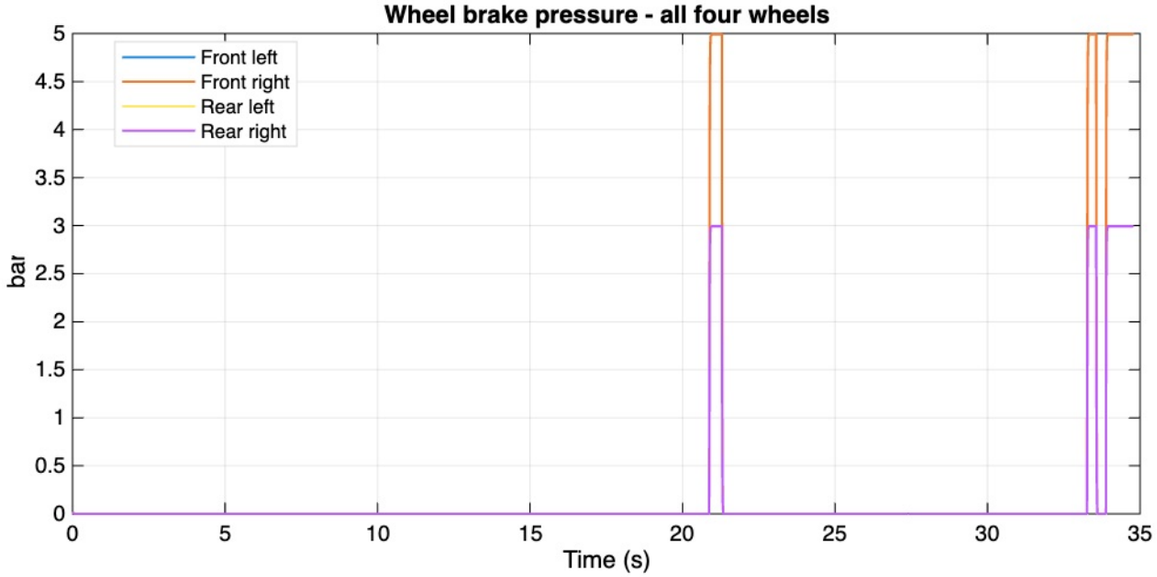

Effective tire rolling radius - rear left
mm
Time (s)


Effective tire rolling radius - rear right
mm
Time (s)

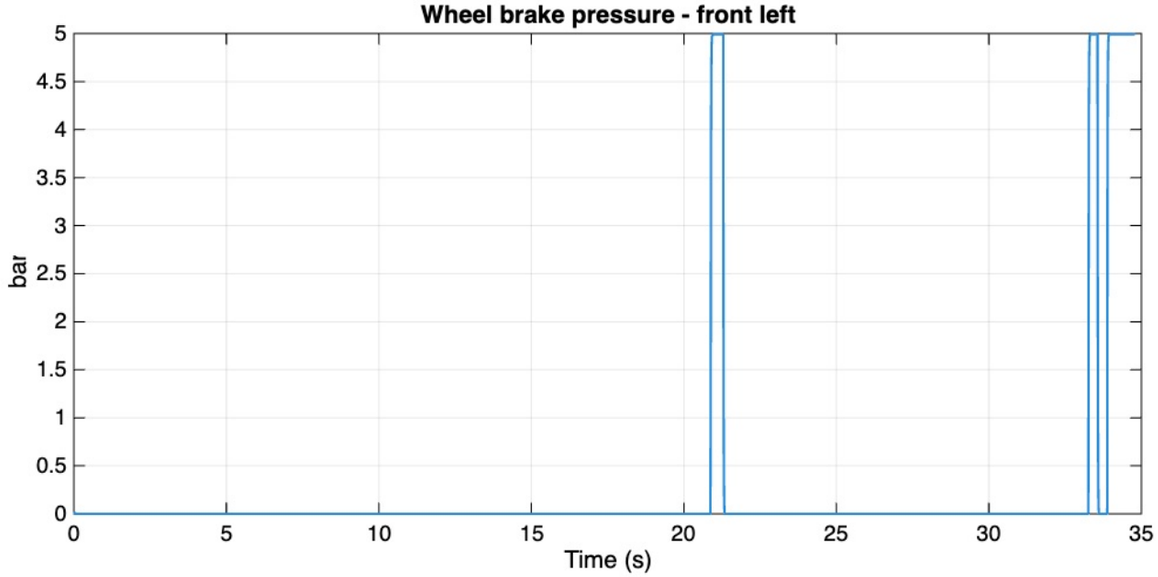

Effective tire rolling radius - first 3 seconds
Front left
Front right
Rear left
Rear right
mm
Time (s)

Simple signal plots for tuning - plot page 24

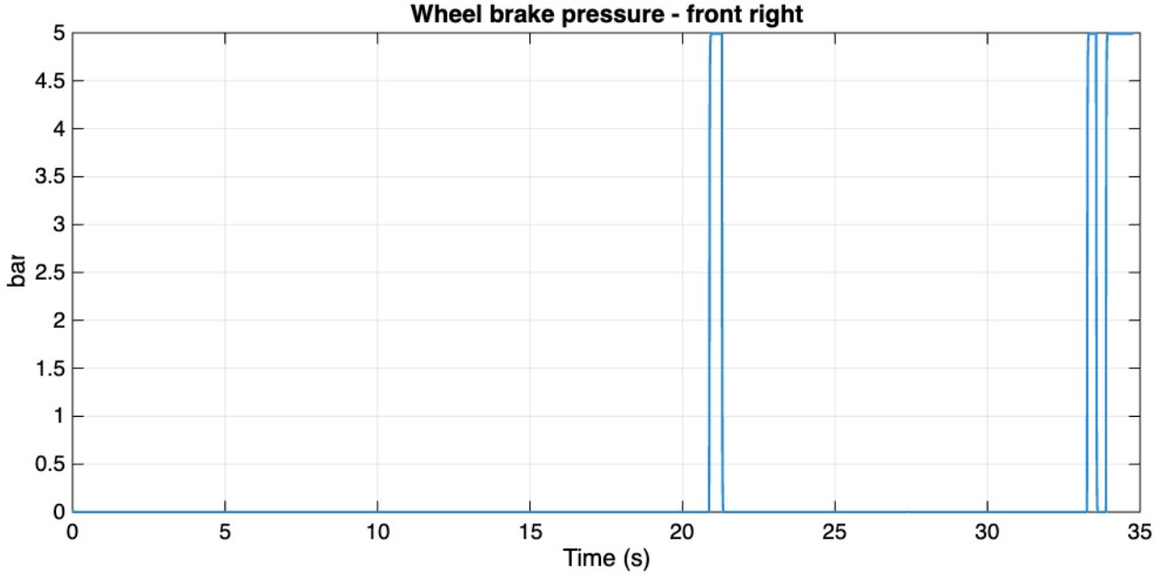
**Wheel brake pressure - all four wheels**



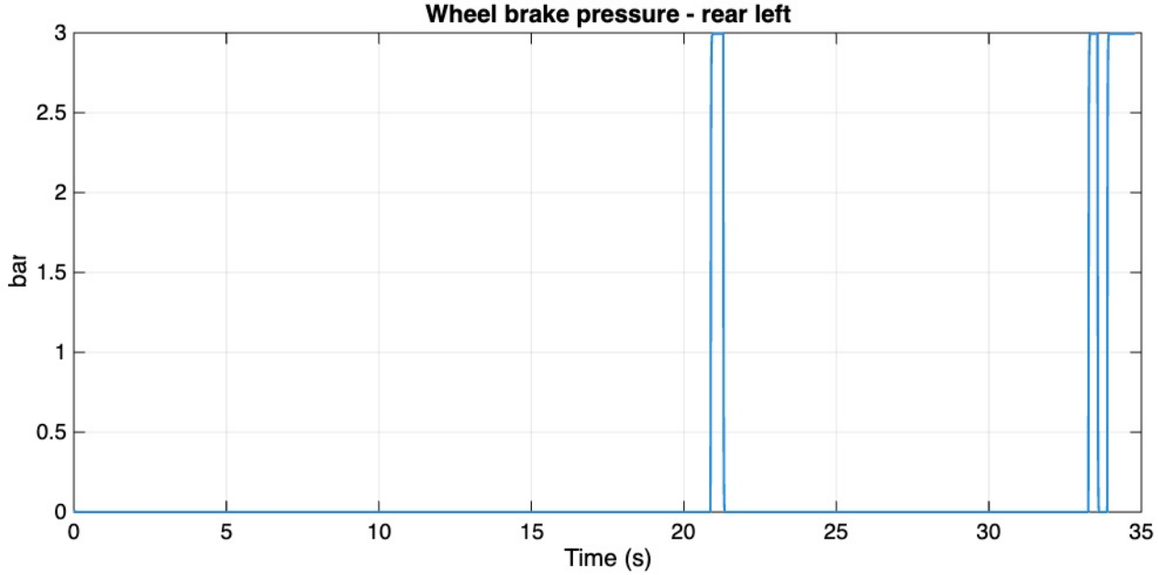

**Wheel brake pressure - front left**



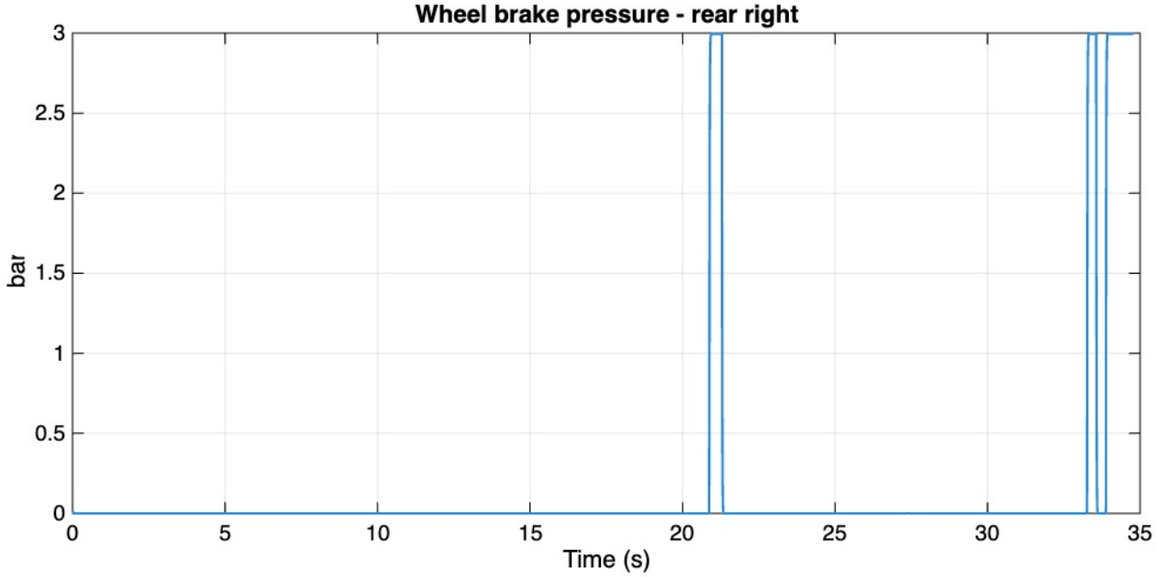

**Wheel brake pressure - front right**




**Wheel brake pressure - rear left**




**Wheel brake pressure - rear right**




**Wheel brake pressure - first 3 seconds**

